

The Physics of Physiology

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Lecture 1: *Physical units for physiology & introduction to multiscale modeling*

Lecture 1: *Physical units for physiology*

Life depends on the **flow of energy and increase in entropy** associated with mechanical, electromagnetic and chemical processes. By the end of this lecture you should understand why we choose the **six fundamental units** needed for all the biophysical mechanisms that underpin physiology. We also show how processes at the macroscale of physiology are linked with atomic level physics and that the small number of physical constants needed for physiology (e.g. Faraday's constant and the gas constant) use Avogadro's number to bridge the scale from atoms to cells and tissues.

The overall approach to multiscale modeling of human physiology is also described briefly here, in order to motivate the modeling techniques presented in upcoming lectures.

7 units:

- **Joule (J)**
 - **Second (s)**
 - **Meter (m)**
- } space-time + energy
- **Coulomb (C)** - count electrons
 - **Candela (Cd)*** - count photons
 - **Mole (mol)** - count atoms
 - **Entropy (e)** - count probable states

*A photon, with energy $h\nu$, is a quantum of an electromagnetic field. The Candela is only needed when a model requires the number of discrete photons, rather than just their summed energy. Mammalian eyes can detect 1 photon ([ref](#)).

Physiological processes

1. **Mechanics (J,s,m,e):** (i) Solids; (ii) Fluids
2. **Heat transfer (J,s,e)**
3. **Electro-physiology (J,s,C)**
4. **Electro-magnetic (J,s,m,C)**
5. **Membrane transporters (J,s,mol,C)**
6. **Metabolic pathways (J,s,mol)**

Physical constants

$$c \approx 3.10^8 \text{ m.s}^{-1} (=299\,792\,458 \text{ m.s}^{-1})$$

$$N_A = 6.022 \times 10^{23} \text{ particles.mol}^{-1} + k_B = 1.38 \times 10^{-23} \text{ J.K}^{-1} \Rightarrow R = k_B \cdot N_A = 8.314 \text{ J.mol}^{-1}.\text{K}^{-1} \Rightarrow RT \cong 2.5 \text{ kJ.mol}^{-1} \text{ at } 300\text{K}$$

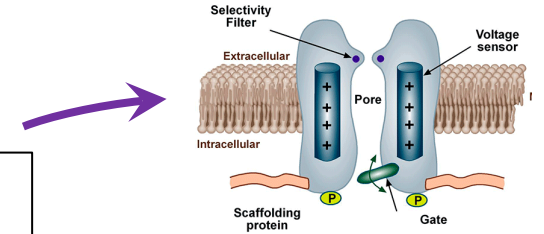
$$+ q_e = 1.602 \times 10^{-19} \text{ C} \Rightarrow F = q_e \cdot N_A = 0.965 \times 10^5 \text{ C.mol}^{-1}$$

For n moles of an ideal gas, RT (J.mol^{-1}) $\times n$ (mol) gives nRT (J), which is expressed by the ideal gas law as $PV=nRT$ (J) – see later.

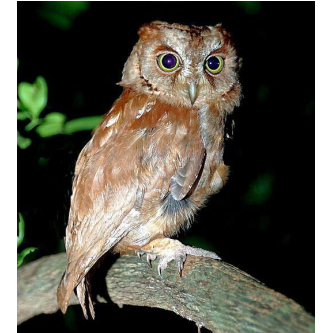
$$\Delta u^q = RT \ln \frac{q_o}{q_i} + z_q F \Delta u^{membrane}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 J.mol^{-1} J.mol^{-1} C.mol^{-1} J.C^{-1}

Equation 5-6 in Boron & Boulpaep textbook (3rd Edⁿ).



Detect 1 electron

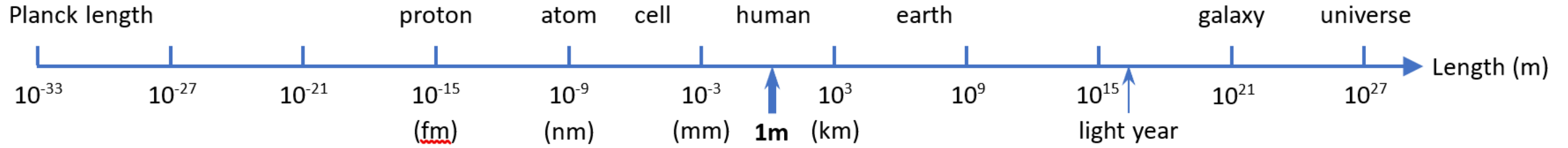


Detect 1 photon

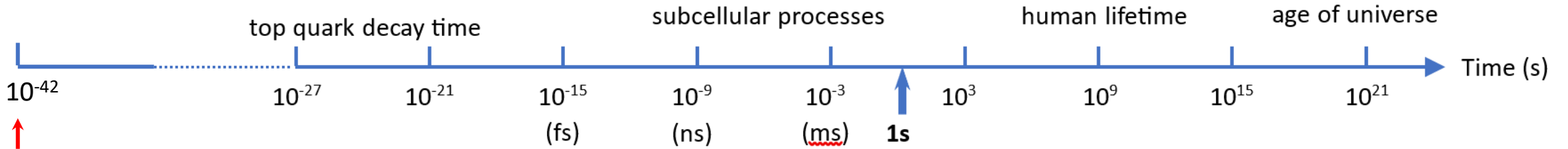


300m olfactory sensors
1 part in 10¹²

Length in meters



Time in seconds



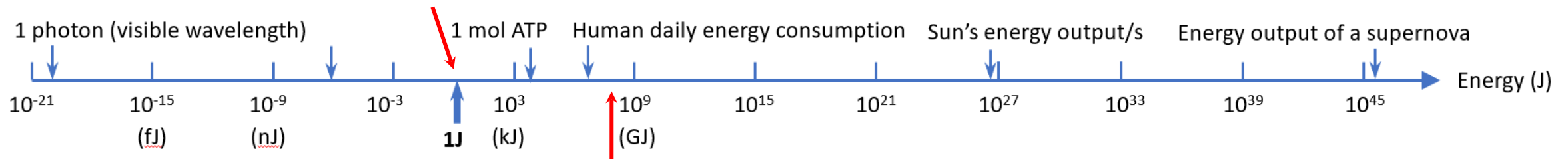
‘Planck time’ (time for photon to traverse a Planck length) = $10^{-33}/3 \times 10^8 \text{ m.s}^{-1}$

1J is energy of a 0.1 kg apple dropping 1m to give Newton a bump on the head.

1J is also the energy output by the heart in one beat.

4J per gram per degree Kelvin is the heat capacity of water (= $4 \text{ kJ.kg}^{-1}.\text{K}^{-1}$).

Energy in Joules



300 MJ is the energy contained in a fully charged Tesla car battery
.. and is the energy stored in 10 kg of body fat (30 MJ/kg)

1 kJ.s^{-1} (1 kW) is power output of an electric kettle (1L or 1kg).
Heating 1 kg of water from 0-100 degK takes 400s ($\rightarrow$ 400kJ).

Respiration: ΔG for ATP $\approx 30 \text{ kJ.mol}^{-1}$. 1 mol glucose (180g) generates 36 mol ATP i.e. $\approx 1 \text{ MJ.mol}^{-1}$, so 180g bar of chocolate is 1MJ.

Note on units, unit prefixes and frequently used constants

<u>Prefix</u>			<u>Useful units in physiology</u>
milli-	10^{-3}		Litres (L) $1 \text{ L} = 10^{-3} \text{ m}^3$
micro-	10^{-6}		
nano-	10^{-9}		$1 \text{ kPa} = 10^3 \text{ J.m}^{-3} = 1 \text{ J.L}^{-1}$ ($1 \text{ kPa} \times 1 \text{ L.s}^{-1} = 1 \text{ J.s}^{-1}$)
pico-	10^{-12}		
femto-	10^{-15}		$1 \text{ mM} = 1 \text{ millimol.L}^{-1} = 10^{-3} \text{ mol.L}^{-1} = 1 \text{ mol.m}^{-3}$

- The volume of a cell is about 1 pL ($=10^{-12} \text{ L}$ or 10^{-15} m^3) which is $(10^{-5} \text{ m})^3$ or $(10 \text{ }\mu\text{m})^3$.

A concentration of 1 mM (= millimoles per Liter) is $10^{-3} \text{ mol.L}^{-1}$ or 1 mol.m^{-3} .

$1 \text{ mM} \times 1 \text{ pL}$ is $1 \text{ mol.m}^{-3} \times 10^{-15} \text{ m}^3$ or 10^{-15} mol or 1 fmol (femto is 10^{-15}).

So if we assume a volume of **1 pL**, then **1 mM** is equivalent to a molar quantity of **1 fmol**

- $R = k_B \cdot N_A = 1.38 \times 10^{-23} \text{ J.K}^{-1} \times 6.022 \times 10^{23} \text{ mol}^{-1} = 8.314 \text{ J.mol}^{-1}.\text{K}^{-1}$

Therefore at 300K, **$RT \approx 2.5 \text{ kJ.mol}^{-1}$** $\times 10^{12} \text{ nJ.kJ}^{-1} \times 10^{-15} \text{ mol.fmol}^{-1}$ gives **$RT \approx 2.5 \times 10^{-3} \text{ nJ.fmol}^{-1}$**

This is thermal energy due to Brownian motion.

For a gas at atmospheric pressure $pV = nRT$, where $p \approx$ **100 kPa** (the *energy density*),

$$V = \frac{2.5 \text{ kJ.mol}^{-1}}{100 \text{ J.L}^{-1}} = \textbf{25 L.mol}^{-1} \quad (22.4 \text{ L.mol}^{-1} \text{ at STP})$$

- Faraday's constant is approx $F = 0.965 \times 10^5 \text{ C.mol}^{-1}$ or $0.965 \times 10^{-10} \text{ C.fmol}^{-1}$ or $0.0965 \text{ nC.fmol}^{-1}$

1 V is 1 J.C^{-1} or 1 nJ.nC^{-1} and 1 nA is 1 nC.s^{-1} .

$$\frac{RT}{F} \approx 2.5 \text{ kJ.mol}^{-1} / 10^5 \text{ C.mol}^{-1} = 0.025 \text{ J.C}^{-1} \quad \text{or} \quad \frac{RT}{F} \approx \textbf{25mV}$$

Thermodynamics[†]:

Work and heat are two different faces of a single concept - *energy*.

.. the English physicist James Joule figured this out and hence the unit of energy is the Joule (J).

Energy is the *capacity to do work*.

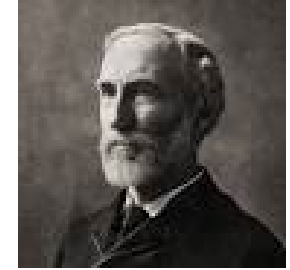
Work is *motion against an opposing force*.

Heat (thermal vibration of atoms) is intimately linked with entropy (see next slide).

An **open system** can exchange both energy and matter with its surroundings.

A **closed system** can exchange energy but not matter with its surroundings

An **isolated system** is cannot exchange either energy or matter with its surroundings



Josiah Willard Gibbs

The conservation laws of physics:

Conservation of mass, **Conservation of charge** and **Conservation of energy**

First law of thermodynamics: *The internal energy of an isolated system is constant* (Conservation of energy)

Second law of thermodynamics: *The entropy of an isolated system tends to increase* (Constraint on energy flow)

[†] Peter Atkins, Physical Chemistry: Thermodynamics, Structure, and Change. 11th Edn 2018

Energy and entropy (measures distribution of energy)



$$\text{Energy} = h\nu$$

0.5 μm
Highly directed
(low entropy)

1.3 $\text{kJ.s}^{-1}.\text{m}^{-2}$
(3.5 10^{14} kJ.s^{-1})



10 μm
Undirected
(high entropy)

For every 1 photon at 0.5 μm reaching the earth, 20 photons at 10 μm radiate away. Energy is conserved, but entropy increases.

Energy flow

q_i - quantity (qty) in units of m or m³, C, mol, or entropy

$v_i = \frac{dq_i}{dt}$ - rate of change ('flow' or 'flux'). Quantity per second (qty.s⁻¹).

u_i - potential (drives the flow v_i), and is measured in J.(qty)⁻¹

mechanical potential (J.m ⁻¹ , J.m ⁻² , J.m ⁻³)	drives movement (m.s ⁻¹ , m ² .s ⁻¹ , m ³ .s ⁻¹)
electrical potential (J.C ⁻¹)	drives electrical current (C.s ⁻¹)
chemical potential (J.mol ⁻¹)	drives molar flux (mol.s ⁻¹)
thermal potential (J.entropy ⁻¹ = K)	drives heat flow (entropy.s ⁻¹)

The product: potential (J.qty⁻¹) x flow (qty.s⁻¹) = power (J.s⁻¹)

We will study some protein families

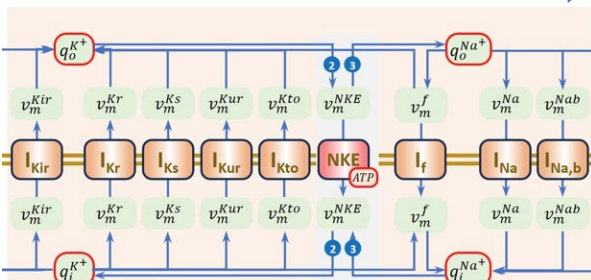
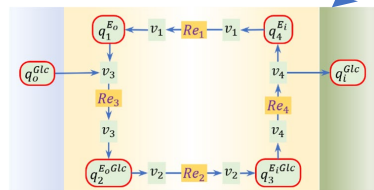
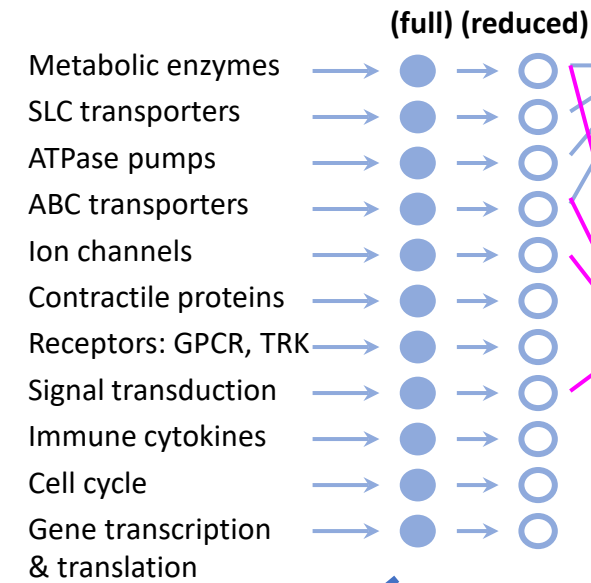
- **SLC transporters**
 - transport of small molecules across membranes via ‘facilitated’ diffusion or using an ion gradient (usually sodium)
- **ATPase pumps**
 - energy-dependent transport
- **ABC transporters**
 - (ATP-Binding Cassette) energy-dependent transport
- **Ion channels**
 - high volume driven by concentration gradient but usually transient with voltage-dependent gating
- **Metabolic enzymes**
 - tissue-specific control of metabolism
- **Contractile proteins**
 - actin & myosin interact to generate mechanical force
- **Aquaporins**
 - water transport through membrane channels
- **Receptors: GPCR, TRK, ...**
 - convey external signals to intracellular signalling pathways

And functional cell units (FCUs)

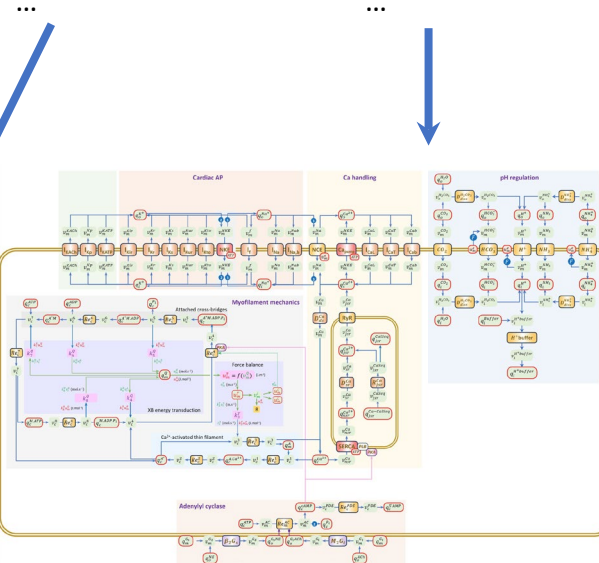
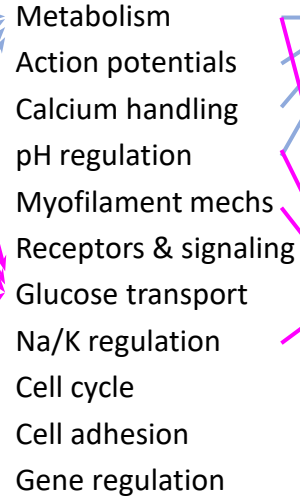
- **Glucose transport & Na/K regulation**
- **Intracellular pH regulation**
- **Metabolic pathways**
- **Action potentials**
- **Myofilament mechanics & calcium handling**

Overview of physics-based multiscale modelling

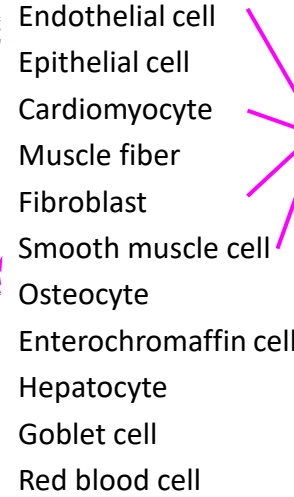
1. BG protein templates



2. FCUs

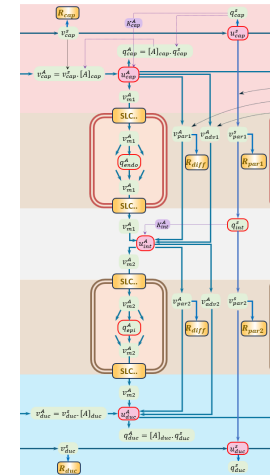
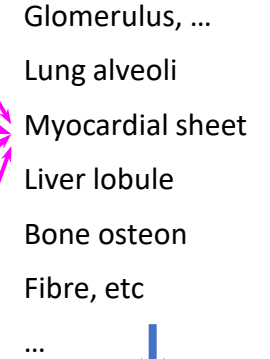


3. Cells



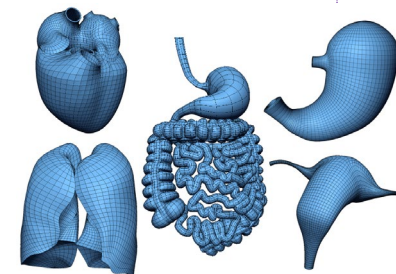
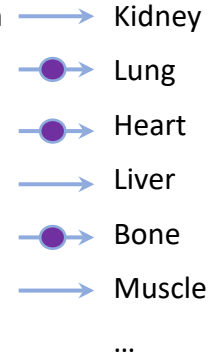
4. FTUs

(including 3D structure)



5. Organs

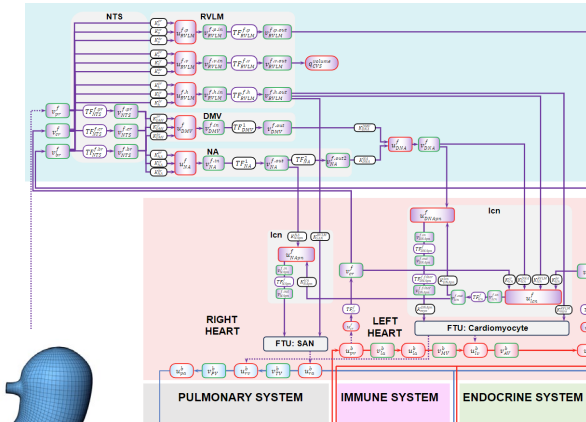
● =surrogate



6. Systems

(3D anatomical model of the body)

Regulation of fluid volume
Regulation of arterial blood pressure
Regulation of pH
Regulation of electrolyte balance
Regulation of glucose
Regulation of O₂ and CO₂
Regulation of calcium
Regulation of phosphate
Regulation of body temperature
...



Finite element models that are subsequently reduced (with AI methods) to surrogate models.

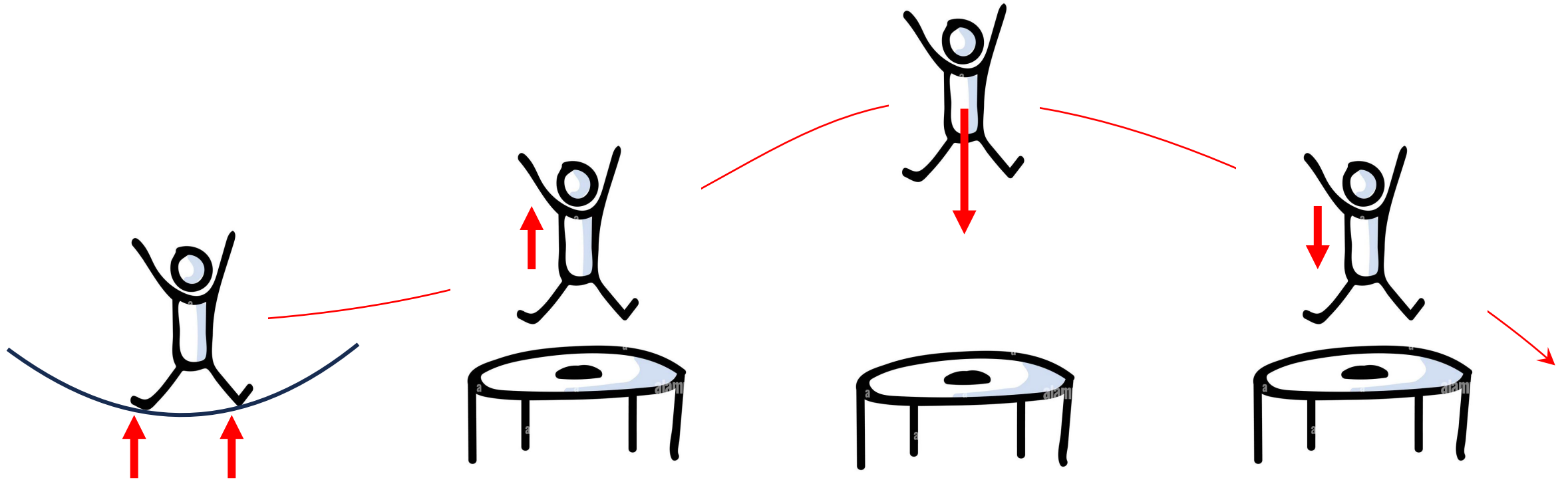
Lecture 2: *Physical laws and bond graphs*

Lecture 2: *Physical laws and bond graphs*

Energy can be **transmitted**, **stored**, and **converted** between **mechanical energy**, **electromagnetic energy** and **chemical energy** (including heat) ... nothing else! Almost every physiology process uses ALL of these 3 energetic mechanisms ... think of a muscle cell: chemical processes generate an electrical current to activate the cell, the mechanical energy for contraction (and the generated heat from all three helps to maintain body temperature). By the end of this lecture you should understand the physical laws (**conservation of mass**, **conservation of charge**, and **conservation of energy** – all coming from one underlying principle .. **conservation of power**) that govern the flow and storage of energy and its conversion between the 3 physical processes.

You should also understand how all physiological processes can be described by one simple but very powerful way of modelling them, called 'bond graphs'. You will learn that energy can be stored mechanically with a spring, electrically with a capacitor, or chemically as a solute in a solution – but that these are just variations on the common process of *energy storage*. The wide variety of ad-hoc descriptions of physical processes often presented in physiology text books, such as Fick's law of diffusion, Ohms' law & Fourier's law of heat conduction, are just variations on the underlying principle of *a potential gradient driving a flow of a quantity (that has the units described in Lecture 1)*, through a resistive element that is either a mechanical damper, an electrical resistance, or a chemical reaction. But they unhelpfully confuse laws of physics with empirical material properties.

Energy is either potential (due to position) or kinetic (due to motion)



Potential energy
(elastic trampoline)

Kinetic energy
 $\left(\frac{1}{2}mv^2\right)$

Potential energy
(gravity .. mgh)

Kinetic energy
 $\left(\frac{1}{2}mv^2\right)$

Energy can be **transmitted**, **stored** and **converted** between **mechanical energy**, **electromagnetic energy** & **chemical energy** (including **heat**) ... nothing else! (at the physiology scale).

Energy transmission

mechanical potential (J.m ⁻¹ , J.m ⁻² , J.m ⁻³)	drives movement (m.s ⁻¹ , m ² .s ⁻¹ , m ³ .s ⁻¹)
electrical potential (J.C ⁻¹)	drives electrical current (C.s ⁻¹)
chemical potential (J.mol ⁻¹)	drives molar flux (mol.s ⁻¹)
thermal potential (J.entropy ⁻¹)	drives heat flow (entropy.s ⁻¹)

Let's represent that transmission by a line 

We need to also represent the flow v_i and the potential driving that flow u_i :  $= \left(\frac{dq_i}{dt} \right)$

where v_i is in **m.s⁻¹** **m³.s⁻¹** **C.s⁻¹** (Amp) **mol.s⁻¹** or **e.s⁻¹**

and u_i is in **J.m⁻¹** (Newton) **J.m⁻³** (Pascal) **J.C⁻¹** (Volt) **J.mol⁻¹** or **J.e⁻¹** (deg Kelvin)

$u_i \times v_i$ is **J.s⁻¹** **J.s⁻¹** **J.s⁻¹** **J.s⁻¹** **J.s⁻¹**

$u_i \times v_i$ is always power!


is the 'bond' in a 'bond graph'.

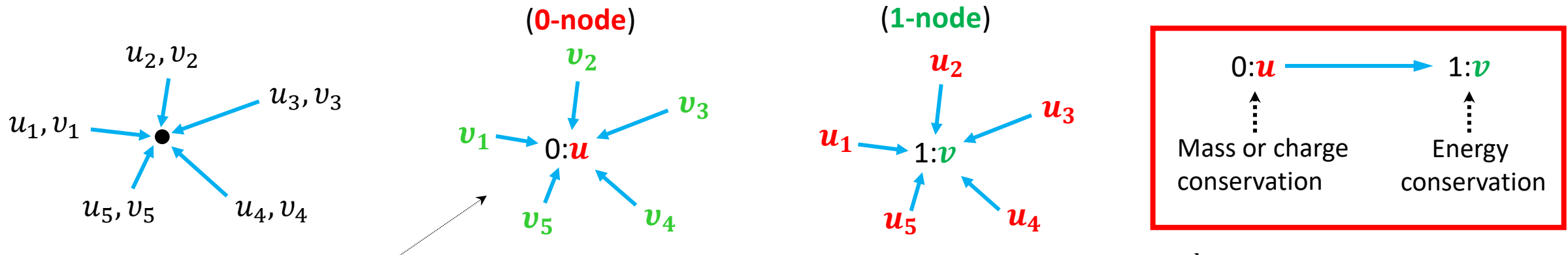
Bond graph 0:node and 1:node



Henry Paynter, MIT

Conservation of power: $u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 + u_5 v_5 + \dots = 0$

$$\sum u_i v_i = 0 \left\{ \begin{array}{l} \text{If } u_1 = u_2 = u_3 = u_4 = u_5 = \mathbf{u} \text{ then } \sum \mathbf{v}_i = 0 \text{ .. This is conservation of mass or charge}^\# \\ \text{If } v_1 = v_2 = v_3 = v_4 = v_5 = \mathbf{v} \text{ then } \sum \mathbf{u}_i = 0 \text{ .. This is conservation of energy} \end{array} \right.$$



Note: To sum to zero, some of these flows must be negative! $^\# \sum v_i = \frac{d}{dt} (\sum q_i) = 0$, so $\sum q_i$ is constant.

Note: $\sum \mathbf{v}_i = 0$ only contains $\mathbf{v}_i$
And $\sum \mathbf{u}_i = 0$ only contains $\mathbf{u}_i$

We also need to find a relationship between $\mathbf{u}_i$ and $\mathbf{v}_i$ (or $\mathbf{q}_i$ or $\frac{d\mathbf{v}_i}{dt}$)

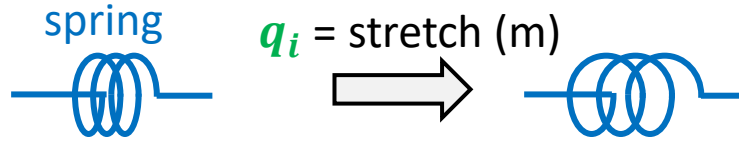
Material property or 'constitutive law' or 'phenomenological relation'

Energy storage

Potential energy (static)

Kinetic energy (dynamic)

Mechanical



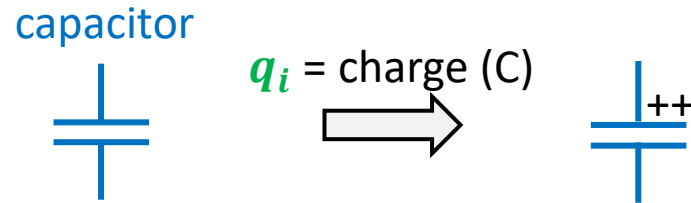
potential $u_i = f(q_i)$

$u_i = k \cdot q_i$

or $u_i = k \cdot \left(\frac{q_i - q_0}{q_0} \right) = k \bar{q}_i$

Hooke's law ' $f=kx$ '
where k is stiffness

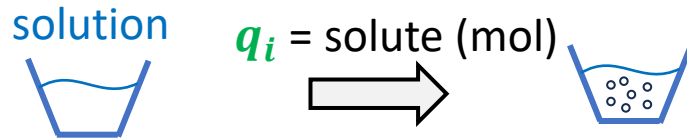
Electrical



$u_i = \frac{1}{C} \cdot q_i$

C is capacitance

Chemical



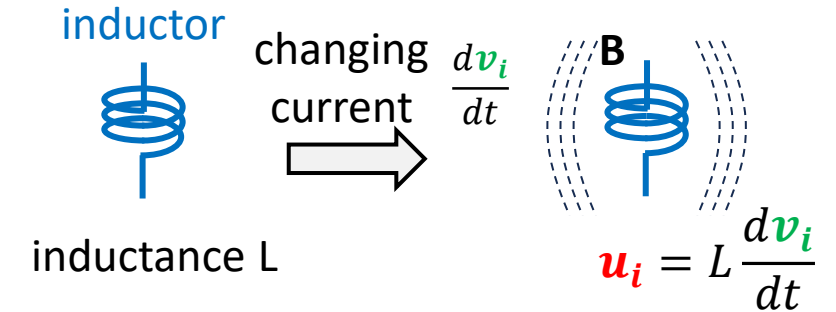
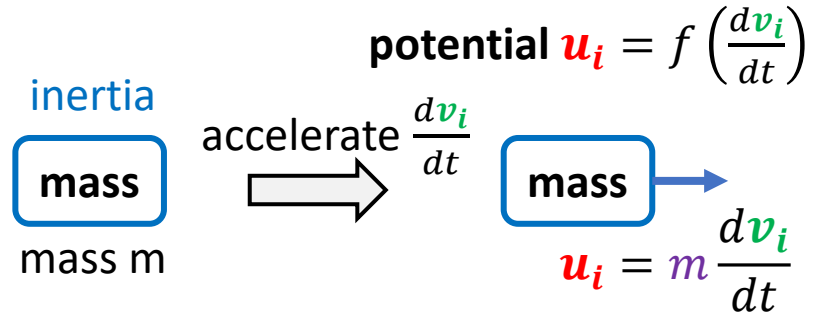
$u_i = RT \ln(K_i q_i)$

or $u_i = RT \ln\left(\frac{q_i}{q_0}\right)$

or $u_i = RT \ln(\bar{q}_i)$

Boltzmann eqn

Note that the static potential u_i is always defined relative to a reference.



heat

Heat is kinetic energy of particles
i.e. reflects average velocity of particles

Gas law: $PV=nRT$

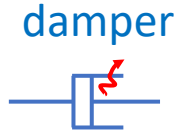
These are 'constitutive laws' and 'material constants' – found by experiment
(e.g. elasticity, electrical capacitance, thermodynamic constant)

Energy dissipation (conversion to heat)



Aharon Katchalsky

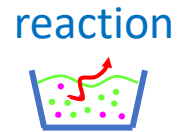
Mechanical



Electrical

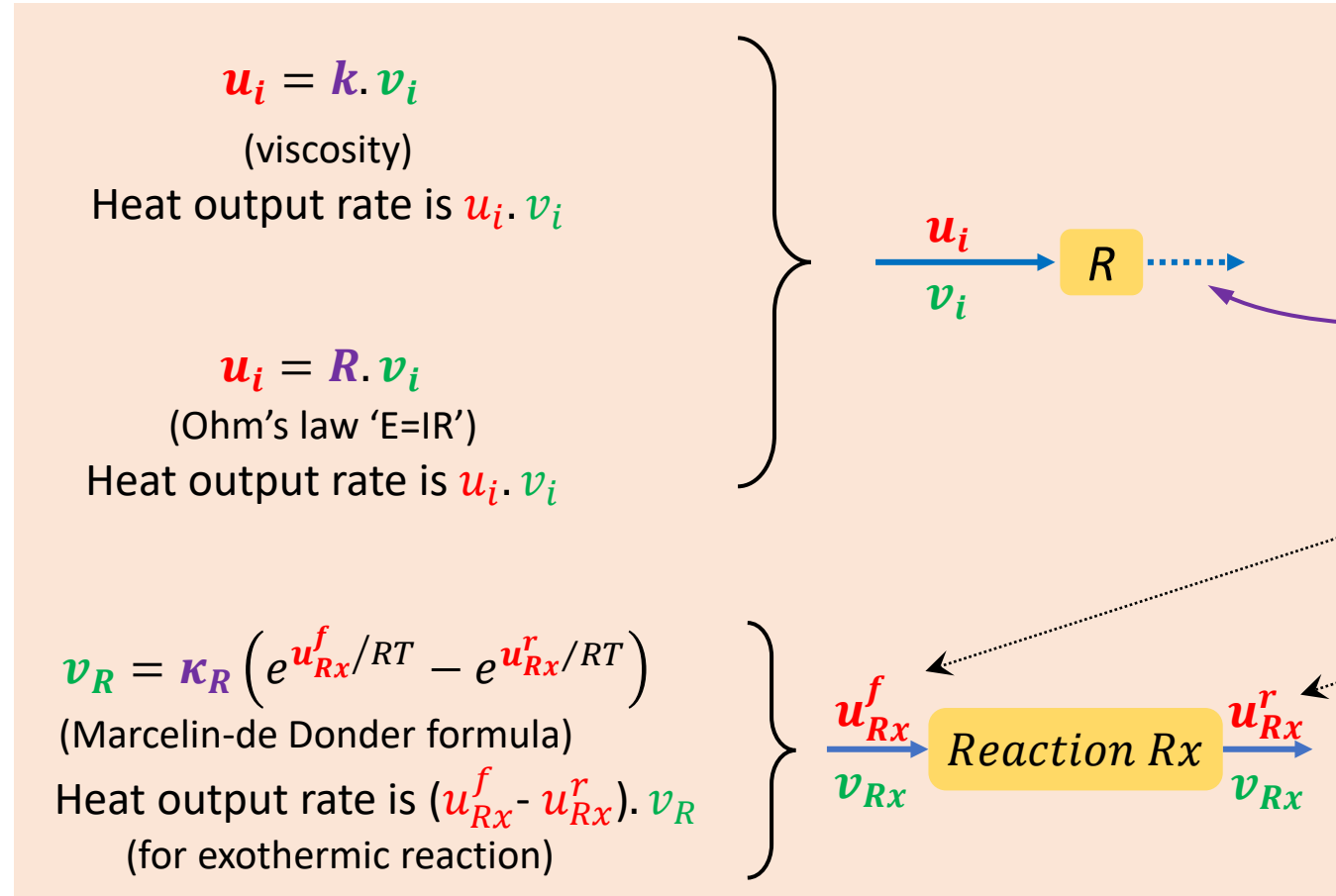


Chemical



Potential $u_i = f(v_i)$

Bond graph



(1-port node)

.. but only because we ignore the heat output!

forward potential
(sum of reactant potentials)

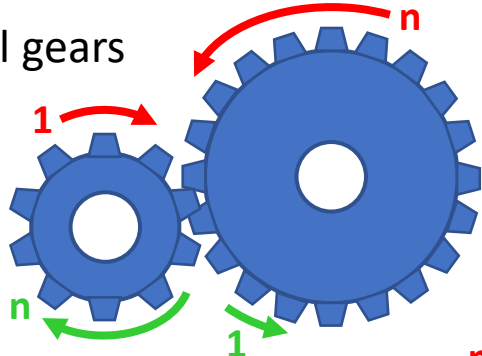
reverse potential
(sum of product potentials)

(2-port node)

These are 'constitutive laws' and 'material constants' – found by experiment
(e.g. viscosity, resistance, reaction rate)

Energy conversion: Bond graph 'transformers'

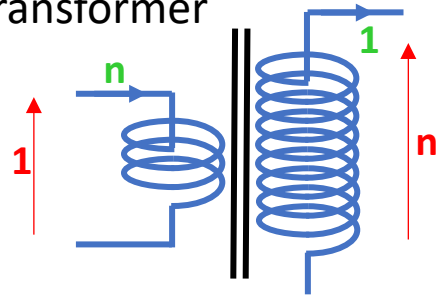
Mechanical gears
(ratio n)



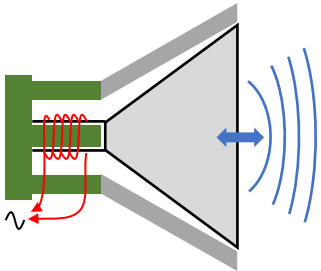
Mechanical lever
(ratio n)



Electrical transformer
(ratio n)

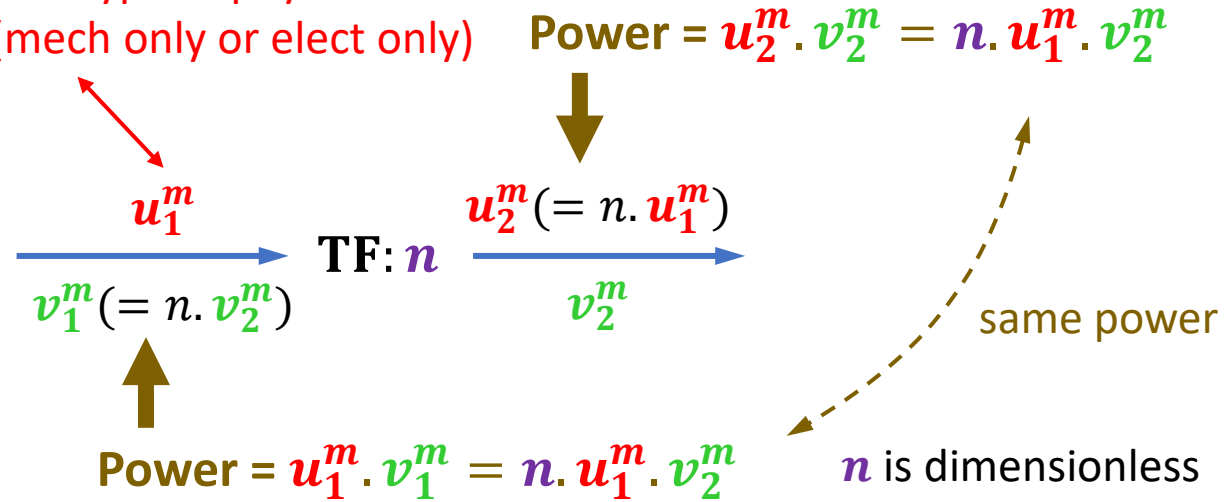


Mechano-electric coupling

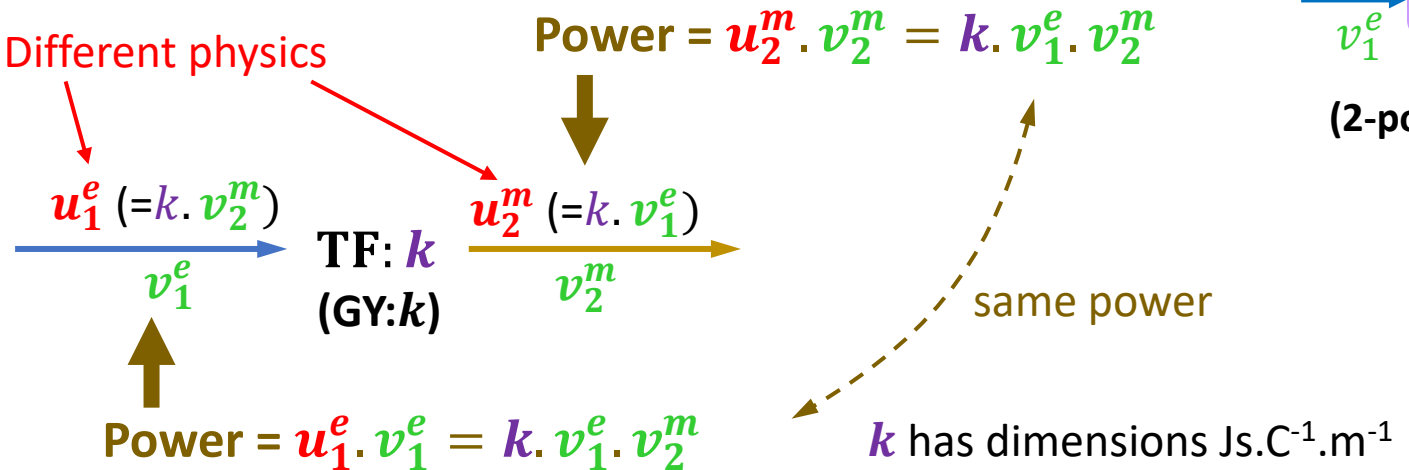


(voice coil, coupling via magnetic field)

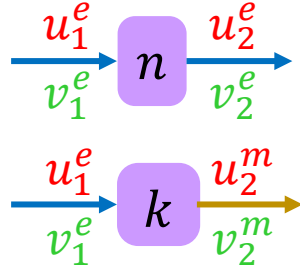
One type of physics
(mech only or elect only)



Different physics



Bond graph



(2-port node)

Summary

Energy is either *potential energy* (associated with position) or *kinetic energy* (associated with movement).

Energy *transmission* is indicated by a *bond* $\xrightarrow[v_i]{u_i}$ that carries the *flow* $v_i = \left(\frac{dq_i}{dt}\right)$, driven by *potential* u_i .

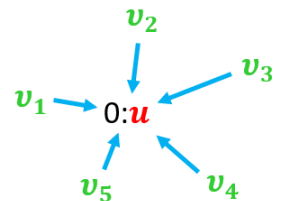
The quantity q_i has units m (or m³), C, mol, or e, encompassing *mechanics* (solid & fluid), *electromagnetics* and *chemistry*. Sometimes useful to express as a nondimensional quantity. E.g. strain $\bar{q}_i = \frac{q_i - q_0}{q_0}$ in mechanics, or $\bar{q}_i = \frac{q_i}{q_0}$ for biochemistry.

The potential (energy gradient) u_i has units J.m⁻¹ (N), J.m⁻³ (Pa), J.C⁻¹ (V), J.mol⁻¹, J.e⁻¹ (K).

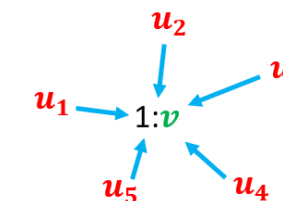
The product $u_i.v_i$ has units of power (J.s⁻¹). i.e. J.m⁻¹.m.s⁻¹, J.m⁻³.m³.s⁻¹, J.C⁻¹.C.s⁻¹, J.mol⁻¹.mol.s⁻¹, or J.e⁻¹.e.s⁻¹.

Power conservation is *mass or charge conservation* at a 0:node (0: u) and *energy conservation* at a 1:node (1: v).

These are both in multiport junctions:


$$\sum u_i v_i = u. \sum v_i = 0$$

Mass/charge conservation


$$\sum u_i v_i = v. \sum u_i = 0$$

Energy conservation

The potential u_i and flow v_i are connected by an empirical *material law* for energy storage, dissipation or conversion.

Energy *storage* is either *static* (stretched *spring*, charged *capacitor*, or solute in *solution*) or dynamic (*mass*, *inductance*)

Energy *dissipation* (conversion to heat) is via an electrical or mechanical *resistance* R (a 1-port device): $\longrightarrow R$

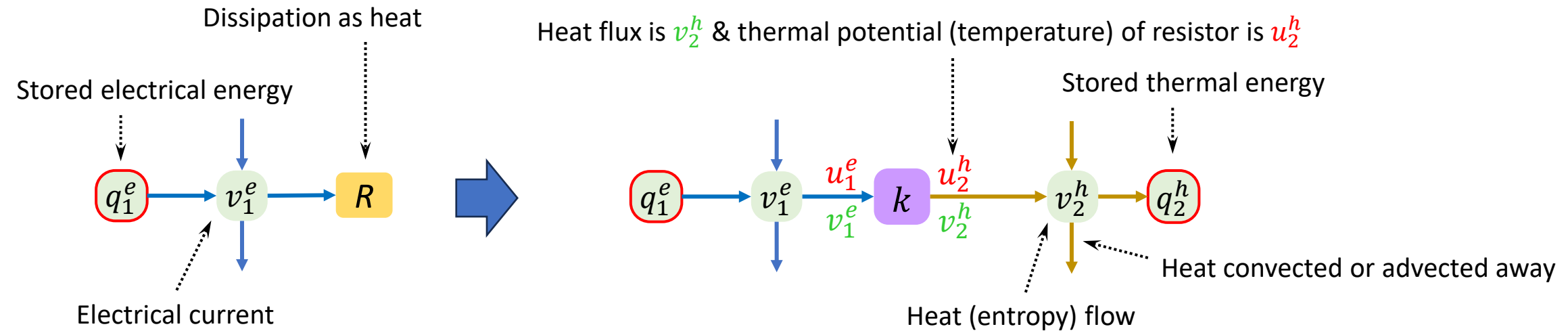
or via a chemical *reaction* R (a 2-port device): $\longrightarrow \text{Reaction Rx} \longrightarrow$

Energy *conversion* (mechanical-electrical-chemical) is handled via a 2-port *transforming factor* (TF): $\longrightarrow k \longrightarrow$

Accounting explicitly for heat flow

The heat output from a 1-port dissipation mechanism is carried away into the surroundings. Usually there is no need to further consider this heat (other than knowing its magnitude).

But if necessary, the 1-port electrical resistor or mechanical damper can be replaced by a 2-port TF node in order to explicitly model the heat flow, as shown below.



The conversion factor k depends both on the material through which the current is passing and how the heat is carried away via conduction, advection or radiation.

Heat from a biochemical reaction in a cell is *conducted* by diffusion through the tissue to the blood stream, where it is then *advected* away and either stored elsewhere in the body or conveyed to the skin where it is *radiated* away (often using the latent heat of evaporation).

For a black body radiator the radiated power is proportional to the 4th power of temperature. i.e. $u_2^h \cdot v_2^h = \sigma \cdot (u_2^h)^4$ (Stefan's law). The electrical resistance of most metals increases linearly with temperature: $R = R_0(1 + \alpha(T - T_0))$, where $T = u_2^h$ is temperature in Kelvin.

Lecture 3 *Simple bond graph models*

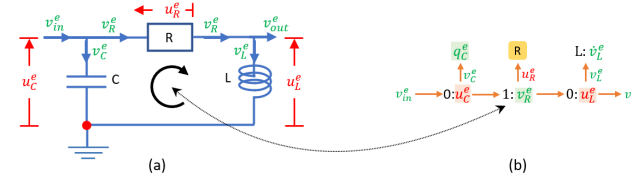
Lecture 3: *Simple bond graph models*

To show the power of **bond graphs** we use them to model a wide range of simple processes used in physiology, including reactions with multiple reactants and products, enzyme-catalysed reactions, and the electro-diffusion of a charged species (such as H^+ or HCO_3^-). For enzyme catalysed reactions we focus on the steady state approximations that are commonly used in physiology texts, and we discuss when those approximations are valid.

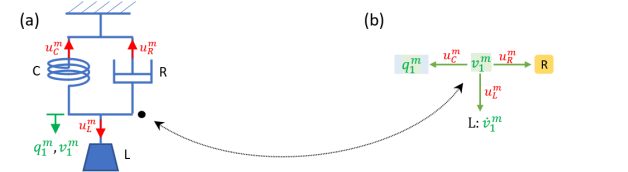
We show you how to solve the bond graph models using an open-source software package called **OpenCOR**. By the end of this lecture and the accompanying tutorials you should understand how to build simple models using bond graphs and how to solve them and interpret the results using OpenCOR.

Simple bond graph models

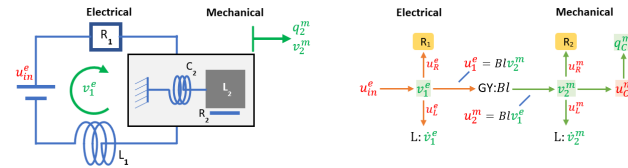
1. An electrical circuit



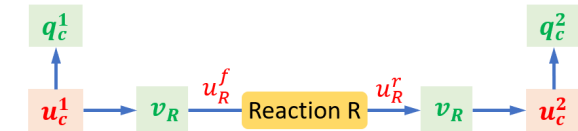
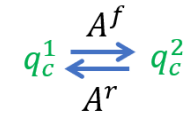
2. A mechanical system



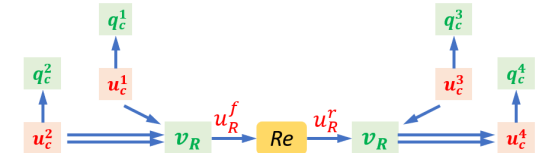
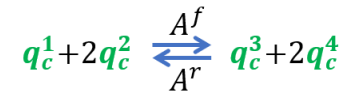
3. An electromechanical system



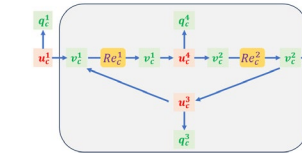
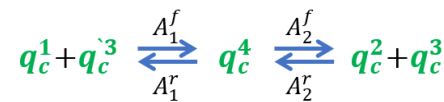
4. A simple biochemical reaction



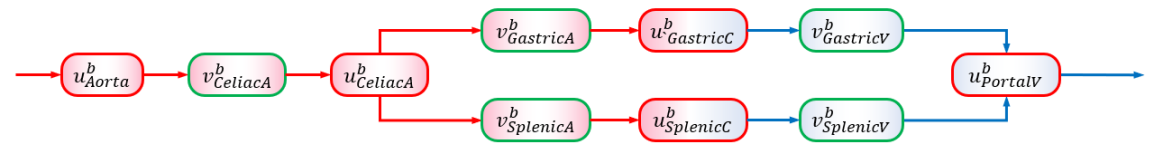
5. A reaction with mixed stoichiometry



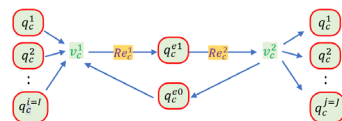
6. An enzyme-catalyzed reaction



7. Blood flow in the cardiovascular system



8. A reaction with multiple reactants & products



Defining units in CellML

$J.C^{-1}$ { `def unit J_per_C as`
 `unit joule;`
 `unit coulomb {expo: -1};`
 `enddef;`

$C.S^{-1}$ { `def unit C_per_s as`
 `unit coulomb;`
 `unit second {expo: -1};`
 `enddef;`

$C.S^{-2}$ { `def unit C_per_s2 as`
 `unit coulomb;`
 `unit second {expo: -2};`
 `enddef;`

$C^2.J^{-1}$ { `def unit C2_per_J as`
 `unit coulomb {expo: 2};`
 `unit joule {expo: -1};`
 `enddef;`

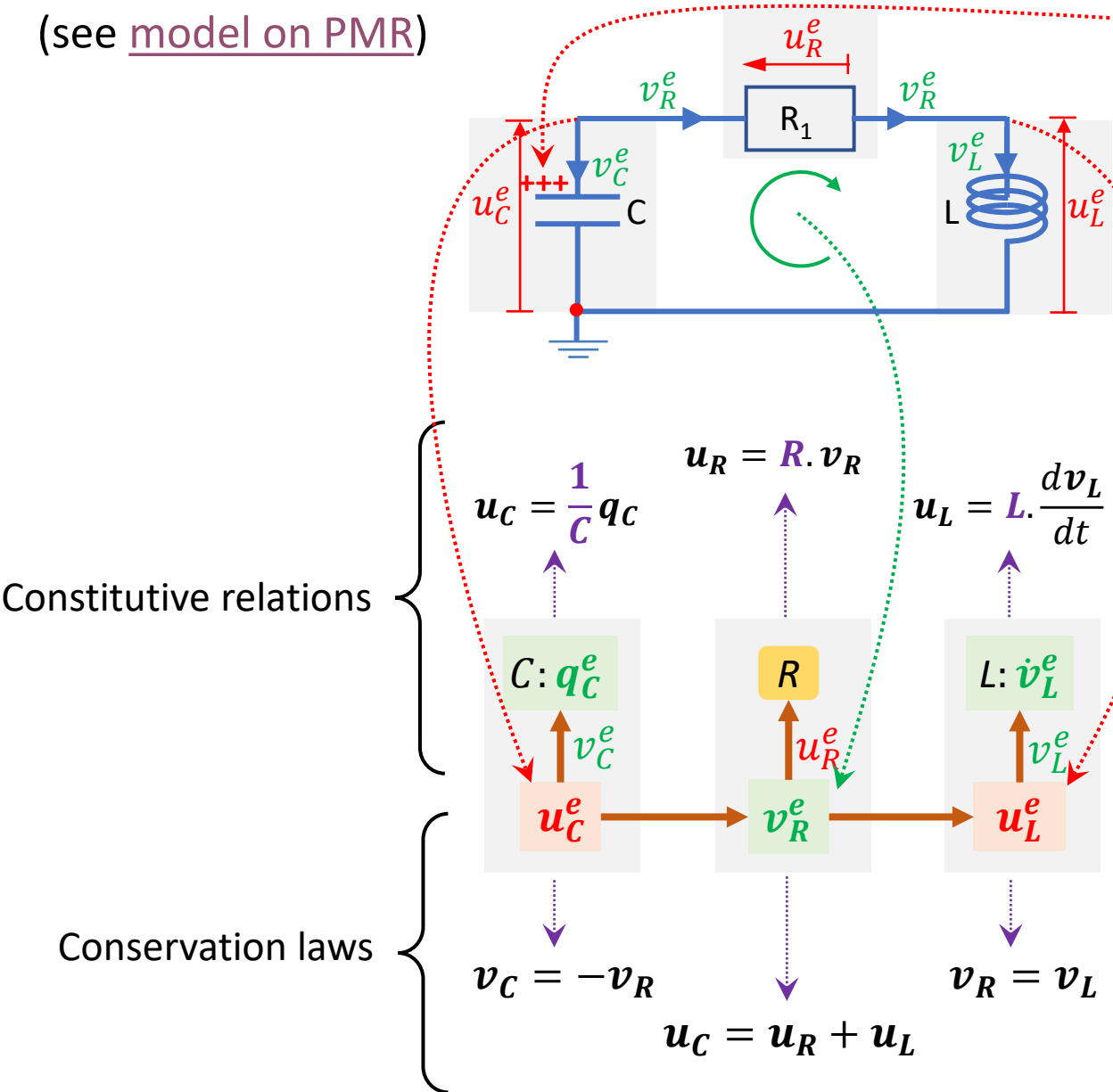
$J.S.C^{-2}$ { `def unit Js_per_C2 as`
 `unit joule;`
 `unit second;`
 `unit coulomb {expo: -2};`
 `enddef;`

$J.S^2.C^{-2}$ { `def unit Js2_per_C2 as`
 `unit joule;`
 `unit second {expo: 2};`
 `unit coulomb {expo: -2};`
 `enddef;`

`def unit dim as`
 `unit dimensionless;`
 `enddef;`

1. An electrical circuit

(see [model on PMR](#))



```
def comp main as
  var t: second {init: 0};

  // state variables
  var q_C: coulomb {init: 1};
  var v_C: C_per_s;
  var v_R: C_per_s;
  var v_L: C_per_s {init: 0};
  var a_L: C_per_s2;
  var u_C: J_per_C;

  var u_R: J_per_C;
  var u_L: J_per_C;

  // constitutive parameters
  var C: C2_per_J {init: 20};
  var R: Js_per_C2 {init: 2};
  var L: Js2_per_C2 {init: 10};

  // constitutive relations
  u_C = q_C/C;
  u_R = v_R*R;
  u_L = a_L*L;

  // conservation laws
  ode(q_C, t) = v_C;
  ode(v_L, t) = a_L;

  v_C = -v_R;
  v_R = v_L;
  u_C = u_R+u_L;

enddef;
```

Power & energy

Capacitor

$$Power = \frac{dE}{dt} = u \cdot v = \frac{1}{C} q \cdot \frac{dq}{dt} = \frac{1}{C} \cdot \frac{1}{2} \frac{dq^2}{dt} \quad \text{gives } E = \frac{1}{2} \cdot \frac{1}{C} \cdot q^2$$

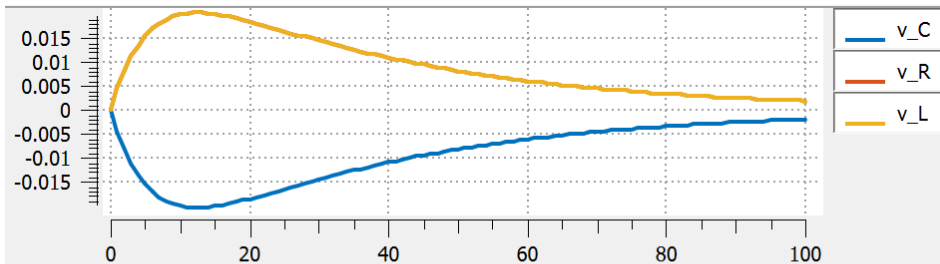
Inductor

$$Power = \frac{dE}{dt} = u \cdot v = L \frac{dv}{dt} \cdot v = L \cdot \frac{1}{2} \frac{dv^2}{dt} \quad \text{gives } E = \frac{1}{2} L \cdot v^2$$

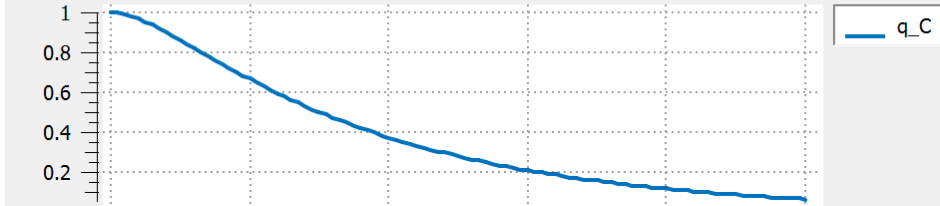
Resistor

$$Power = \frac{dE}{dt} = u \cdot v = R \cdot v \cdot v = R \cdot v^2 \quad \text{gives } E = \int R \cdot v^2 dt$$

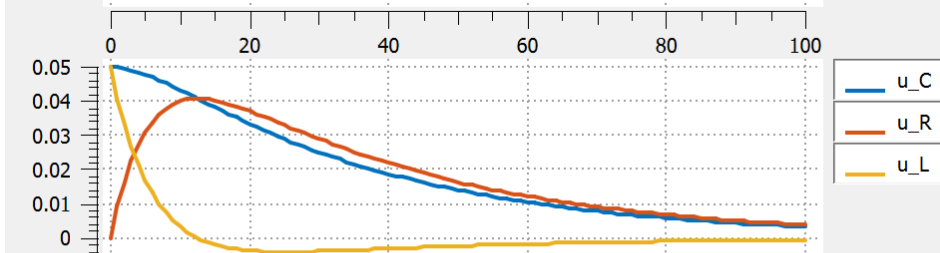
Currents



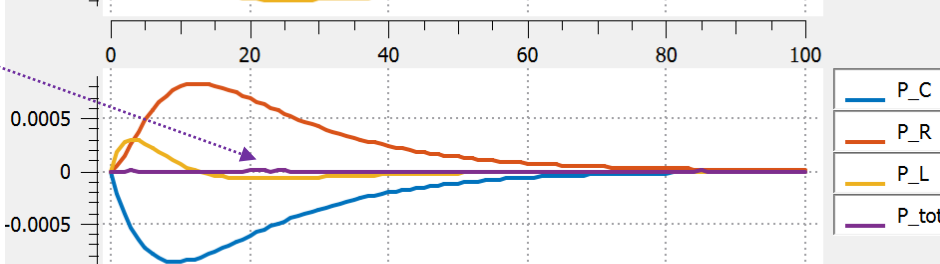
Charge



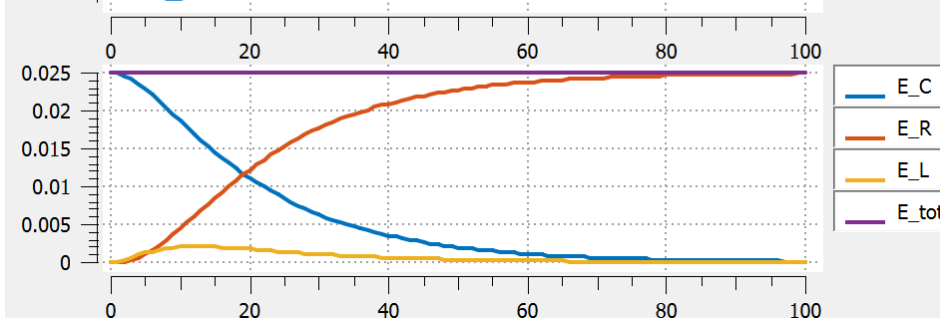
Potentials



Power



Energy



// conservation laws

```
ode(v_L, t) = a_L;  
v_C = -v_R;  
v_R = v_L;
```

```
ode(q_C, t) = v_C;
```

```
u_C = u_R+u_L;
```

// constitutive relations

```
u_C = q_C/C;  
u_R = v_R*R;  
u_L = a_L*L;
```

// Energy calculations

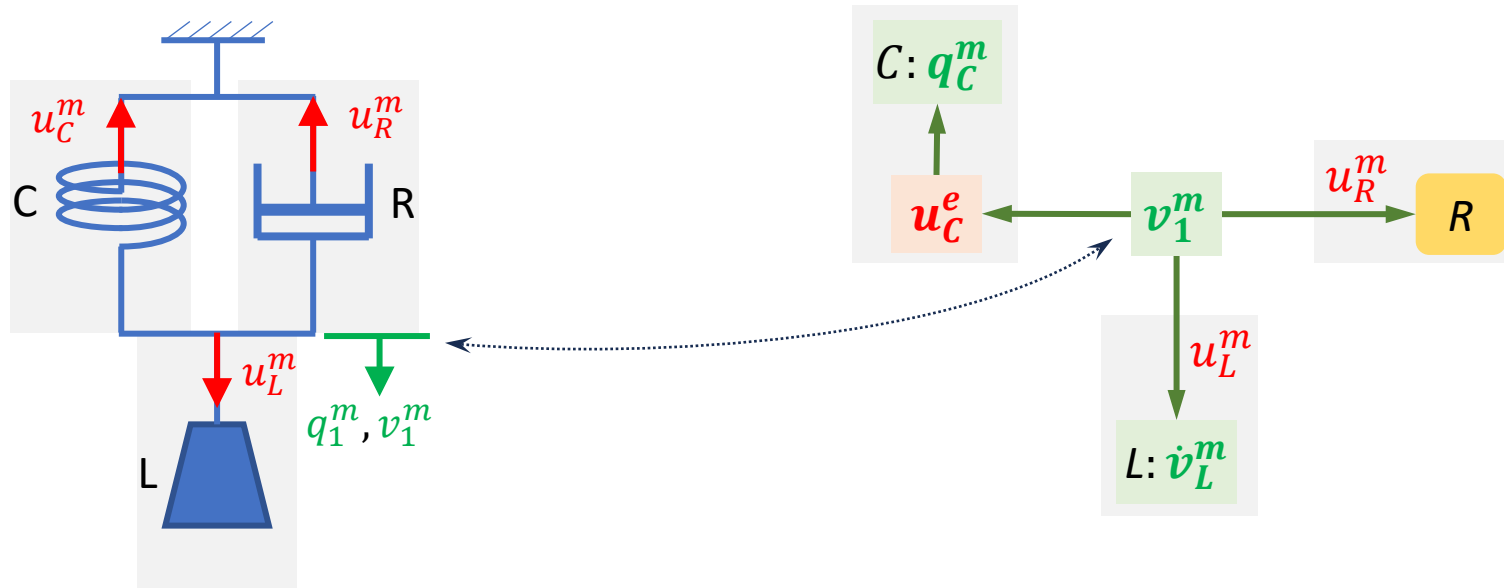
```
P_C = u_C*v_C;  
P_R = u_R*v_R;  
P_L = u_L*v_L;  
P_tot = P_C+P_R+P_L;
```

```
ode(E_R, t) = P_R;  
E_C = 0.5*1/C*sqr(q_C);  
E_L = 0.5*L*sqr(v_L);  
E_tot = E_C+E_L+E_R;
```

Model on PMR

2. A mechanical system

(see model on PMR)



Conservation of energy at 1:node: $u_C^m + u_R^m + u_L^m = 0$

Constitutive relations $u_C^m = \frac{1}{C} q_1^m$; $u_R^m = R v_1^m$; $u_L^m = L \frac{dv_1^m}{dt}$.

Note that substituting the constitutive relations into the conservation eqn with $v_1^m = \frac{dq_1^m}{dt}$ gives the 2nd order eqn $L \frac{d^2 q_1^m}{dt^2} + R \frac{dq_1^m}{dt} + \frac{1}{C} q_1^m = 0$, but bond graphs always uses the first order eqns.

3. An electromechanical system

(see [model on PMR](#))

Electromechanical energy exchange

Faraday's law

$$u_1^e (=k \cdot v_2^m)$$

Lorentz force

$$u_2^m (=k \cdot v_1^e)$$

TF: k

v_2^m

(GY: k)

v_1^e

Units of k : $\text{J.C}^{-1}.\text{s.m}^{-1} = \text{J.m}^{-1}.\text{s.C}^{-1}$

Fixed potential (battery)
delivers current into system

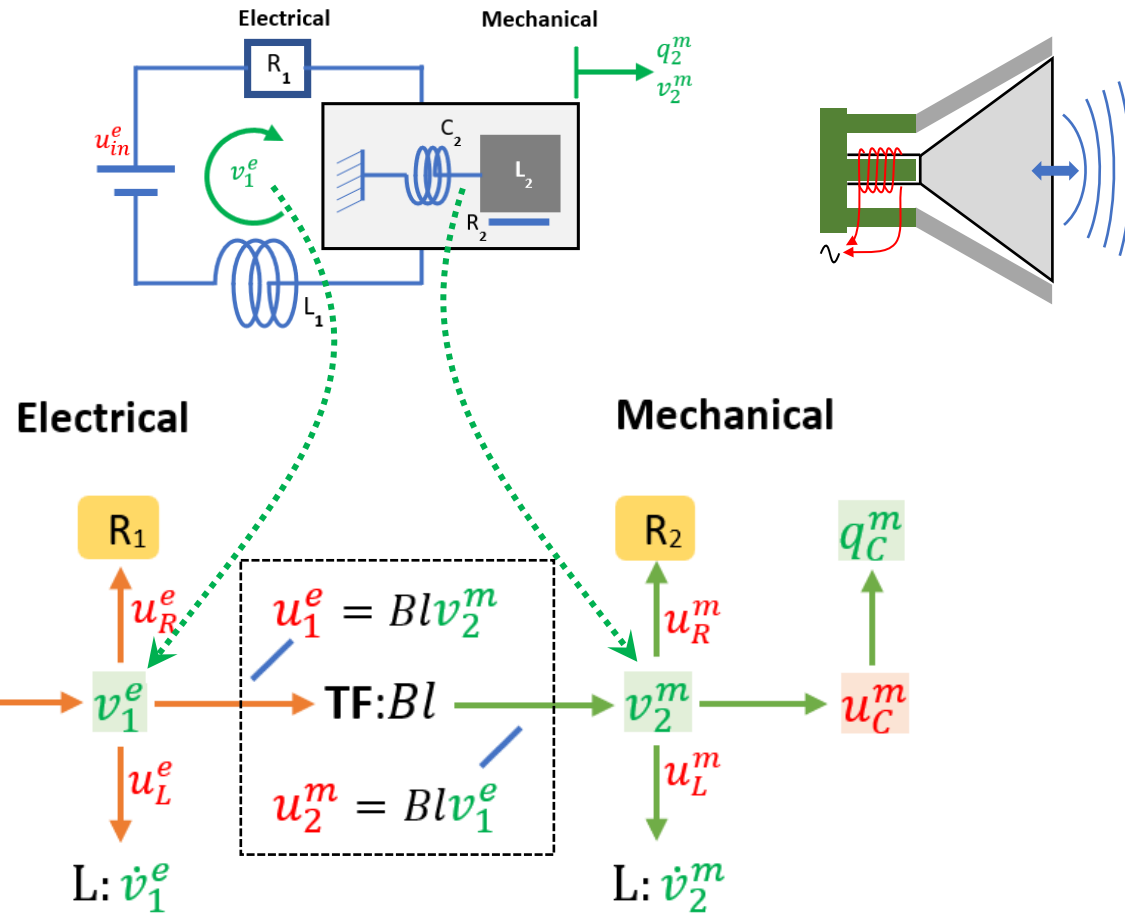
$$(u_C^m) \quad \frac{dq_C^m}{dt} = v_2^m$$

$$(v_1^e) \quad u_{in}^e = u_R^e + u_L^e + u_1^e$$

$$(v_2^e) \quad u_2^m = u_R^m + u_L^m + u_C^m$$

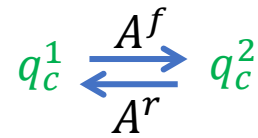
$$(TF) \quad u_1^e = Blv_2^m; \quad u_2^m = Blv_1^e$$

$$(CRs) \quad u_R^e = R_1 v_1^e; \quad u_L^e = L_1 \frac{dv_1^e}{dt}; \quad u_R^m = R_2 v_2^m; \quad u_L^m = L_2 \frac{dv_2^m}{dt}; \quad u_C^m = q_C^m / C_2;$$



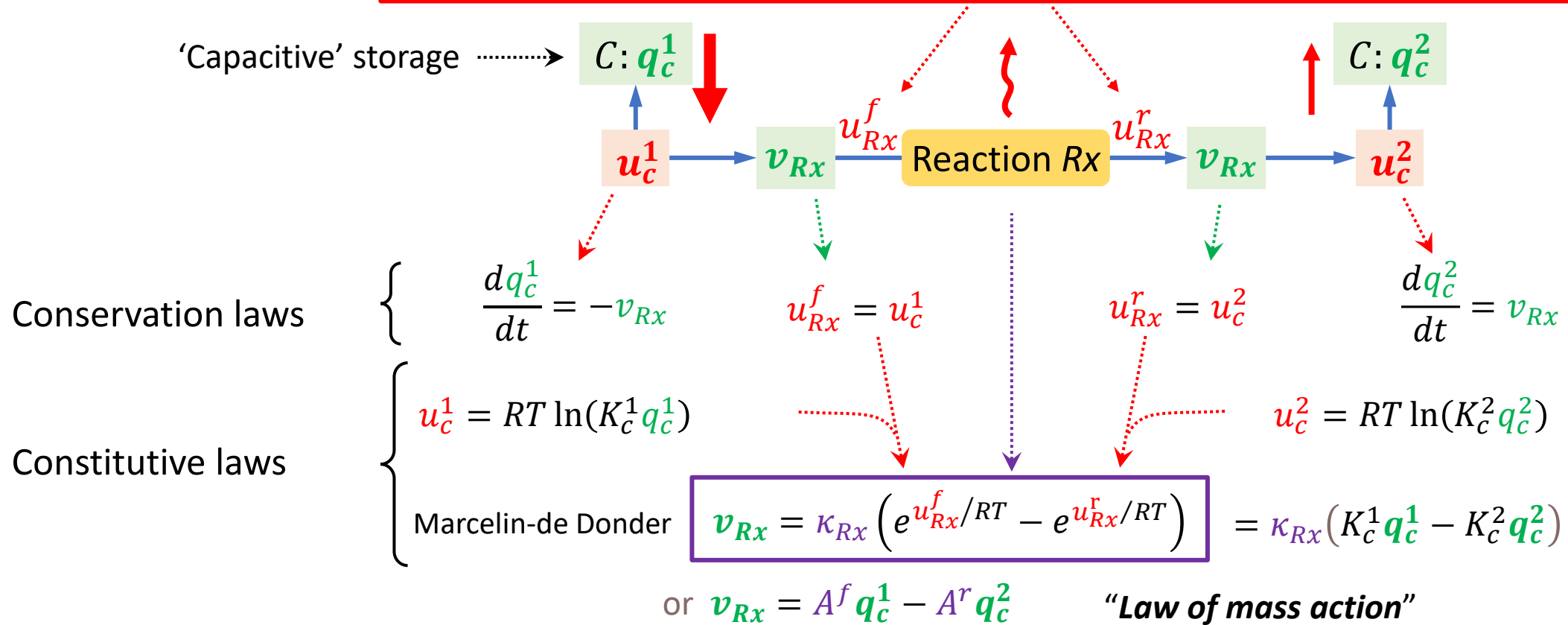
4. A simple biochemical reaction

(see [model on PMR](#))



The change in **Gibbs 'free energy'** (per mole) is the difference between these two chemical potentials and is the energy available to do work (i.e. convert the internal energy of the reactant stored as q_c^1 into the internal energy of the product stored as q_c^2). The heat production rate from the reaction is $(u_{Rx}^f - u_{Rx}^r) v_{Rx}$.

i.e. Reactant internal energy flux $u_{Rx}^f \cdot v_{Rx}$ = Product internal energy flux $u_{Rx}^r \cdot v_{Rx}$ + Heat flux $\Delta u \cdot v_{Rx}$.



where the parameters $A^f = \kappa_{Rx} K_c^1$ and $A^r = \kappa_{Rx} K_c^2$ combine thermodynamic and reaction kinetic parameters.

Note on thermodynamic parameters

For a dilute system the chemical potential is given by the Boltzmann thermodynamic relation

$$u_c^i = u_0^i + RT \ln \frac{q_c^i}{q_c^{tot}} \quad (\text{J.mol}^{-1})$$

where q_c^i is the number of moles of chemical species i and q_c^{tot} is the total number of moles of all substances in the mixture in compartment c (Sauro, 2012). u_0^i is the (reference) potential when $q_c^i = q_c^{tot}$.

More compactly, $u_c^i = RT \ln K_c^i q_c^i$ (J.mol⁻¹), where $K_c^i = \frac{1}{q_c^{tot}} e^{u_0^i/RT}$ (mol⁻¹),

is the constitutive law for biochemical energy storage, and K_c^i (mol⁻¹) is a thermodynamic parameter.

Note: $pH = -\log_{10}[H^+]$ where $[H^+]$ is expressed in mol.L⁻¹. i.e. pH of 7 is $[H^+] = 10^{-7}$ mol.L⁻¹

Note on concentrations

q_1^A represents the quantity in moles of solute A at location 1

$u_1^A = RT \ln(K_1^A q_1^A)$ is the corresponding chemical potential

K_1^A is the thermodynamic constant that provides a reference for u_1^A

i.e. when $q_1^A = \frac{1}{K_1^A}$, $u_1^A = RT \ln(1) = 0$

The concentration of solute A in a volume V_1 at location 1 is $c_1^A = \frac{q_1^A}{V_1}$.

K_1^A is proportional to $\frac{1}{V_1}$ since a larger volume V_1 requires a larger molar amount q_1^A to maintain the same u_1^A

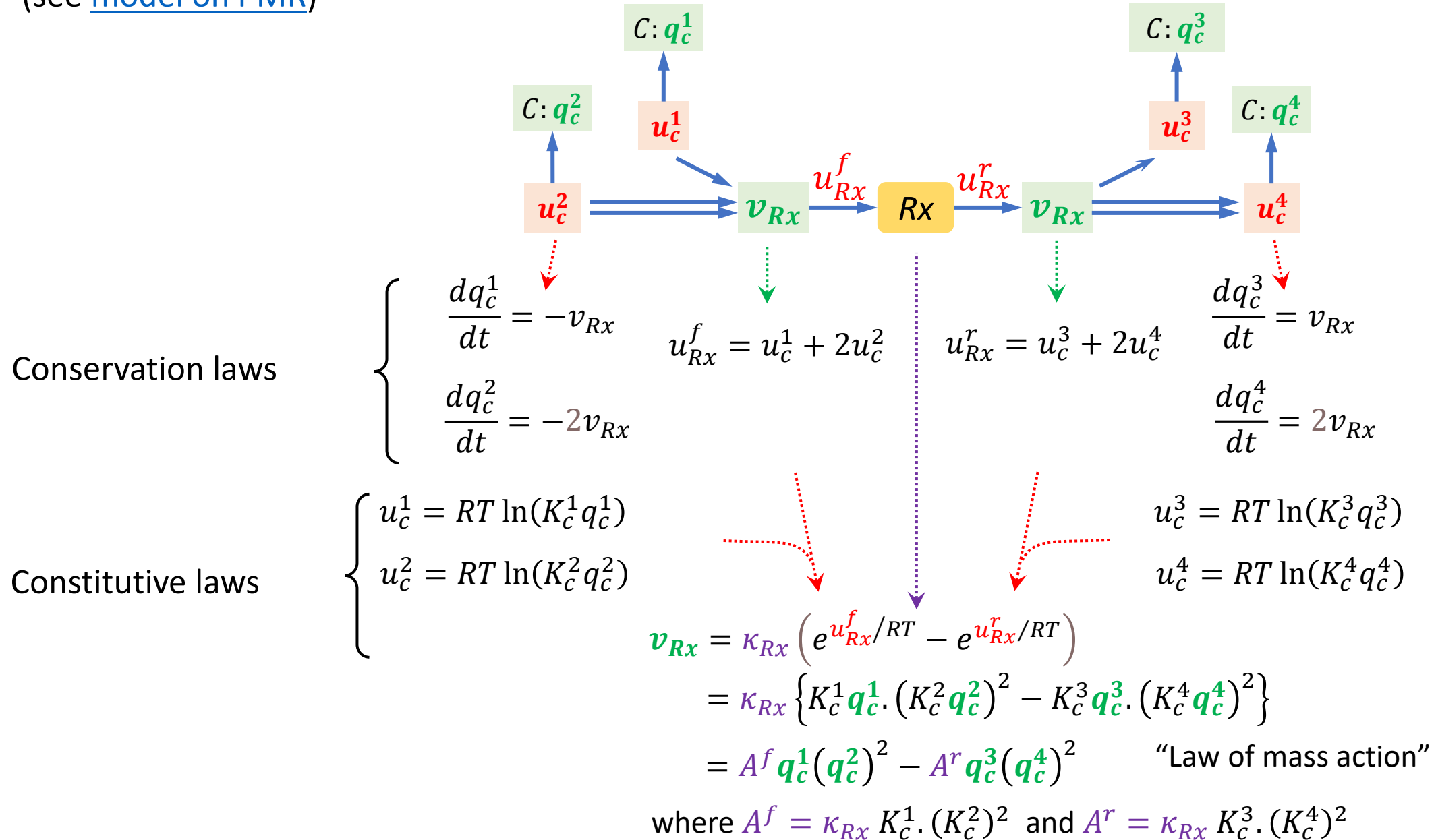
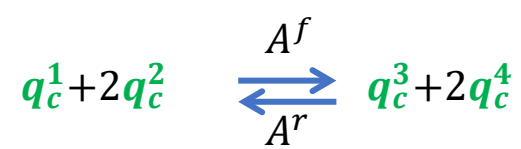
So $\frac{K_2^A}{K_1^A} = \frac{V_1}{V_2}$.

And hence $\frac{K_2^A q_2^A}{K_1^A q_1^A} = \frac{V_1 q_2^A}{V_2 q_1^A} = \frac{q_2^A/V_2}{q_1^A/V_1} = \frac{c_2^A}{c_1^A}$.

We later use this to express the Nernst potential in terms of concentrations: $u_m^e = \frac{RT}{z_q F} \ln \left(\frac{K_2^i q_2^i}{K_1^i q_1^i} \right) = \frac{RT}{z_q F} \ln \left(\frac{c_2^i}{c_1^i} \right)$.

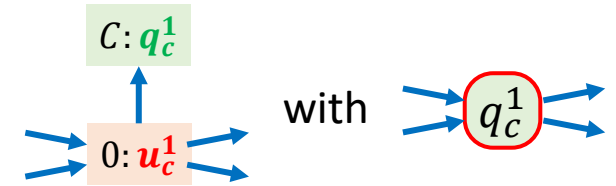
5. A biochemical reaction with mixed stoichiometry

(see [model on PMR](#))



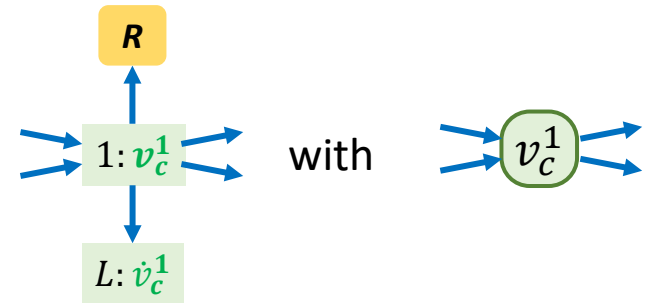
Simplification of bond graph diagrams

It is often convenient to combine (1-port) static storage with a 0:node and replace



The red border indicates the 0:node (mass or charge balance) and the q_c^1 term reflects both the static storage and that the potential u_c^1 is given by a storage relationship $u_c^1 = f(q_c^1)$.

Similarly, 1-port dissipation can sometimes be combined with a 1:node to replace



The green border indicates the 1:node (energy balance) together with the dissipation function $f(v_c^1)$ (e.g. $f(v_c^1) = Rv_c^1$) and the kinetic energy storage term $L\dot{v}_c^1$. All of these depend on v_c^1 .

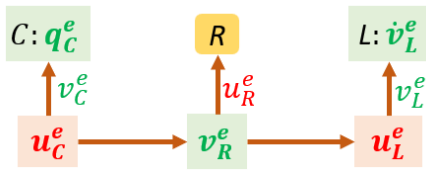
Note that the 1:node v_{Rx} is retained in its current form in chemical reactions where the stoichiometry needs to be shown.

We also replace $\Rightarrow v_{Rx}$ with $\textcircled{2} \rightarrow v_{Rx}$

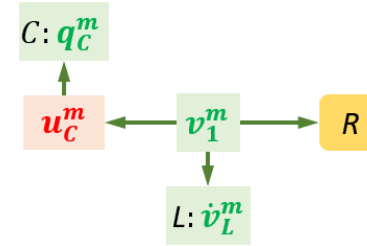
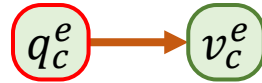
A further simplification is to put $\bar{q}_c^i = K_c^i q_c^i$ where $\bar{q}_c^i$ is now a dimensionless quantity and is the relative molar quantity that provides the energy to drive reactions (i.e. $u_c^i = RT \ln \bar{q}_c^i$).

Note that from now on we use the superscript i and subscript j on q_j^i , v_j^i & u_j^i to denote species i and location j , respectively.

Examples ..



becomes

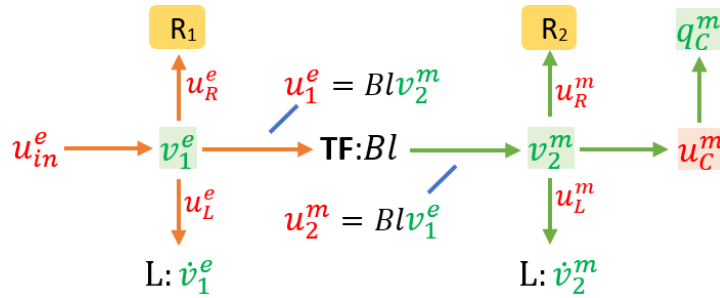


becomes

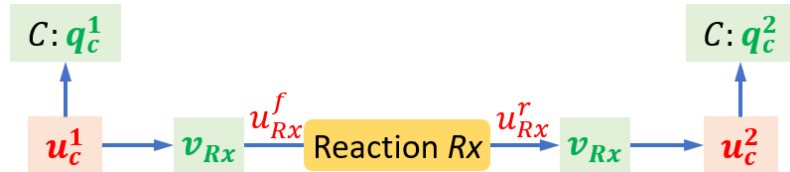


Electrical

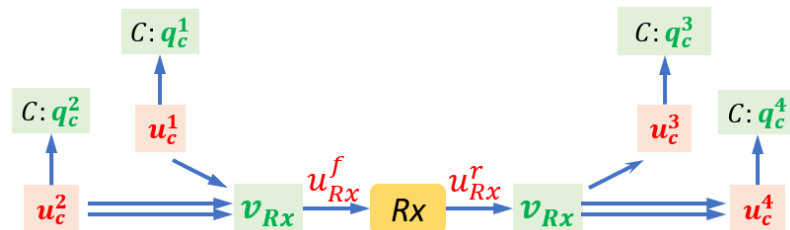
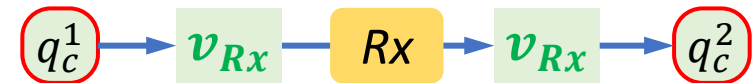
Mechanical



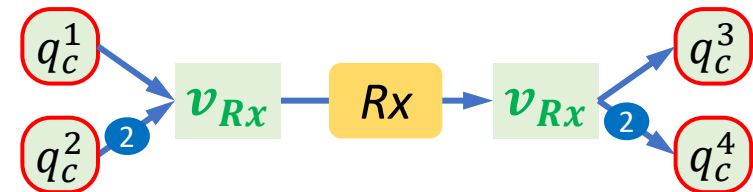
becomes



becomes

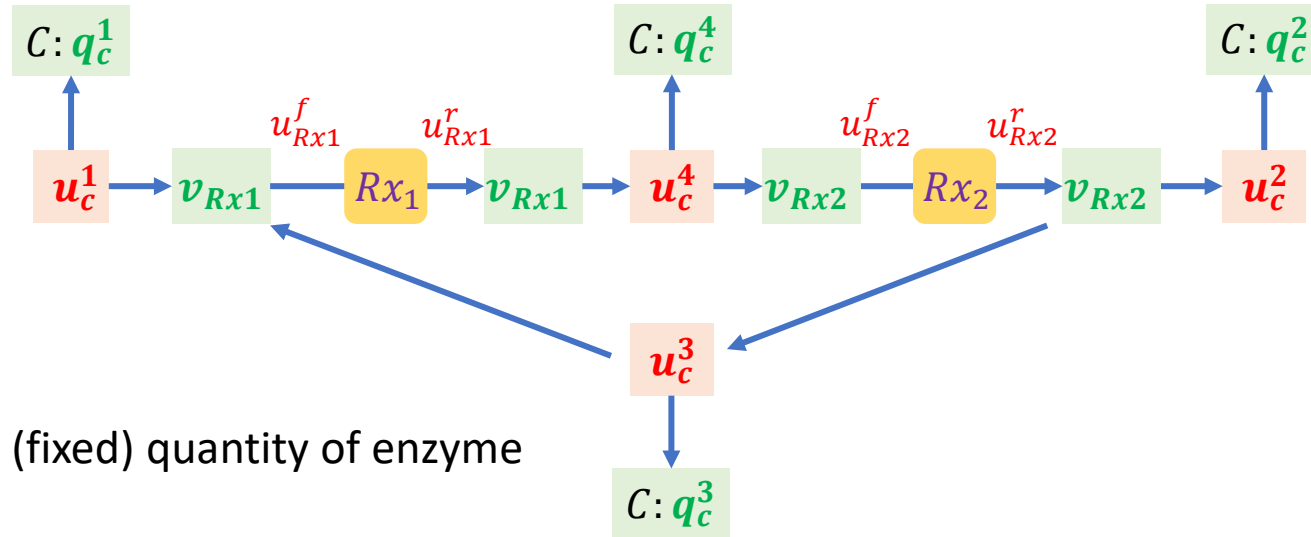
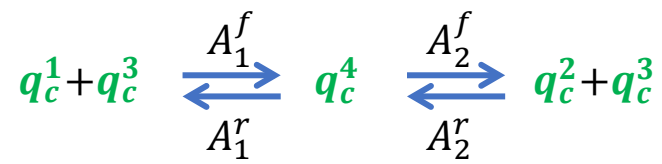


becomes



6. An enzyme-catalysed reaction

(see [model on PMR](#))



$q_c^3 + q_c^4 = E_0$, where E_0 is the initial (fixed) quantity of enzyme

Mass conservation at the 0:nodes:

$$(u_c^1) \quad \frac{d}{dt} q_c^1 = -v_{Rx1}; \quad (u_c^2) \quad \frac{d}{dt} q_c^2 = v_{Rx2}; \quad (u_c^3) \quad \frac{d}{dt} q_c^3 = v_{Rx2} - v_{Rx1}; \quad (u_c^4) \quad \frac{d}{dt} q_c^4 = v_{Rx1} - v_{Rx2};$$

$$(v_{Rx1}) \quad u_{Rx1}^f = u_c^1 + u_c^3; \quad u_{Rx1}^r = u_c^4; \quad (v_{Rx2}) \quad u_{Rx2}^f = u_c^4; \quad u_{Rx2}^r = u_c^2 + u_c^3.$$

Substitute for u_{Rx}^f & u_{Rx}^r and use Boltzmann equations

$$(Rx_1) \quad v_{Rx1} = \kappa_{Rx1} \left(e^{u_{Rx1}^f/RT} - e^{u_{Rx1}^r/RT} \right) = \kappa_{Rx1} (K_c^1 q_c^1 K_c^3 q_c^3 - K_c^4 q_c^4) = A_1^f q_c^1 q_c^3 - A_1^r q_c^4,$$

$$\text{where } A_1^f = \kappa_{Rx1} \cdot K_c^1 K_c^3 \text{ (mol}^{-1} \cdot \text{s}^{-1}\text{); } A_1^r = \kappa_{Rx1} \cdot K_c^4 \text{ (s}^{-1}\text{)}$$

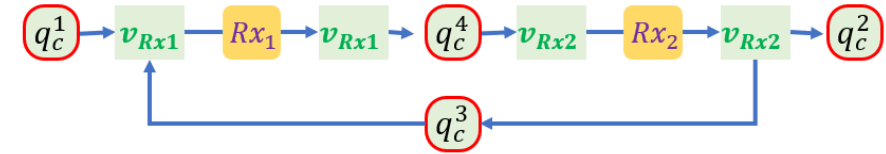
$$(Rx_2) \quad v_{Rx2} = \kappa_{Rx2} \left(e^{u_{Rx2}^f/RT} - e^{u_{Rx2}^r/RT} \right) = \kappa_{Rx2} (K_c^4 q_c^4 - K_c^2 q_c^2 K_c^3 q_c^3) = A_2^f q_c^4 - A_2^r q_c^2 q_c^3$$

$$\text{where } A_2^f = \kappa_{Rx2} \cdot K_c^4 \text{ (s}^{-1}\text{) and } A_2^r = \kappa_{Rx2} \cdot K_c^2 K_c^3 \text{ (mol}^{-1} \cdot \text{s}^{-1}\text{)}$$

Rewriting using more condensed notation:

$$(Rx_1) \quad v_{Rx1} = \kappa_{Rx1} (K_c^1 q_c^1 K_c^3 q_c^3 - K_c^4 q_c^4) = \kappa_{Rx1} (\bar{q}_c^1 \bar{q}_c^3 - \bar{q}_c^4)$$

$$(Rx_2) \quad v_{Rx2} = \kappa_{Rx2} (K_c^4 q_c^4 - K_c^2 q_c^2 K_c^3 q_c^3) = \kappa_{Rx2} (\bar{q}_c^4 - \bar{q}_c^2 \bar{q}_c^3)$$



The **Briggs-Haldane assumption** is that the amount of enzyme ($q_c^3 + q_c^4$) is much less than the amounts of substrate (q_c^1 & q_c^2), so that the unbound enzyme q_c^3 and the complex q_c^4 therefore quickly reach a steady state ($\frac{d}{dt} q_c^3 = 0$ and $\frac{d}{dt} q_c^4 = 0$).

Hence $\frac{d}{dt} q_c^3 = v_{Rx2} - v_{Rx1} = 0$ gives $v_{Rx1} = v_{Rx2}$ and $\frac{d}{dt} q_c^4 = v_{Rx1} - v_{Rx2} = 0$ gives $v_{Rx1} = v_{Rx2}$

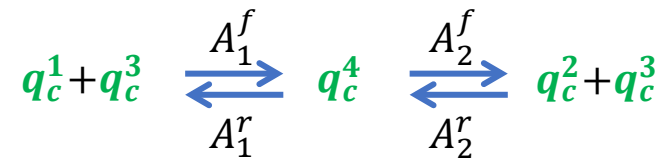
The other two mass-balance relations: $\frac{d}{dt} q_c^1 = -v_{Rx1} = \kappa_{Rx1} (\bar{q}_c^1 \bar{q}_c^3 - \bar{q}_c^4)$ and $\frac{d}{dt} q_c^2 = v_{Rx2} = \kappa_{Rx2} (\bar{q}_c^4 - \bar{q}_c^2 \bar{q}_c^3)$,

together with the constraint $q_c^3 + q_c^4 = E_0$, can be solved to eliminate q_c^3 & q_c^4 and give a steady-state flux:

$$v = v_{Rx1} = v_{Rx2} = E_0 \cdot \frac{\kappa_{Rx1} \kappa_{Rx2} (\bar{q}_c^1 - \bar{q}_c^2)}{\frac{\kappa_{Rx1} + \kappa_{Rx2}}{K_c^3} + \frac{\kappa_{Rx1}}{K_c^4} \bar{q}_c^1 + \frac{\kappa_{Rx2}}{K_c^4} \bar{q}_c^2}$$

or $v = v_m \frac{\bar{q}_c^1 - \bar{q}_c^2}{1 + \frac{\bar{q}_c^1}{k_m^1} + \frac{\bar{q}_c^2}{k_m^2}}$ where $v_m = E_0 \cdot \frac{K_c^3 \cdot \kappa_{Rx1} \kappa_{Rx2}}{\kappa_{Rx1} + \kappa_{Rx2}}$, $k_m^1 = \frac{K_c^4}{K_c^3} \cdot \frac{\kappa_{Rx1} + \kappa_{Rx2}}{\kappa_{Rx1}}$, $k_m^2 = \frac{K_c^4}{K_c^3} \cdot \frac{\kappa_{Rx1} + \kappa_{Rx2}}{\kappa_{Rx2}}$

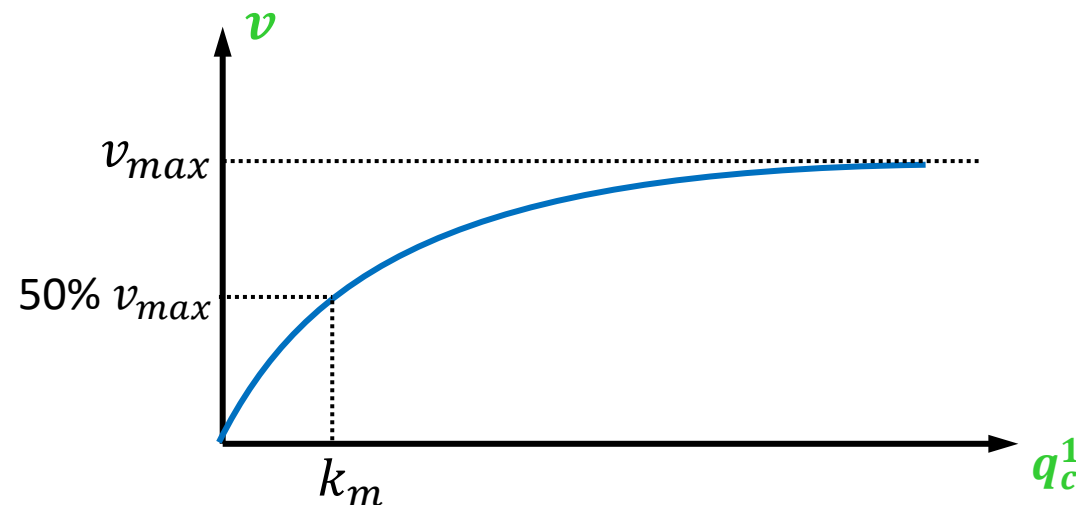
Michaelis-Menten (MM)



If $q_c^2 \ll q_c^1$, the MM assumption puts $q_c^2 = 0$ (which violates the laws of thermodynamics!).

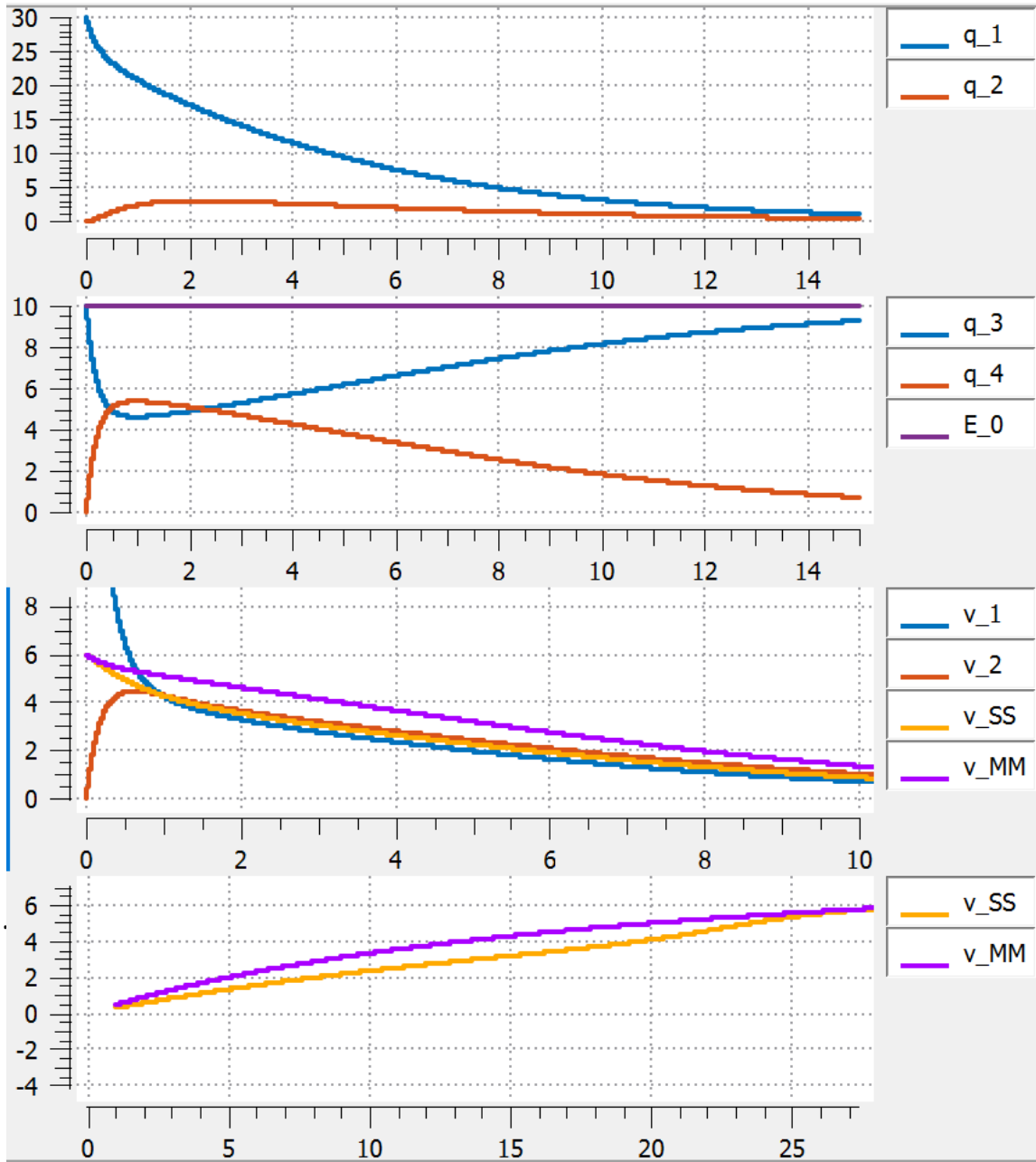
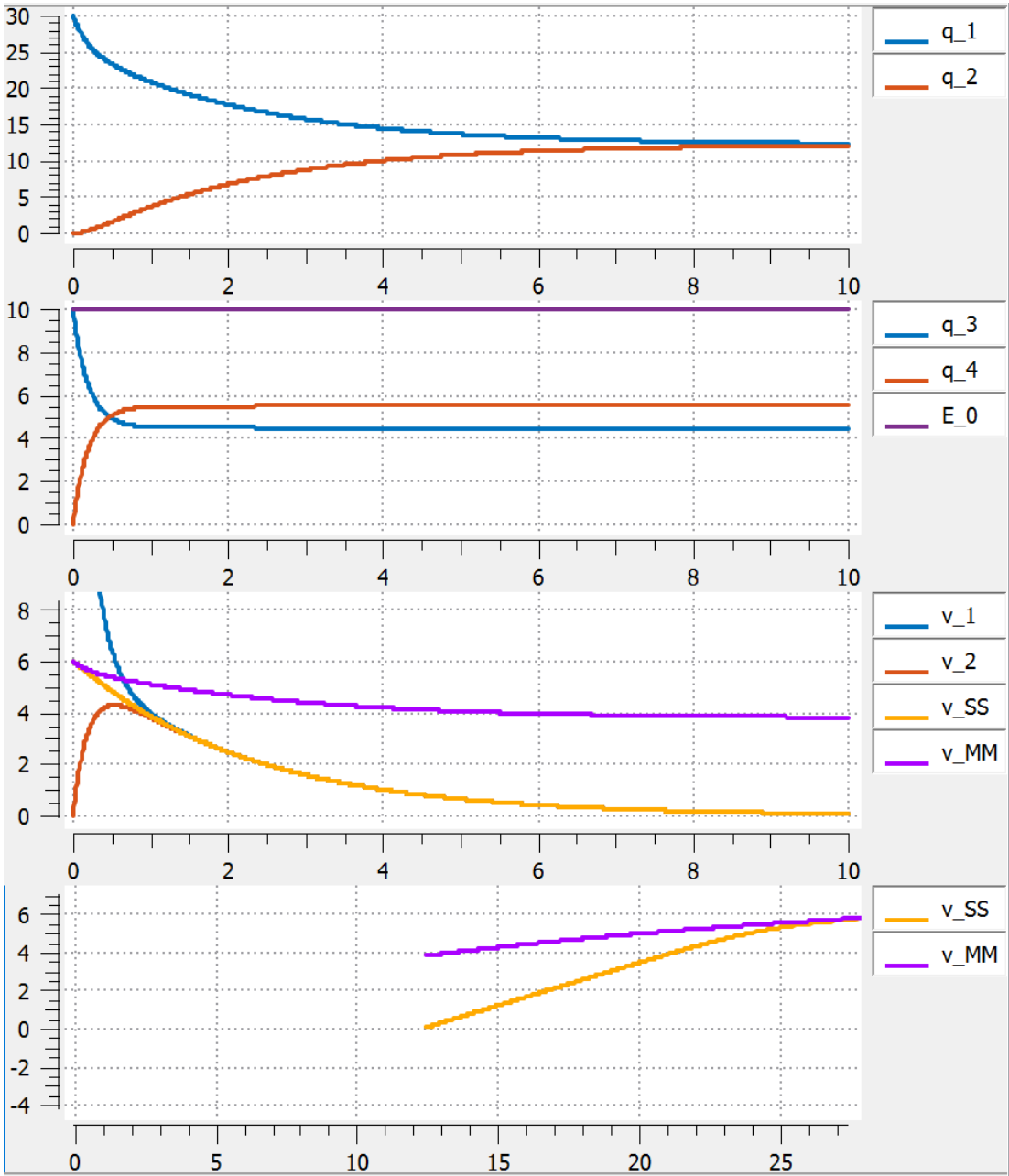
With that assumption, $v = v_m \frac{\bar{q}_c^1 - \bar{q}_c^2}{1 + \frac{\bar{q}_c^1}{k_m^1} + \frac{\bar{q}_c^2}{k_m^2}}$ reduces to $v = v_m \frac{\bar{q}_c^1}{1 + \frac{\bar{q}_c^1}{k_m^1}} = v_m k_m^1 \cdot \frac{q_c^1}{\frac{k_m^1}{K_c^1} + q_c^1}$,

which can be rewritten as $v = v_{max} \cdot \frac{q_c^1}{k_m + q_c^1}$ (where $v_{max} = v_m \cdot k_m^1$ and $k_m = \frac{k_m^1}{K_c^1}$)



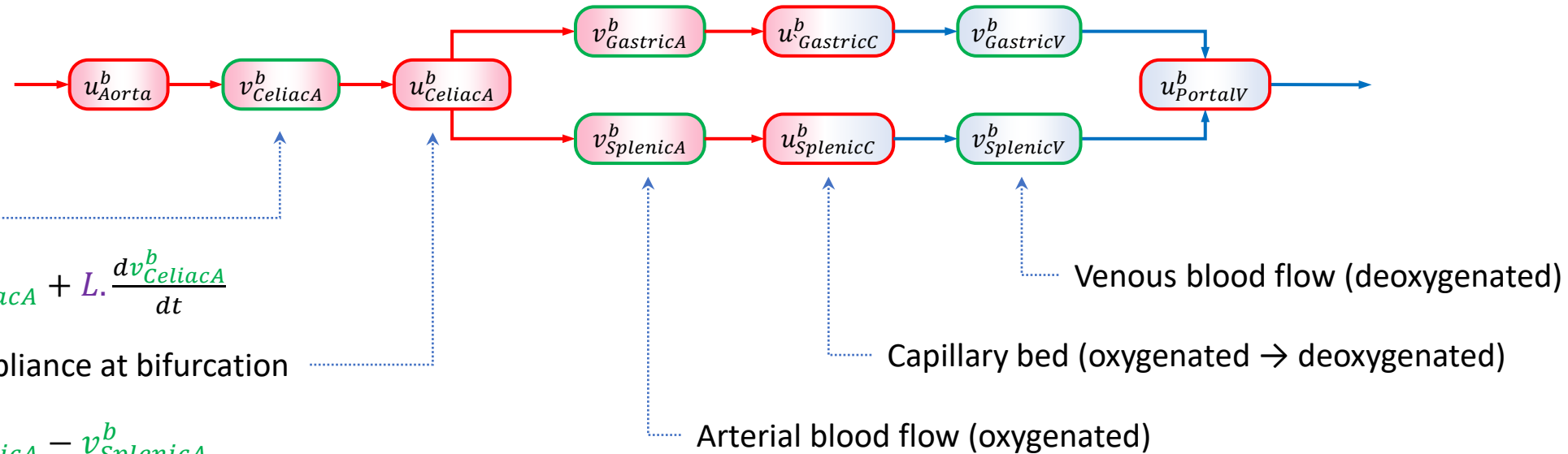
Comparing solutions under different conditions

Add a sink for q_c^2



7. Blood flow in the cardiovascular system

The following example of blood flow through the stomach & spleen illustrates the use of both of these new symbols:



Energy balance*

$$u_{Aorta}^b = u_{CeliacA}^b + R \cdot v_{CeliacA}^b + L \cdot \frac{dv_{CeliacA}^b}{dt}$$

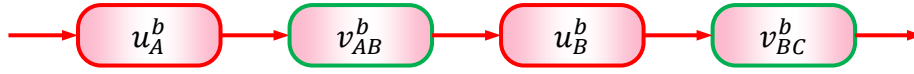
Mass balance & storage compliance at bifurcation

$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

$$u_{CeliacA}^b = (q_{CeliacA}^b - \hat{q}_{CeliacA}^b) / C_{CeliacA}^b$$

* To solve rewrite as $\frac{dv_{CeliacA}^b}{dt} = (u_{Aorta}^b - u_{CeliacA}^b - R \cdot v_{CeliacA}^b) / L$

Calculation of material constants for blood flow model



Mass conservation at the 0:node u_B^b is $\frac{dq_B^b}{dt} = v_{AB}^b - v_{BC}^b$

with the vascular compliance given by $u_B^b = q_B^b / C_B$

To calculate the compliance C_B we assume a thin vascular wall (thickness h).

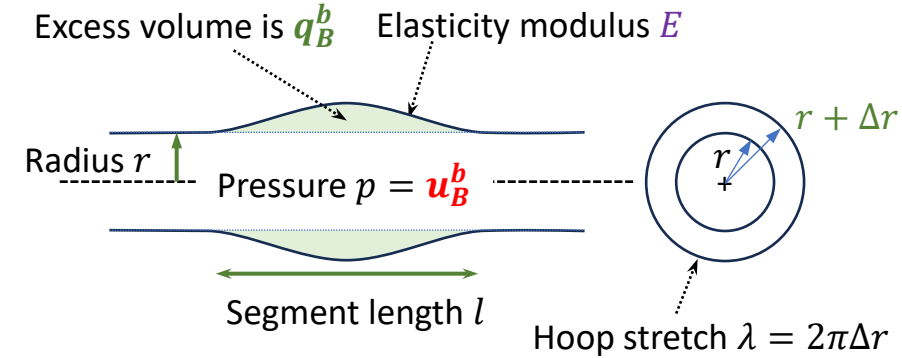
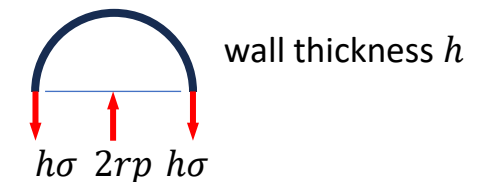
At pressure p the radius r increases to $r + \Delta r$, giving a hoop stretch $\lambda = 2\pi(r + \Delta r) - 2\pi r = 2\pi\Delta r$.

Assuming a linear elastic material (modulus E), this generates a stress $\sigma = E\lambda = E \cdot 2\pi\Delta r$, giving $\Delta r = \frac{\sigma}{2\pi E}$

and the increase in volume for the cylinder is $q_B^b = \Delta V = 2\pi(r + \Delta r)l - 2\pi r l = 2\pi\Delta r l = 2\pi \cdot \frac{\sigma}{2\pi E} l = \frac{\sigma l}{E}$.

The force balance for a thin-walled cylinder (Laplace) is: $2rp = 2h\sigma$ or $\sigma = \frac{r}{h}p = \frac{r}{h}$,

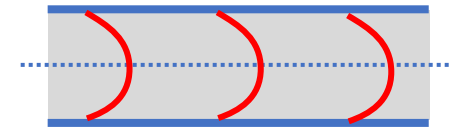
and therefore $q_B^b = \frac{rl}{Eh} p \cdot u_B^b$ or $u_B^b = q_B^b / C_B$, where the compliance $C_B = \frac{rl}{Eh}$.



Finally energy conservation at the 1:node v_{AB}^b is given by $v_{AB}^b = (u_A^b - u_B^b) / R_{AB}$

where, assuming steady fully developed Poiseuille flow, the viscous resistance is $R_{AB} = \frac{8\mu l}{\pi r^4}$.

Derivation of Poiseuille relation for steady flow in a tube



Steady fully-developed flow

The Navier-Stokes (NS) equations are for incompressible fluid with density ρ and dynamic viscosity μ :

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} + \frac{1}{\rho} \nabla p = \frac{\mu}{\rho} \nabla^2 \mathbf{v} \quad \text{with} \quad \nabla \cdot \mathbf{v} = 0.$$

Assuming $v = v_x(r, x, t)$ in the x -direction only and putting $p = p(x, t)$, the NS eqn in cylindrical polar coordinates is

$$\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\mu}{\rho} \left\{ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) + \frac{\partial^2 v_x}{\partial x^2} \right\} \quad \text{with incompressibility condition} \quad \frac{\partial v_x}{\partial x} = 0.$$

For steady flow in a straight tube there is no change in x or t and therefore $v(r)$ is determined by $\frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{\mu}{\rho} \cdot \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v}{\partial r} \right)$,

where the constant pressure gradient balances the viscous forces.

Putting $\frac{1}{\mu} \frac{\partial p}{\partial x} = k$, this becomes $\frac{1}{r} \frac{d}{dr} \left(r \frac{dv}{dr} \right) = k$ or $r \frac{dv}{dr} = \frac{1}{2} k r^2 + a$ or $\frac{dv}{dr} = \frac{1}{2} k r + \frac{a}{r}$.

But, since $\left. \frac{dv}{dr} \right|_{r=0} = 0$, $a = 0$ and $\frac{dv}{dr} = \frac{1}{2} k r$ gives $v(r) = \frac{1}{4} k r^2 + b$

At the tube inner radial boundary, where $r = R$, $v(R) = \frac{1}{4} k R^2 + b = 0$. Therefore $b = -\frac{1}{4} k R^2$, and $v(r) = \frac{1}{4} k (r^2 - R^2)$

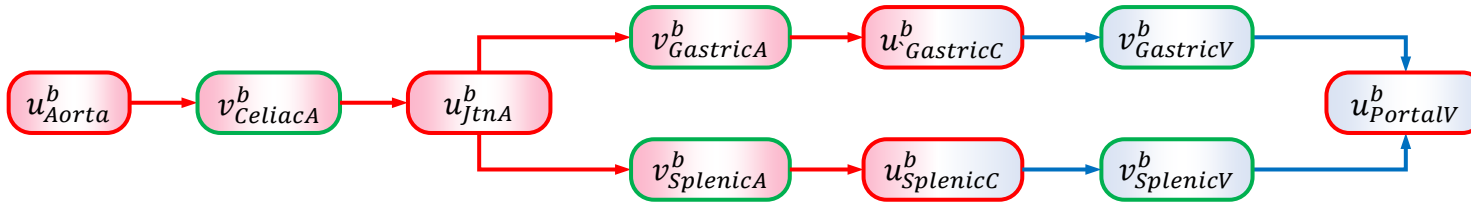
If we put $v(0) = V = -\frac{1}{4} k R^2$. With $k = \frac{1}{\mu} \frac{dp}{dx}$, $\frac{dp}{dx} = -\frac{4\mu V}{R^2}$ and the flow through the tube is

$$Q = \int_0^R v(r) \cdot 2\pi r \cdot dr = \frac{\pi}{2} k \int_0^R (r^3 - R^2 r) \cdot dr = \frac{\pi}{2} k \left(\frac{R^4}{4} - \frac{R^4}{2} \right) = -\frac{\pi R^4}{8\mu} \cdot \frac{dp}{dx}, \quad \text{or} \quad k = -\frac{8}{\pi R^4} Q \quad \text{and} \quad \frac{dp}{dx} = -\frac{8\mu}{\pi R^4} Q.$$

i.e. the pressure drop between the two ends of the constant radius tube of length l is $\Delta p = \frac{8l\mu}{\pi R^4} Q$

or $\mathbf{u}_A^b - \mathbf{u}_B^b = R_{AB} \cdot \mathbf{v}_{AB}^b$, where $R_{AB} = \frac{8l\mu}{\pi r^4}$ is the resistance for the vessel AB of length l and radius r with blood of viscosity μ .

Bond graph equations and solution for blood flow model



Assuming pressures u_{Aorta}^b and $u_{PortalV}^b$ are specified as bdy conditions and ignore inertial terms.

Energy balance at 1-nodes

$$v_{CeliacA}^b = (u_{Aorta}^b - u_{CeliacA}^b) / R_{CeliacA}^b$$

$$v_{GastricA}^b = (u_{CeliacA}^b - u_{GastricC}^b) / R_{GastricA}^b$$

$$v_{GastricV}^b = (u_{GastricC}^b - u_{PortalV}^b) / R_{GastricV}^b$$

$$v_{SplenicA}^b = (u_{CeliacA}^b - u_{SplenicC}^b) / R_{SplenicA}^b$$

$$v_{SplenicV}^b = (u_{SplenicC}^b - u_{PortalV}^b) / R_{SplenicV}^b$$

Mass balance & compliance relations at 1-nodes

$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

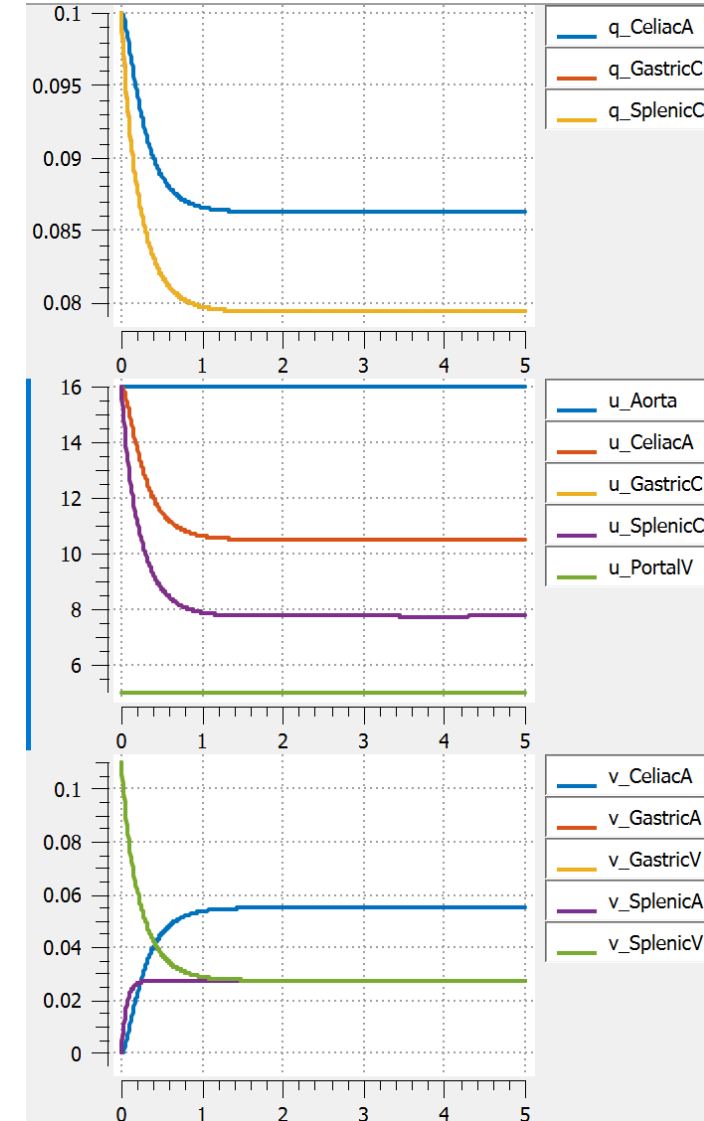
$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$

$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$

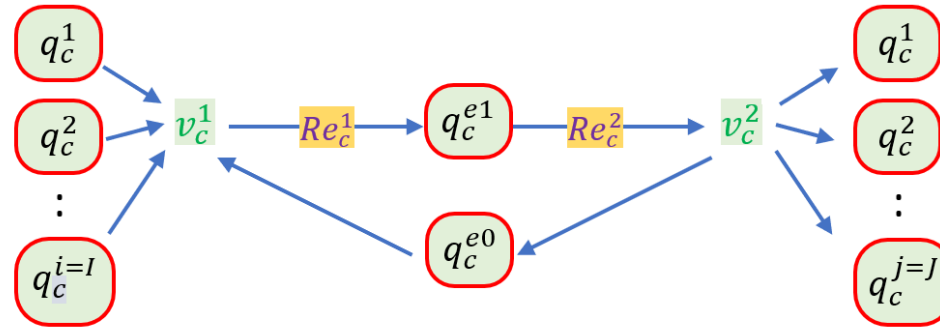
$$\frac{dq_{CeliacA}^b}{dt} = v_{CeliacA}^b - v_{GastricA}^b - v_{SplenicA}^b$$

$$u_{CeliacA}^b = q_{CeliacA}^b / C_{CeliacA}^b$$



8. An enzyme-catalysed reaction with multiple reactants and products

(see [model on PMR](#))



$$v = \frac{A^f \prod_i q_c^i - A^r \prod_j q_c^j}{1 + B^f \prod_i q_c^i + B^r \prod_j q_c^j} \quad \text{where } \prod_i q_c^i = q_c^1 \cdot q_c^2 \cdot q_c^3 \cdot \dots \cdot q_c^I$$

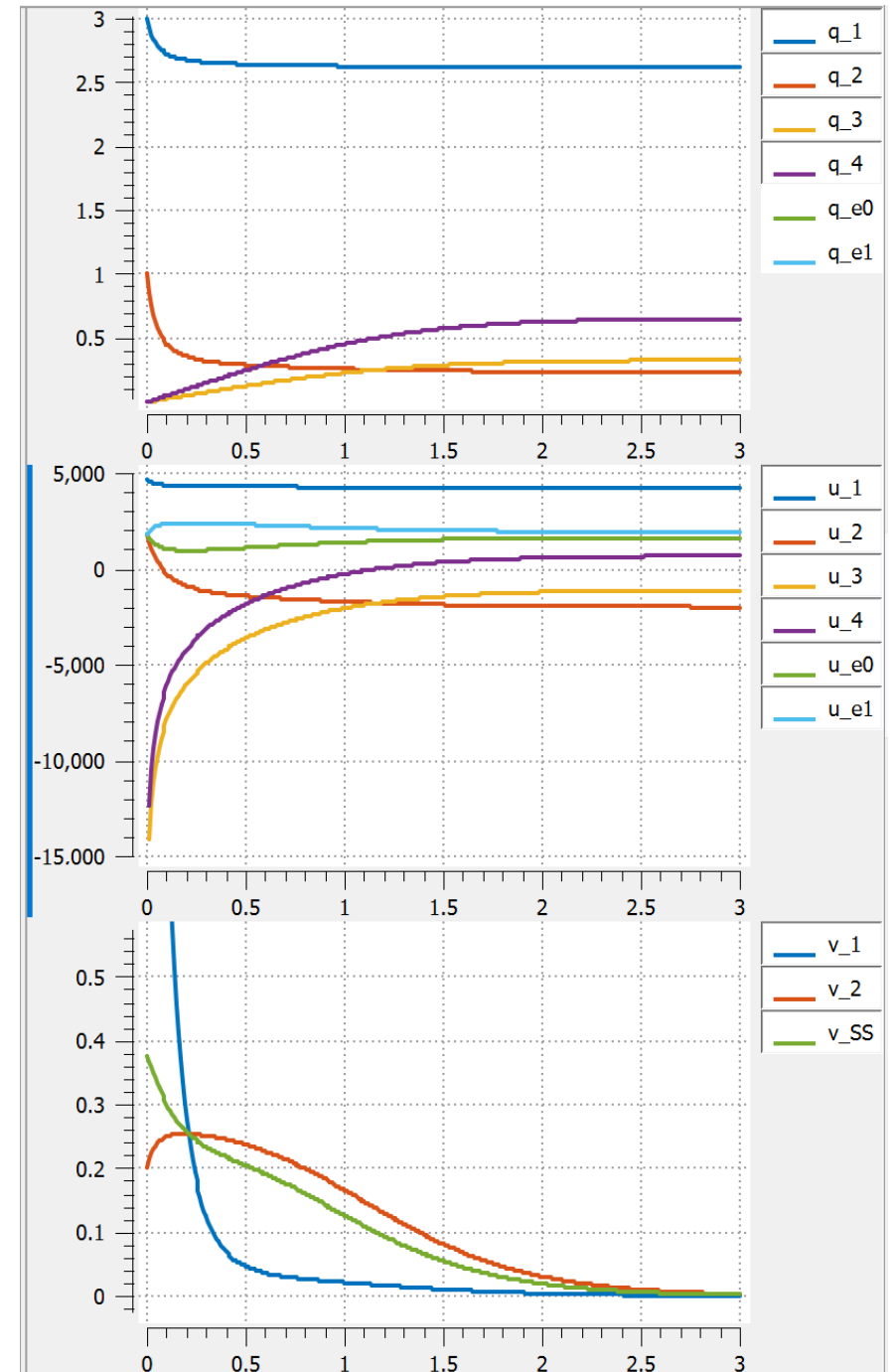
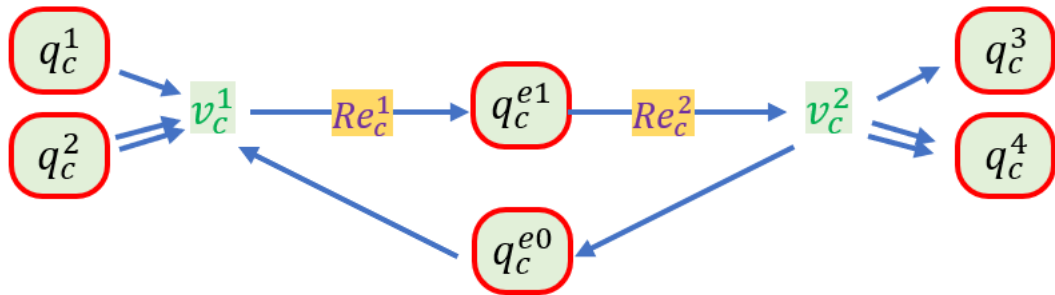
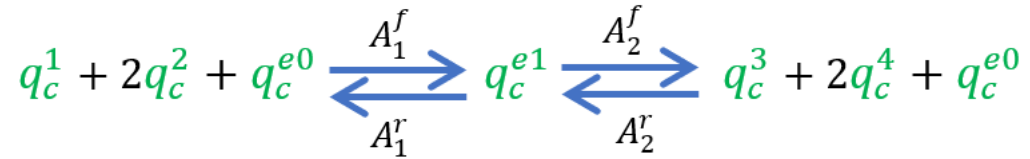
and $\prod_j q_c^j = q_c^1 \cdot q_c^2 \cdot q_c^3 \cdot \dots \cdot q_c^J$

where $A^f = E_0 K_c^{e0} \cdot \frac{\kappa_1 \kappa_2}{\kappa_1 + \kappa_2} \cdot \prod_i K_c^i \quad (\text{mol}^{-I} \cdot \text{s}^{-1});$ $A^r = E_0 K_c^{e0} \cdot \frac{\kappa_1 \kappa_2}{(\kappa_1 + \kappa_2)} \cdot \prod_j K_c^j \quad (\text{mol}^{-J} \cdot \text{s}^{-1});$

$B^f = \frac{K_c^{e0}}{K_c^{e1}} \cdot \frac{\kappa_1}{(\kappa_1 + \kappa_2)} \prod_i K_c^i \quad (\text{mol}^{-I});$ $B^r = \frac{K_c^{e0}}{K_c^{e1}} \cdot \frac{\kappa_2}{(\kappa_1 + \kappa_2)} \prod_j K_c^j \quad (\text{mol}^{-J});$

An example

(see [model on PMR](#))



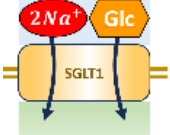
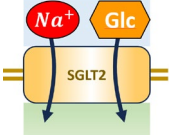
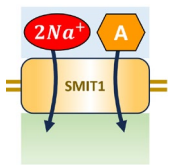
Lecture 4: The SLC family

(Transport of small molecules across cell membranes)

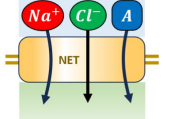
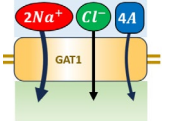
The primary goal of this lecture is to show you how to model **facilitated diffusion** – the process by which a small molecule passes into a cell – and how to handle **electrogenic transporters**. The notes describe many more **transporters** (**from the SLC family**), and once you understand the bond graph approach to modelling the reactions involved in these transporters, with enough practice you should be able to tackle any of these. The other key message in this lecture is how to **reduce the complexity of a model in a way that continues to satisfy the physical conservation laws**. The model for steady state flux of a small molecule through a cell membrane can usefully be reduced to depend on a small number of parameters, but unlike standard Hill-type models that use only 2 parameters (v_{max} and k_m), these can now be understood in terms of the underlying chemical process parameters. Most importantly, the model deals with situations where the Michaelis-Menten assumption that the concentration of a solute on one side of a reaction (whether in the cytosol or extracellular space) is much less than on the other side (and therefore can be ignored) is not appropriate – as it violates thermodynamic principles.

The SLC (solute carrier) group of proteins, encoded by 458 mammalian genes, mediate the passive transport of small molecules across cell membranes in nearly all types of tissues (i.e., does not include ATPase pumps). They are categorised under 66 gene families, labelled SLC1 to SLC66 ([SLCtables \(bioparadigms.org\)](http://bioparadigms.org)). Each family deals with a specific type of transported molecule.

SLC5 family: Na⁺/glucose cotransporters

SLC5A1	P13866	SGLT1	Na ⁺ -driven glucose transporter	
SLC5A2	P31639	SGLT2		
SLC5A3	P53794	SMIT1	Na ⁺ /myo-inositol cotransporter A = myo-inositol	

SLC6 family: Na⁺- and Cl⁻-dependent neurotransmitter symporters

SLC6A1	P30531	GAT1	A = Aminobutanoate	
SLC6A2	P23975	NET	A = Norepinephrine	

SLC transporters: 458 genes in 66 gene families

SLC1: High-affinity glutamate and neutral amino acid transporters
SLC2: Facilitative GLUT transporters
SLC3: Heavy subunits heterodimeric amino acid transporters
SLC4: Bicarbonate cotransporters
SLC5: Na⁺/glucose cotransporters
SLC6: Na⁺- and Cl⁻-dependent neurotransmitter symporters
SLC7: Cationic AA transporter/glycoprotein - assoc AA transporters
SLC8: Na⁺/Ca²⁺ exchangers
SLC9: Na⁺/H⁺ exchangers
SLC10: Na⁺/bile salt co-transporters
SLC11: Proton-coupled metal ion transporters
SLC12: Electroneutral cation/Cl⁻ co-transporters
SLC13: Na⁺/SO₄²⁻/carboxylate co-transporters
SLC14: Urea transporters
SLC15: Proton-driven oligopeptide cotransporters
SLC16: Monocarboxylate transporters
SLC17: Type I Na⁺/phosphate cotransporters
& vesicular glutamate transporters
SLC18: Vesicular amine transporters
SLC19: Folate/thiamine transporters
SLC20: Type-III Na⁺/phosphate co-transporters
SLC21: Organic anion and cation transporters
SLC22: Organic cation/anion/zwitterion transporters
SLC23: Na⁺-dependent ascorbic acid transporters
SLC24: Na⁺/(Ca²⁺-K⁺) exchangers
SLC25: Mitochondrial carriers
SLC26: Multifunctional anion exchangers
SLC27: Fatty acid transporters
SLC28: Na⁺-coupled nucleoside transporters
SLC29: Facilitative nucleoside transporters
SLC30: Zinc efflux transporters
SLC31: Copper transporters
SLC32: Vesicular inhibitory amino acid transporters
SLC33: Acetyl-CoA transporters

SLC34: Type II Na⁺-phosphate cotransporters
SLC35: Nucleoside-sugar transporters
SLC36: Proton-coupled amino acid transporters
SLC37: Sugar-phosphate/phosphate exchangers
SLC38: System A and System N Na⁺-coupled neutral AA transporters
SLC39: Metal ion transporters
SLC40: Basolateral Fe²⁺ transporters
SLC41: MgtE-like magnesium transporters
SLC42: NH₃/NH₄⁺, CO₂ channels
SLC43: Na⁺-independent, system-L-like amino acid transporters
SLC44: Choline-like transporters
SLC45: H⁺/sugar cotransporters
SLC46: Folate transporters
SLC47: Multidrug and toxin extrusion (MATE) transporters
SLC48: Heme transporters
SLC49: FLVCR-related transporters
SLC50: Sugar efflux transporters
SLC51: Steroid-derived molecules transporters
SLC52: Riboflavin transporters
SLC53: Phosphate carriers
SLC54: Mitochondrial pyruvate carriers
SLC55: Mitochondrial cation/proton exchangers
SLC56: Sideroflexins
SLC57: NiPA-like magnesium transporters
SLC58: MagT-like magnesium transporters
SLC59: Sodium-dependent lysophosphatidylcholine symporters
SLC60: Glucose transporters
SLC61: Molybdate transporters
SLC62: Pyrophosphate transporters
SLC63: Sphingosine-phosphate transporters
SLC64: Golgi Ca²⁺/H⁺ exchangers
SLC65: NPC-type cholesterol transporters
SLC66: PQ-loop amino acid transporters

458 genes



66 gene families



bond graph templates

Examples:

SLC1: (EAAT) Glutamate transport

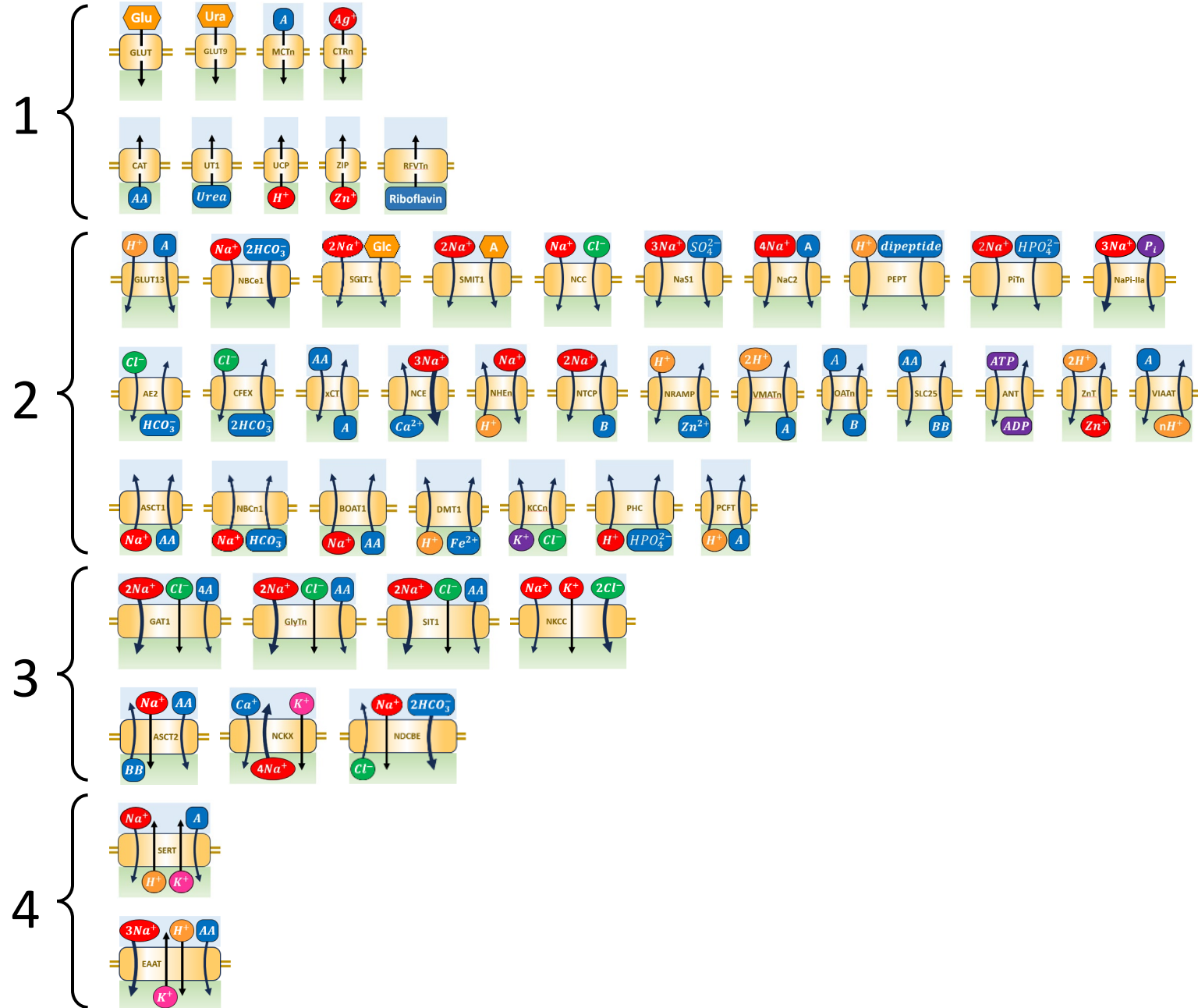
SLC2: (GLUT) Facilitated glucose transport

SLC4: (AE, NBC) Bicarbonate cotransporters

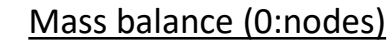
SLC5: (SGLT) Na-glucose cotransport

SLC8: (NCE) Na-Ca exchange

SLC9: (NHE) Na-H exchange



A diagram of the GLUT2 transporter. It is a yellow rounded rectangle with a blue border, embedded in a lipid bilayer. An orange hexagon labeled 'Glc' is shown entering the transporter from the left. A black arrow points from the transporter to the right, indicating the direction of glucose transport.



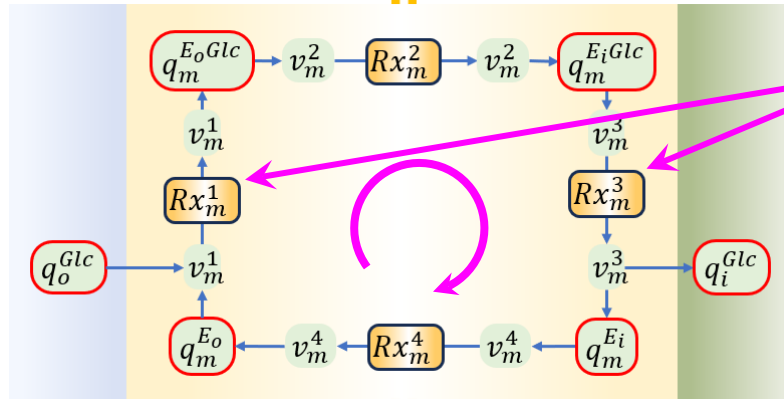
$$\frac{dq_i^{Glc}}{dt} = v_m^3$$

$$\frac{dq_m^{Ei}}{dt} = v_m^3 - v_m^4$$

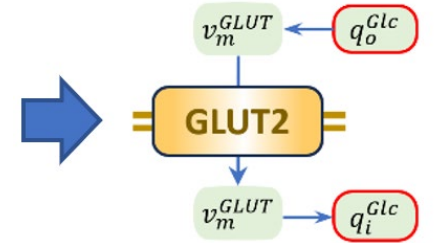
$$\frac{dq_m^{EiGlc}}{dt} = v_m^2 - v_m^3$$

$$\left. \begin{aligned} (Rx_m^1) \quad \mathbf{v}_m^1 &= \kappa_1 \left(e^{u_{Rx1}^f/RT} - e^{u_{Rx1}^r/RT} \right), \text{ where } u_{Rx1}^f = u_m^{Glc} + u_m^{E_o} \quad \text{and } u_{Rx1}^r = u_m^{E_o \cdot Glc}, \quad \text{which gives } \mathbf{v}_m^1 = \kappa_1 \left(\bar{q}_m^{E_o} \cdot \bar{q}_0^{Glc} - \bar{q}_m^{E_o Glc} \right) \\ (Rx_m^2) \quad \mathbf{v}_m^2 &= \kappa_2 \left(e^{u_{Rx2}^f/RT} - e^{u_{Rx2}^r/RT} \right), \text{ where } u_{Rx2}^f = u_m^{E_o \cdot Glc} \quad \text{and } u_{Rx2}^r = u_m^{E_i \cdot Glc}, \quad \text{which gives } \mathbf{v}_m^2 = \kappa_2 \left(\bar{q}_m^{E_o Glc} - \bar{q}_m^{E_i Glc} \right) \\ (Rx_m^3) \quad \mathbf{v}_m^3 &= \kappa_3 \left(e^{u_{Rx3}^f/RT} - e^{u_{Rx3}^r/RT} \right), \text{ where } u_{Rx3}^f = u_m^{E_i \cdot Glc} \quad \text{and } u_{Rx3}^r = u_m^{E_i} + u_i^{Glc}, \quad \text{which gives } \mathbf{v}_m^3 = \kappa_3 \left(\bar{q}_m^{E_i Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \right) \\ (Rx_m^4) \quad \mathbf{v}_m^4 &= \kappa_4 \left(e^{u_{Rx4}^f/RT} - e^{u_{Rx4}^r/RT} \right), \text{ where } u_{Rx4}^f = u_m^{E_i} \quad \text{and } u_{Rx4}^r = u_m^{E_o \cdot Glc}, \quad \text{which gives } \mathbf{v}_m^4 = \kappa_4 \left(\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right) \end{aligned} \right\} \text{ 8 parameters}$$

Reduced models



1. Binding & unbinding is very rapid & assume constant enzyme cycling rate
2. Michaelis-Menten ($q_i^{Glc} \ll q_o^{Glc}$)



Model reduction 1: Fast binding and unbinding ($\kappa_1, \kappa_3 \rightarrow \infty$) and SS ($v_1 = v_2 = v_3 = v_4 = v$)

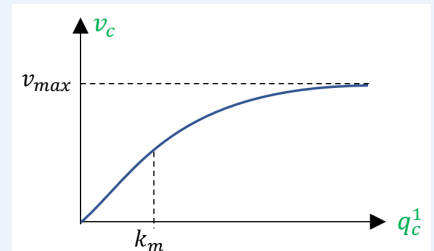
$$v = v_m (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right); \quad 4 \text{ parameters}$$

$$\text{with } k_m^1 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EoGlc})^{-1} + \Lambda (K_m^{Ei})^{-1}}, \quad k_m^2 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EiGlc})^{-1} + \Lambda (K_m^{Eo})^{-1}}, \quad k_m^3 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{\Lambda \left((K_m^{EoGlc})^{-1} + (K_m^{EiGlc})^{-1} \right)}, \quad v_m = \frac{q_{tot} \kappa_2}{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}, \quad \Lambda = \frac{\kappa_2}{\kappa_4}$$

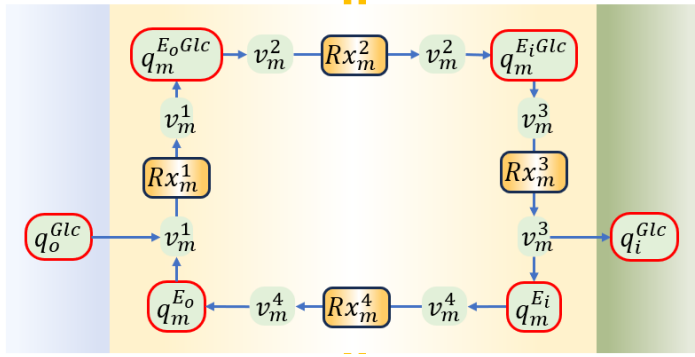
where $\bar{q}_j^i = K_j q_j^i = V_j K_j \cdot c_j^i$ (nondimensional), so k_m^1, k_m^2 & k_m^3 are also nondimensional.

Model reduction 2: Michaelis-Menten (MM) assumption ($q_i^{Glc} \ll q_o^{Glc}$)

$$v = v_{max} \frac{\bar{q}_o^{Glc}}{k_m^1 + \bar{q}_o^{Glc}} \quad 2 \text{ parameters} \quad \dots \text{but thermodynamically invalid}$$



Derivation of reduced model



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.Glc}] \quad \text{As } \kappa_1 \rightarrow \infty, [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.Glc}] \rightarrow 0 \text{ and hence } \bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} = \bar{q}_m^{E_o.Glc} \quad (1)$$

$$v_m^2 = \kappa_2 [\bar{q}_m^{E_o.Glc} - \bar{q}_m^{E_i.Glc}] \quad v_m^2 = v_m^{GLUT2} \text{ gives } \kappa_2 [\bar{q}_m^{E_o.Glc} - \bar{q}_m^{E_i.Glc}] = v_m^{GLUT2} \quad (2)$$

$$v_m^3 = \kappa_3 [\bar{q}_m^{E_i.Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc}] \quad \text{As } \kappa_3 \rightarrow \infty, [\bar{q}_m^{E_i.Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc}] \rightarrow 0 \text{ and hence } \bar{q}_m^{E_i.Glc} = \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \quad (3)$$

$$v_m^4 = \kappa_4 [\bar{q}_m^{E_i} - \bar{q}_m^{E_o}] \quad v_m^4 = v_m^{GLUT2} \text{ gives } \kappa_4 [\bar{q}_m^{E_i} - \bar{q}_m^{E_o}] = v_m^{GLUT2} \quad (4)$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^4 - v_m^1 = 0 \text{ gives } v_m^4 = v_m^1 = v_m^{GLUT2} \quad \frac{d}{dt} q_o^{Glc} = -v_m^1 = -v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_o.Glc} = v_m^1 - v_m^2 = 0 \text{ gives } v_m^1 = v_m^2 = v_m^{GLUT2} \quad \frac{d}{dt} q_i^{Glc} = v_m^3 = v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_i.Glc} = v_m^2 - v_m^3 = 0 \text{ gives } v_m^2 = v_m^3 = v_m^{GLUT2}$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 - v_m^4 = 0 \text{ gives } v_m^3 = v_m^4 = v_m^{GLUT2}$$

$$\text{An additional constraint is needed: } q_m^{E_o} + q_m^{E_o.Glc} + q_m^{E_i.Glc} + q_m^{E_i} = q_{tot} \quad (5)$$

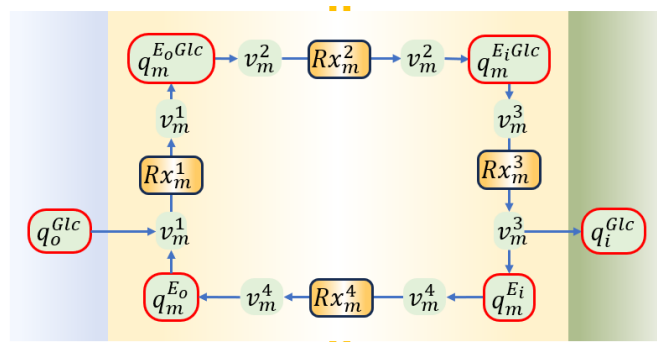
Equations 1..5 are used to eliminate the enzyme states $\bar{q}_m^{E_o}$, $\bar{q}_m^{E_o.Glc}$, $\bar{q}_m^{E_i.Glc}$, $\bar{q}_m^{E_i}$ and hence obtain

$$v_m^{GLUT2} = v_{max} (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right)$$

Note: In the reduced model, at a given enzyme cycling rate v_m^{GLUT2} , the enzyme states are assumed to be in steady-state. With time-varying solutes (ligands) there must be some loss of mass occurring during the transition to steady-state.

Facilitated diffusion: Comparing models

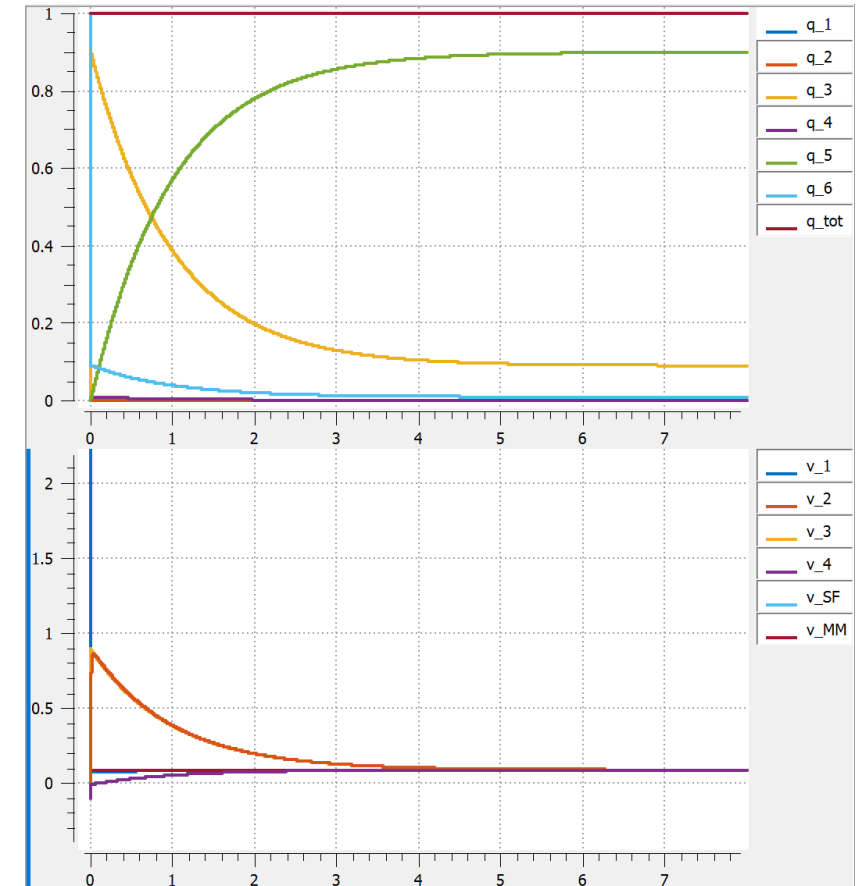
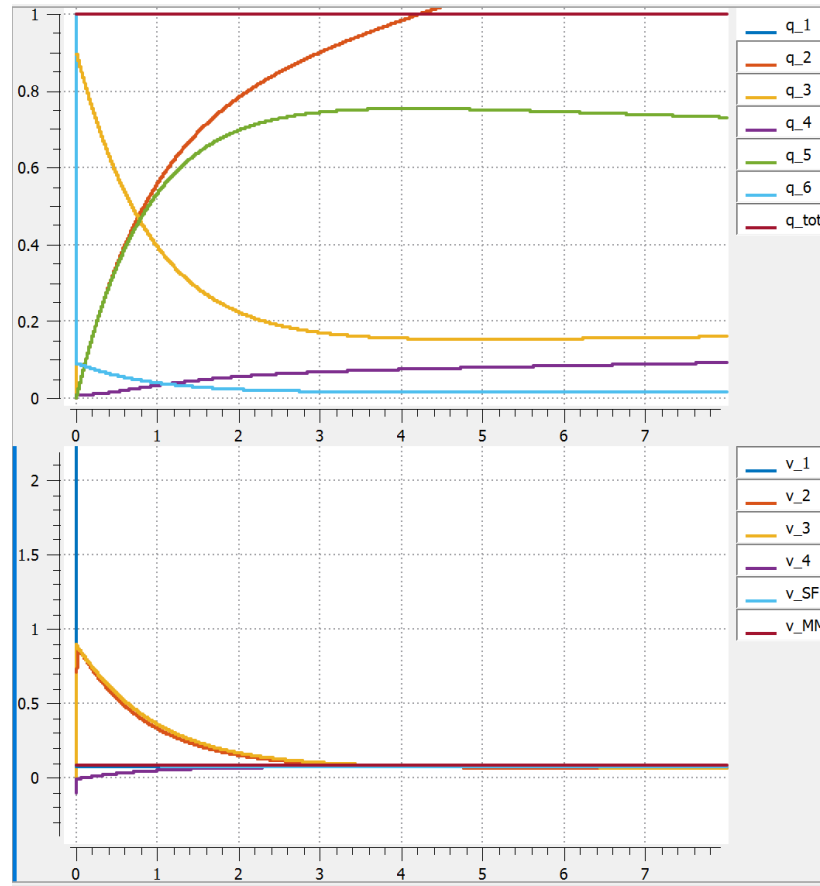
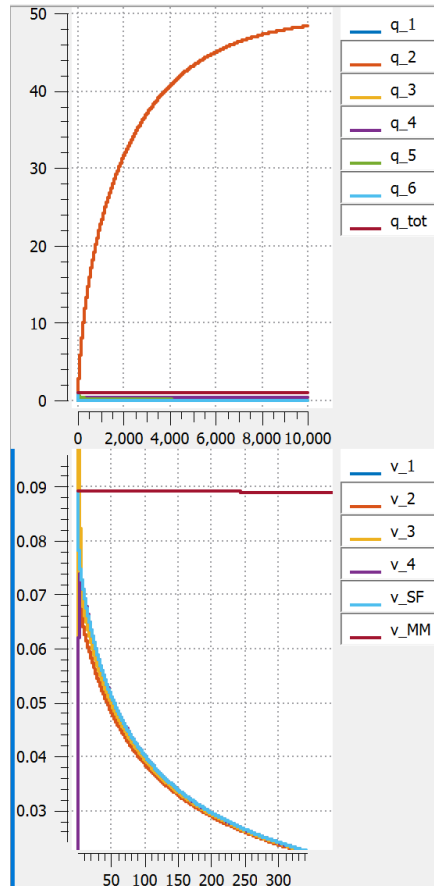
(see [model on PMR](#))



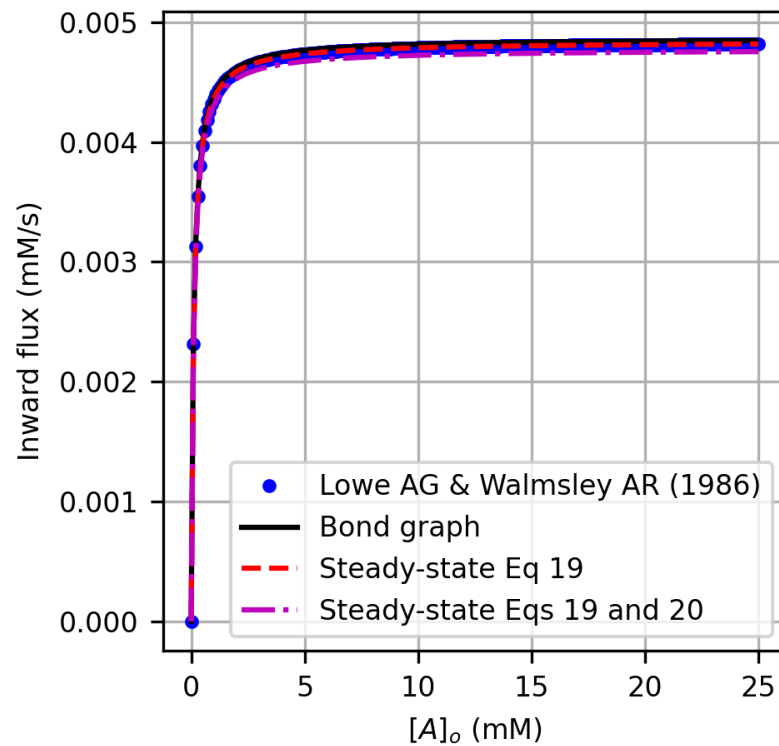
$\kappa_1=100$
 $\kappa_2=1$
 $\kappa_3=100$
 $\kappa_4=0.1$

Initially: $q_o^{Glc} = 100$, $q_i^{Glc} = 0$, $q_{tot}^{enzyme} = 1$

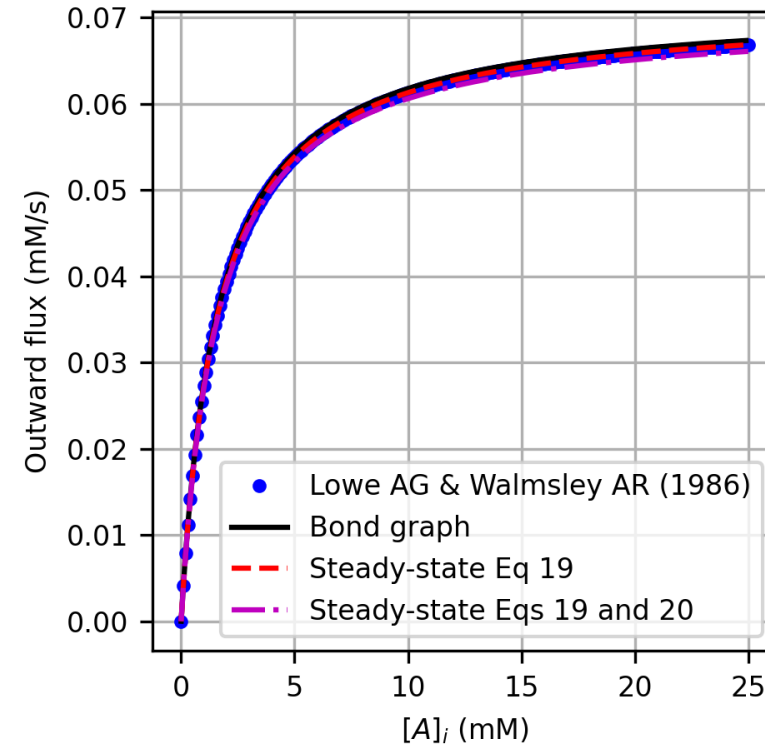
Add a sink for q_2^A ($\frac{d}{dt} q_2^A = v^A - q_2^A/100$)



GLUT2 model parameters fitted to experimental data from Lowe & Walmsley (1986)



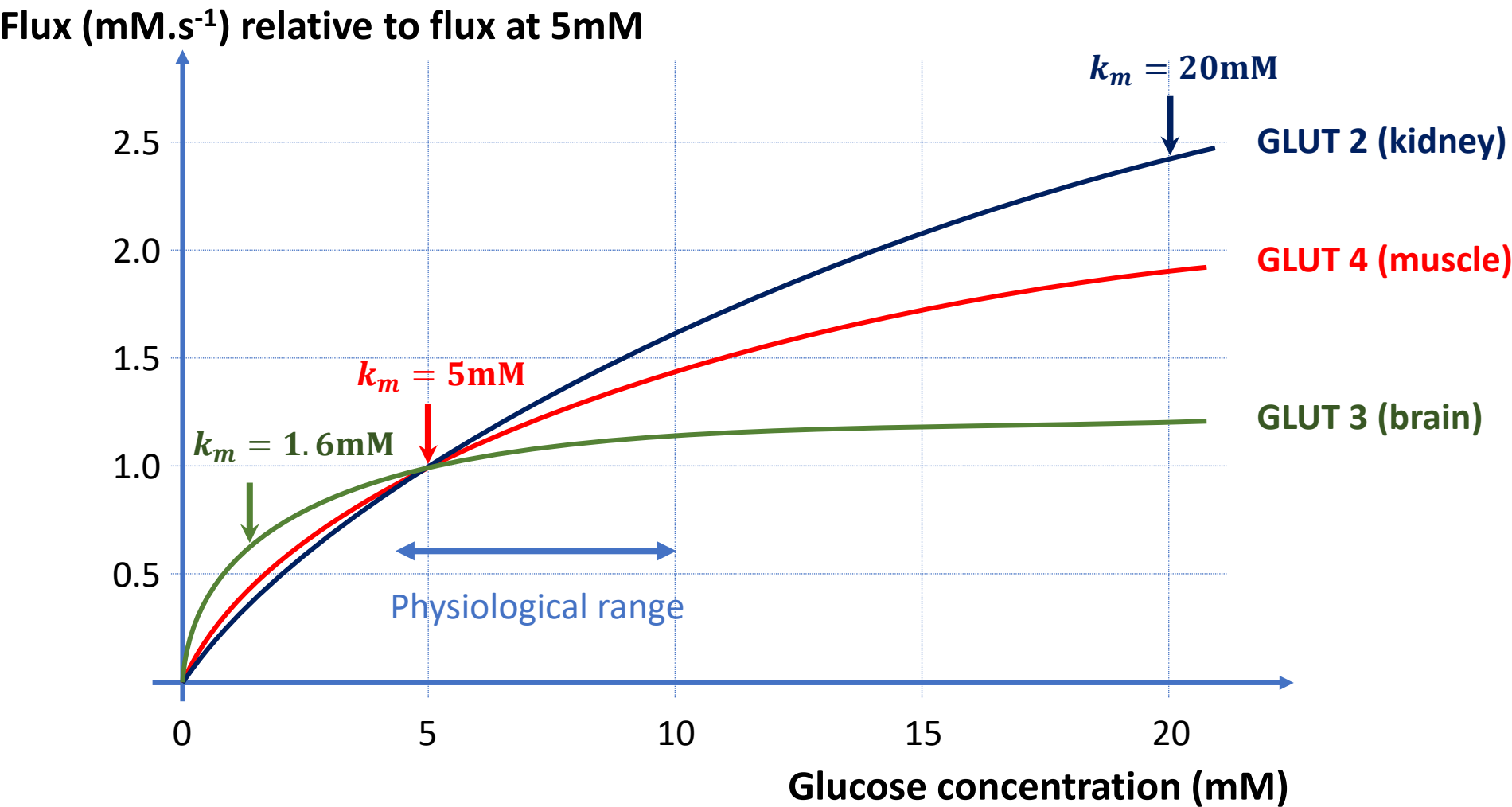
(a)



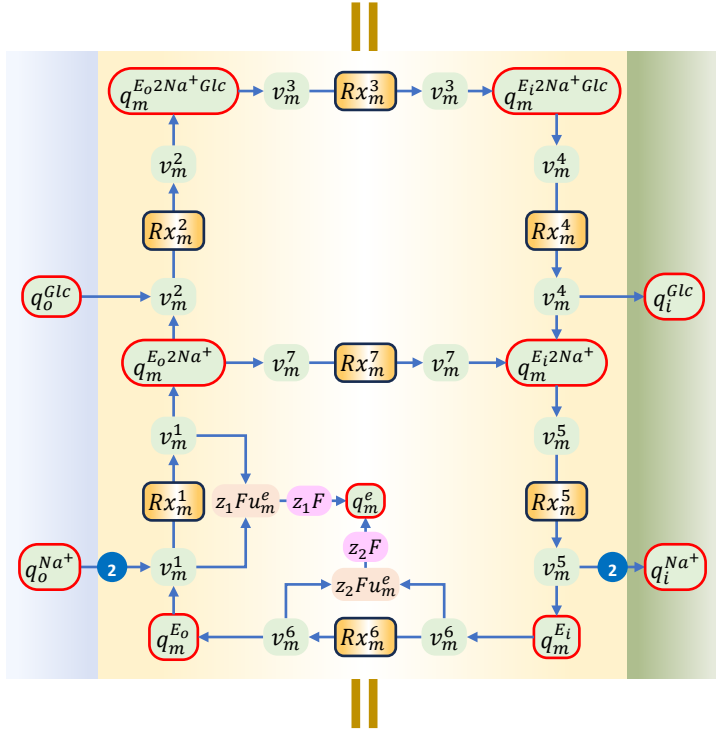
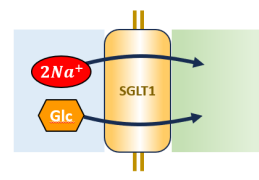
(b)

Figure (a) Inward flux as a function of $[Glc]_o$ when $[Glc]_i = 0$, and (b) outward flux as a function of $[Glc]_i$ when $[Glc]_o = 0$. Note that in order to compare with the kinetic data, the molar amount of glucose in the bond graph model was converted to glucose concentrations.

Note on tissue-specific GLUT transporters



The electrogenic, sodium-driven glucose transporter SGLT1 (*SLC5A1*)



Reaction fluxes

$$v_1 = \kappa_1 \left[(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_m^{E_o} \cdot e^{\frac{-z_1 F u_m^e}{RT}} - \bar{q}_m^{E_o 2Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_o^{Glc} \cdot \bar{q}_m^{E_o 2Na^+} - \bar{q}_m^{E_o 2Na^+ Glc} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_o 2Na^+ Glc} - \bar{q}_m^{E_i 2Na^+ Glc} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i 2Na^+ Glc} - \bar{q}_i^{Glc} \cdot \bar{q}_m^{E_i 2Na^+} \right]$$

$$v_5 = \kappa_5 \left[\bar{q}_m^{E_i 2Na^+} - \bar{q}_i^{Na^+} \cdot \bar{q}_m^{E_i} \right]$$

$$v_6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot e^{\frac{-z_2 F u_m^e}{RT}} - \bar{q}_m^{E_o} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_7 = \kappa_7 \left[\bar{q}_m^{E_o 2Na^+} - \bar{q}_m^{E_i 2Na^+} \right]$$

Mass balance (0:nodes)

$$\frac{dq_o^{Na^+}}{dt} = -2v_m^1$$

$$\frac{dq_i^{Na^+}}{dt} = 2v_m^5$$

$$\frac{dq_o^{Glc}}{dt} = -v_m^2$$

$$\frac{dq_i^{Glc}}{dt} = v_m^4$$

$$\frac{dq_m^{E_o}}{dt} = v_m^6 - v_m^1$$

$$\frac{dq_m^{E_o 2Na^+}}{dt} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{dq_m^{E_o 2Na^+ Glc}}{dt} = v_m^2 - v_m^3$$

$$\frac{dq_m^{E_i 2Na^+ Glc}}{dt} = v_m^3 - v_m^4$$

$$\frac{dq_m^{E_i 2Na^+}}{dt} = v_m^4 - v_m^5 + v_m^7$$

$$\frac{dq_m^{E_i}}{dt} = v_m^5 - v_m^6$$

Energy balance (1:nodes)

$$u_{Rx1}^f = 2u_o^{Na^+} + u_m^{E_o} - z_1 F u_m^e$$

$$u_{Rx3}^f = u_m^{E_o 2Na^+ Glc}$$

$$u_{Rx5}^f = u_m^{E_i 2Na^+}$$

$$u_{R7}^f = u_m^{E_o 2Na^+}$$

$$u_{Rx1}^r = u_m^{E_o 2Na^+} + z_1 F u_m^e$$

$$u_{Rx3}^r = u_m^{E_i 2Na^+ Glc}$$

$$u_{Rx5}^r = 2u_i^{Na^+} + u_m^{E_i}$$

$$u_{R7}^r = u_m^{E_i 2Na^+}$$

$$u_{Rx2}^f = u_o^{Glc} + u_m^{E_o 2Na^+}$$

$$u_{Rx4}^f = u_m^{E_i 2Na^+ Glc}$$

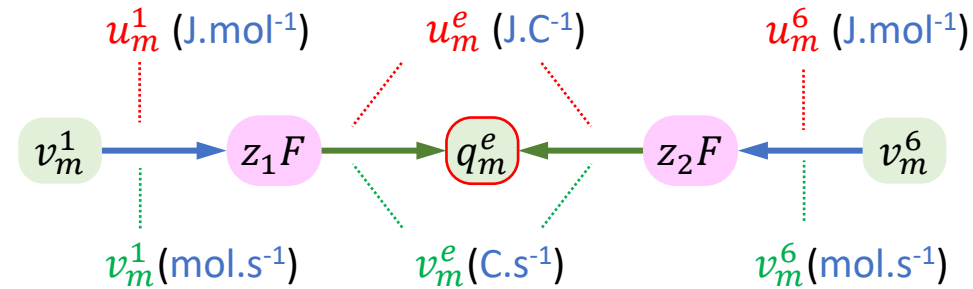
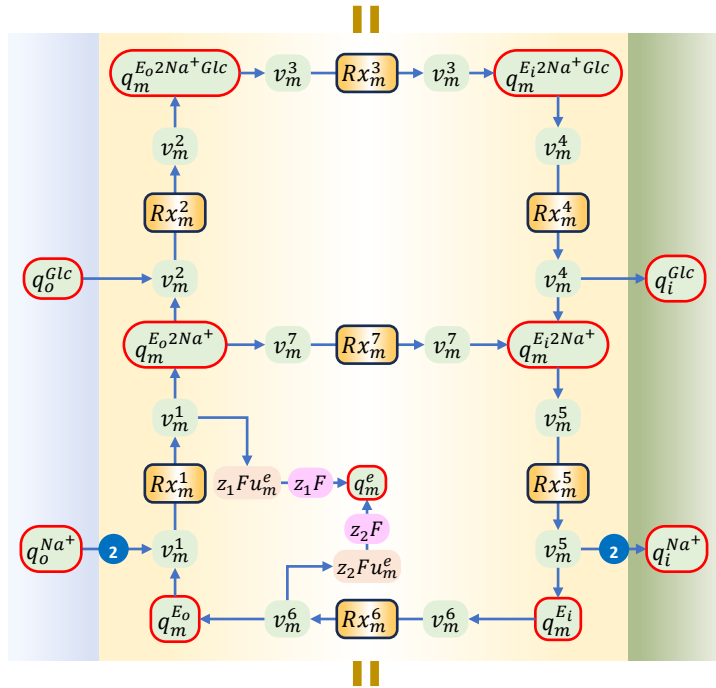
$$u_{Rx6}^f = u_m^{E_i} - z_2 F u_m^e$$

$$u_{Rx2}^r = u_m^{E_o 2Na^+ Glc}$$

$$u_{Rx4}^r = u_i^{Glc} + u_m^{E_i 2Na^+}$$

$$u_{Rx6}^r = u_m^{E_o} + z_2 F u_m^e$$

Modeling charge transfer in electrogenic transporters



Power: $u_m^1 \cdot v_m^1$ (J.s⁻¹) $u_m^e \cdot v_m^e$ (J.s⁻¹) $u_m^6 \cdot v_m^6$ (J.s⁻¹)

where u_m^1 (J.mol⁻¹) = $z_1 F$ (C.mol⁻¹) x u_m^e (J.C⁻¹)

and v_m^e (C.s⁻¹) = $z_1 F$ (C.mol⁻¹) x v_m^1 (mol.s⁻¹)

Note: $u_m^e = q_m^e / C$ with capacitance $C \sim 10^{-6}$ C²J⁻¹ (1μF) per m² of membrane

The sum of potential differences around the loop must be zero (equivalent to adding up potential drops around a circuit).

$$\begin{aligned}
 \text{Sum} &= (u_{Rx1}^f - u_{Rx1}^r) + (u_{Rx2}^f - u_{Rx2}^r) + (u_{Rx3}^f - u_{Rx3}^r) + (u_{Rx4}^f - u_{Rx4}^r) + (u_{Rx5}^f - u_{Rx5}^r) + (u_{Rx6}^f - u_{Rx6}^r) + (u_{Rx7}^f - u_{Rx7}^r) \\
 &= (2u_o^{Na^+} + u_m^{E_o} - u_m^{E_o 2Na^+} - z_1 F u_m^e) + (u_o^{Glc} + u_m^{E_o 2Na^+} - u_m^{E_o 2Na^+ Glc}) + (u_m^{E_o 2Na^+ Glc} - u_m^{E_i 2Na^+ Glc}) + (u_m^{E_i 2Na^+ Glc} - u_i^{Glc} - u_m^{E_i 2Na^+}) \\
 &\quad + (u_m^{E_i 2Na^+} - 2u_i^{Na^+} - u_m^{E_i}) + (u_m^{E_i} - u_m^{E_o} - z_2 F u_m^e) + (u_m^{E_o 2Na^+} - u_m^{E_i 2Na^+}) \\
 &= \underbrace{2(u_o^{Na^+} - u_i^{Na^+})}_{\text{sodium transport}} + \underbrace{(u_o^{Glc} - u_i^{Glc})}_{\text{glucose transport}} + \underbrace{(u_m^{E_o 2Na^+} - u_m^{E_i 2Na^+})}_{\text{slippage}} - \underbrace{(z_1 + z_2) F u_m^e}_{\text{charge transport}}
 \end{aligned}$$

$$0 = 2 \text{ J.mol}^{-1} - 1 \text{ J.mol}^{-1} (\text{against gradient}) - (z_1 + z_2) \text{ J.mol}^{-1} \Rightarrow (z_1 + z_2) = 1 \text{ (ignoring slippage)}$$

Reduced SGLT1 model

Steady-state solution is

$$v_m^{SGLT1} = a_1 \left\{ (\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} - (\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{2Fu_m^e/RT} \right\} / A$$

$$A = \left[e^{2z_1Fu_m^e/RT} + (\bar{q}_o^{Na^+})^2 (a_2 + a_3 \cdot \bar{q}_o^{Glc}) \right] \left[(\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{z_2Fu_m^e/RT} + \Lambda \right] \\ + \left[a_6 + (\bar{q}_i^{Na^+})^2 (a_5 + a_4 \cdot \bar{q}_i^{Glc}) \right] \left[(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} \cdot e^{z_2Fu_m^e/RT} + \Lambda \cdot e^{2Fu_m^e/RT} \right]$$

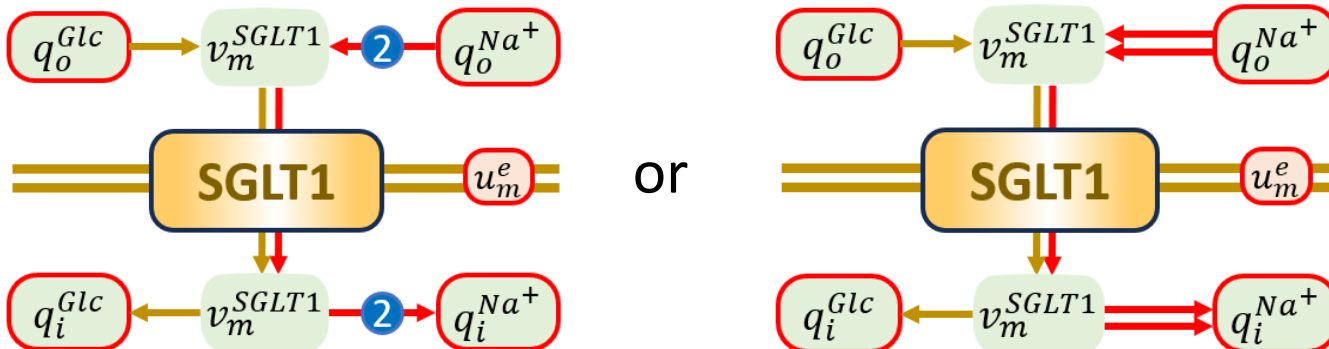
where $a_1 = q_{tot} \kappa_6 K_1$, $a_2 = \frac{K_1}{K_2}$, $a_3 = \frac{K_1}{K_3}$, $a_4 = \frac{K_1}{K_4}$, $a_5 = \frac{K_1}{K_5}$, $a_6 = \frac{K_1}{K_6}$, $\Lambda = \frac{\kappa_6}{\kappa_3}$

Note that the flux is zero when

$$(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} = (\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{\frac{2Fu_m^e}{RT}}$$

i.e. the equilibrium potential is

$$u_m^e = \frac{RT}{2F} \ln \left[\left(\frac{\bar{q}_o^{Na^+}}{\bar{q}_i^{Na^+}} \right)^2 \cdot \frac{\bar{q}_o^{Glc}}{\bar{q}_i^{Glc}} \right]$$



Note different colours for different chemical species.
Each arrow is carrying the same flux v_m^{SGLT1} .

SGLT1 model parameters fitted to experimental data

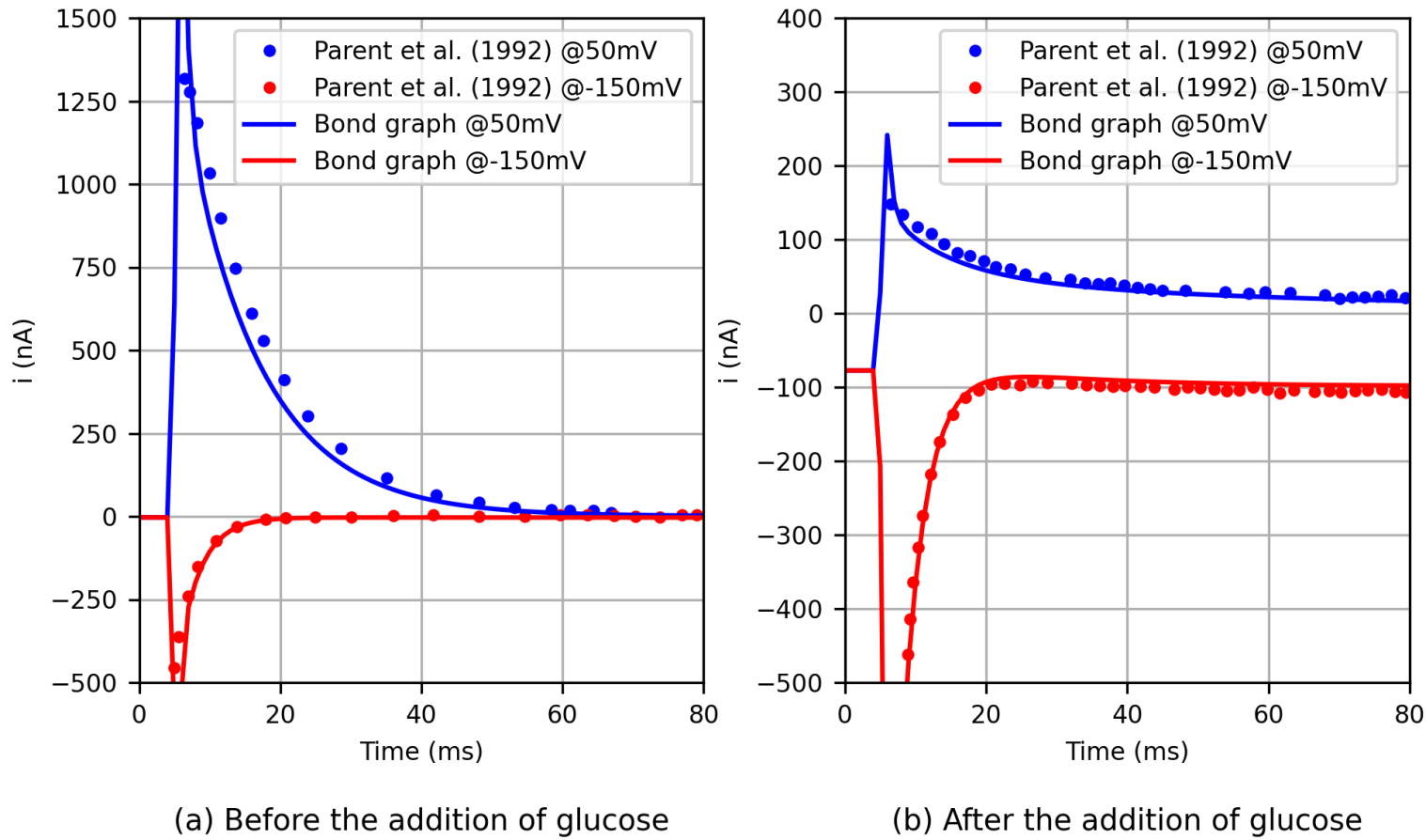


Figure 11 The time course of the carrier-mediated currents. (a) The electrical current when $[Glc]_o = 0$ mM, and (b) the current when $[Glc]_o = 1$ mM; The output of the bond graph model is the current $-I_i$, and the data of Parent et al. [19] are reproduced from Figure 10 of that paper using the digitizing software Engauge. The information needed to reproduce these results can be found in [18].

SLC transporters

In the following slides we show bond graph models for a range of SLC transporters, following the previous two examples. For each one the full kinetic model is presented, with a separate reaction for each ligand binding and unbinding event, and for the outward-facing to inward-facing transition of the protein. We also include all possible 'slippage states' where the transition may occur without completing the full cycle. Note the link to *SLC tables* in the heading for each transporter and the link to a CellML model of the transporter in the *Physiome Model Repository* (PMR).

Note that the bond graph diagrams completely define all the equations needed for the model.

For each full kinetic model we also derive a simplified model for the transporter in which the following assumptions are made:

1. The slippage reactions are ignored.
2. The states of the enzyme are assumed to be at steady-state and the enzyme therefore cycles at a constant rate (the Briggs-Haldane assumption). This is valid when the number of membrane-bound enzymes is small compared with the number of solute (ligand) molecules. The solutes can be time-varying.
3. Ligand binding and unbinding is assumed to be very rapid in comparison to the reaction rates for the membrane outward-inward-facing transitions.

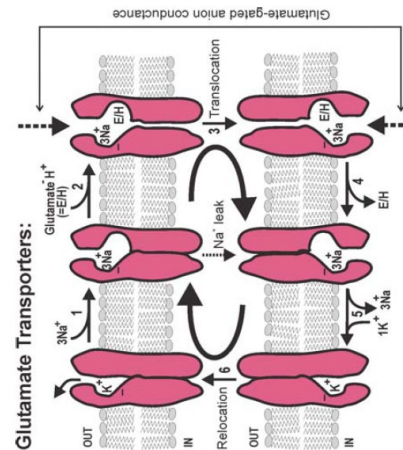
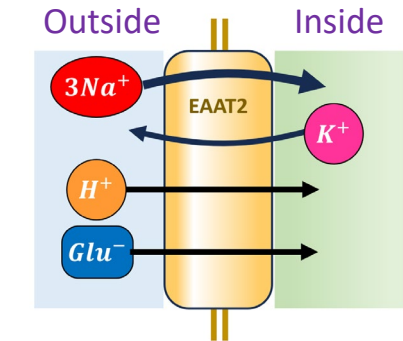
In some cases we consider different levels of complexity for the reduced models, in order to determine the simplest form that is compatible with the experimental data to an acceptable level of precision.

In each case we fit both the full kinetic model and the reduced model(s) to experimental data and compare their behaviours under various conditions.

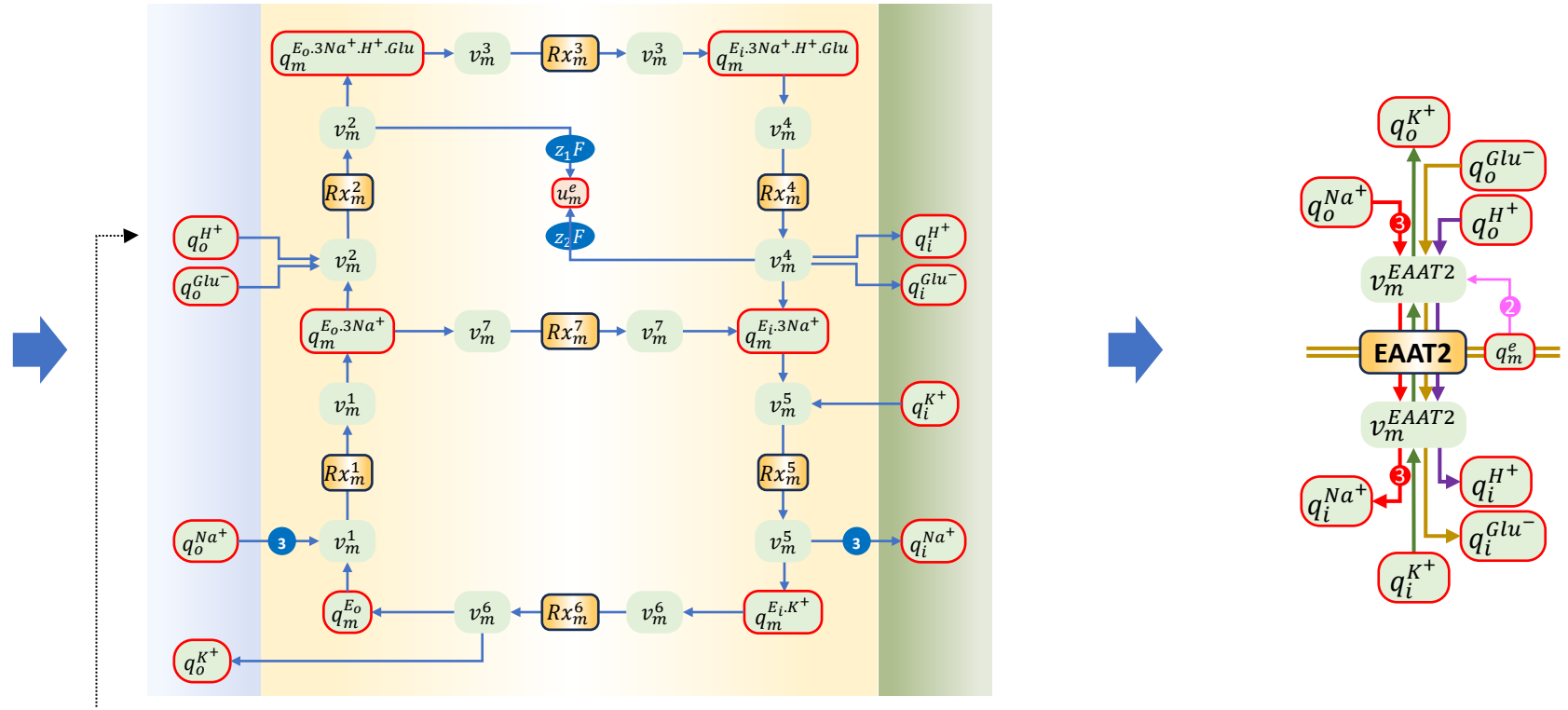
We also provide appropriate diagrams for the reduced models that can be used in the cellular models that combine multiple transporters, pumps and ion channels under dynamic changes of the solutes.

SLC1A2 (EAAT2) Glutamate transport (SLCtables)

(see [model on PMR](#))



Kanai & Hediger (2004)

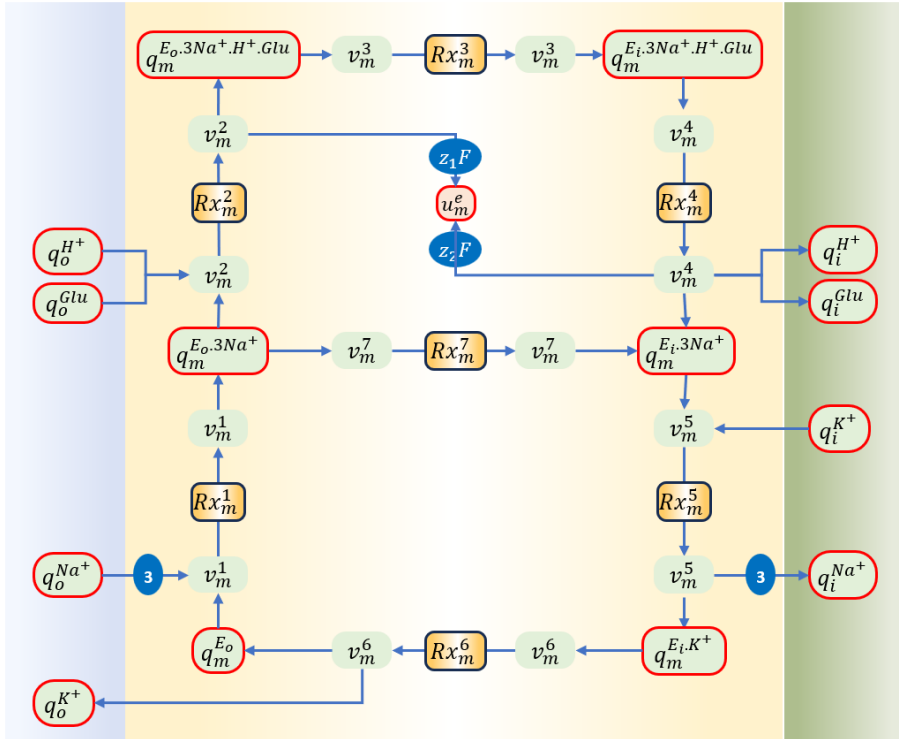


Negatively charged glutamate is often protonated and hence one binding reaction is appropriate.

Note that the order of binding and unbinding needs to be established from experimental data. Generally assume that Na^+ binds first.

Vertical orientation is used for detailed models but horizontal is useful for cell models

SLC1A2 (EAAT2) .. continued



Reaction fluxes

$$v_1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^3 - \bar{q}_m^{E_o.3Na^+} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_m^{E_o.3Na^+} \cdot \bar{q}_o^{H^+} \cdot \bar{q}_o^{Glu} - \bar{q}_m^{E_o.3Na^+.H^+.Glu} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_o.3Na^+.H^+.Glu} - \bar{q}_m^{E_i.3Na^+.H^+.Glu} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i.3Na^+.H^+.Glu} - \bar{q}_m^{E_i.3Na^+} \cdot \bar{q}_i^{H^+} \cdot \bar{q}_i^{Glu} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_5 = \kappa_5 \left[\bar{q}_m^{E_i.3Na^+} \cdot \bar{q}_i^{K^+} - \bar{q}_m^{E_i.K^+} \cdot (\bar{q}_i^{Na^+})^3 \right]$$

$$v_6 = \kappa_6 \left[\bar{q}_m^{E_i.K^+} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{K^+} \right]$$

$$v_7 = \kappa_7 \left[\bar{q}_m^{E_o.3Na^+} - \bar{q}_m^{E_i.3Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o.3Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.3Na^+.H^+.Glu} = v_m^2 - v_m^3$$

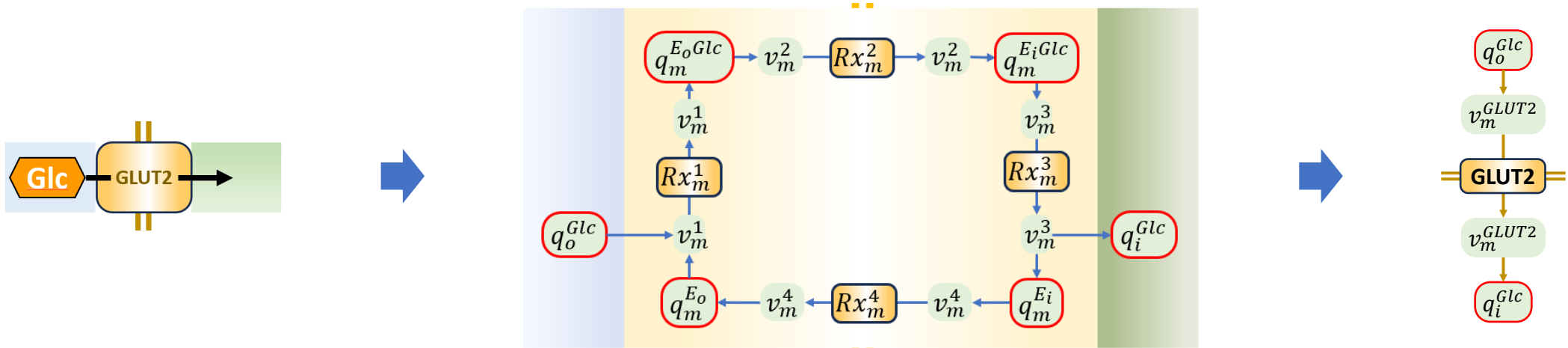
$$\frac{d}{dt} q_m^{E_i.3Na^+.H^+.Glu} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.3Na^+} = v_m^4 + v_m^7 - v_m^5$$

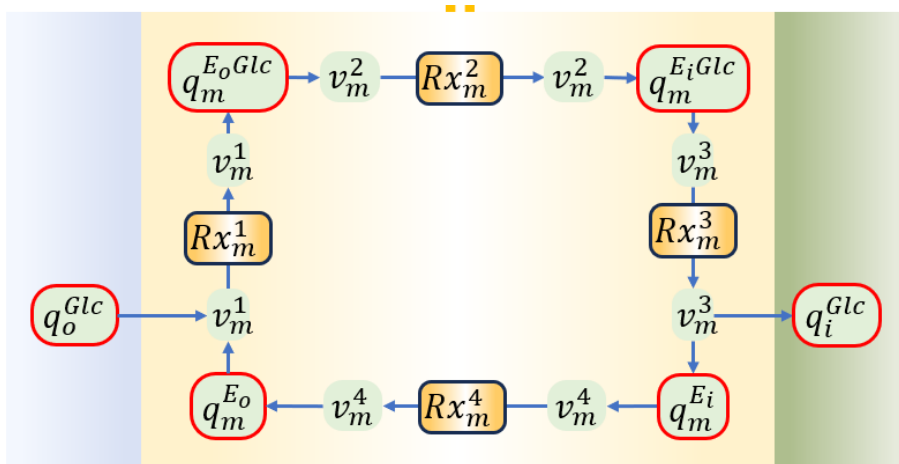
$$\frac{d}{dt} q_m^{E_i.K^+} = v_m^5 - v_m^6$$

SLC2A2 (GLUT2) Glucose transport (SLCtables)

(see [model on PMR](#))



SLC2A2 (GLUT2) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o \cdot Glc} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o \cdot Glc} - \bar{q}_m^{E_i \cdot Glc} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_i \cdot Glc} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Glc} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^4 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o \cdot Glc} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i \cdot Glc} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_o^{Glc} = -v_m^1$$

$$\frac{d}{dt} q_i^{Glc} = v_m^3$$

SLC2A2 (GLUT2) .. Reduced model

$$v_m^{GLUT2} = v_m (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right);$$

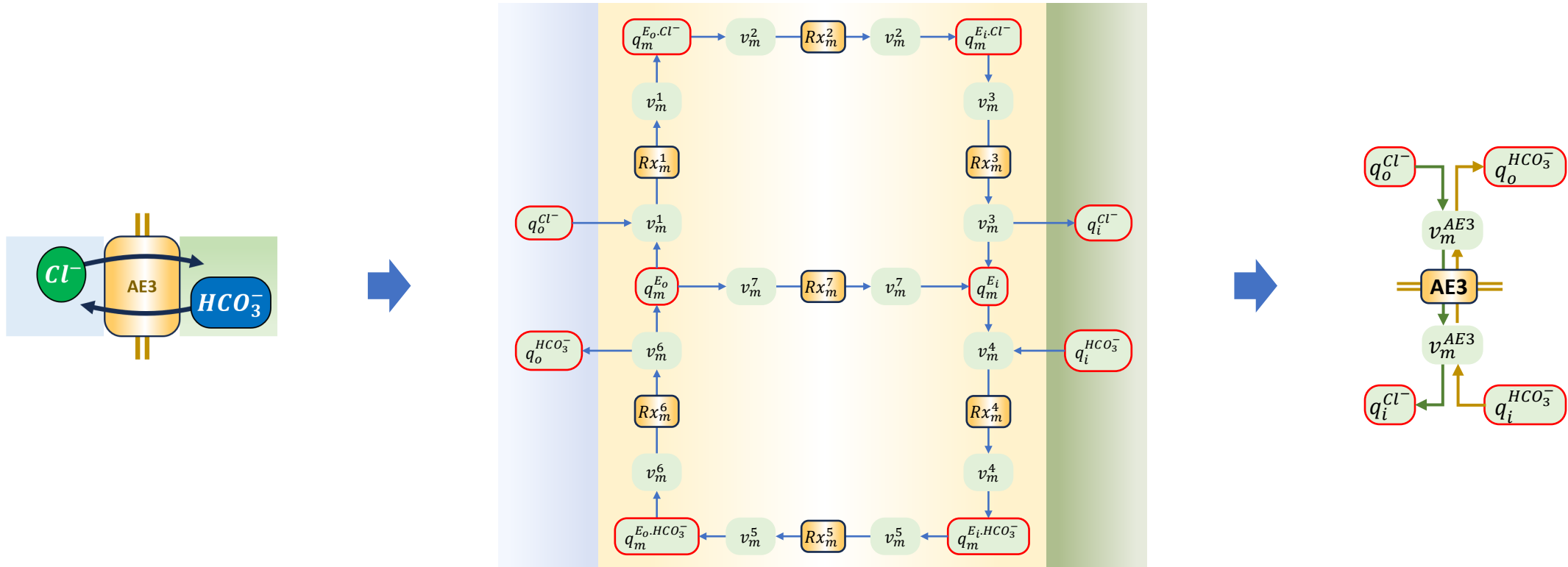
4 parameters

$$\text{with } k_m^1 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EoGlc})^{-1} + \Lambda \cdot (K_m^{Ei})^{-1}}, \quad k_m^2 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{(K_m^{EiGlc})^{-1} + \Lambda \cdot (K_m^{Eo})^{-1}}, \quad k_m^3 = \frac{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}{\Lambda \cdot [(K_m^{EoGlc})^{-1} + (K_m^{EiGlc})^{-1}]}, \quad v_m = \frac{q_{tot} \kappa_2}{(K_m^{Eo})^{-1} + (K_m^{Ei})^{-1}}, \quad \Lambda = \frac{\kappa_2}{\kappa_4}$$

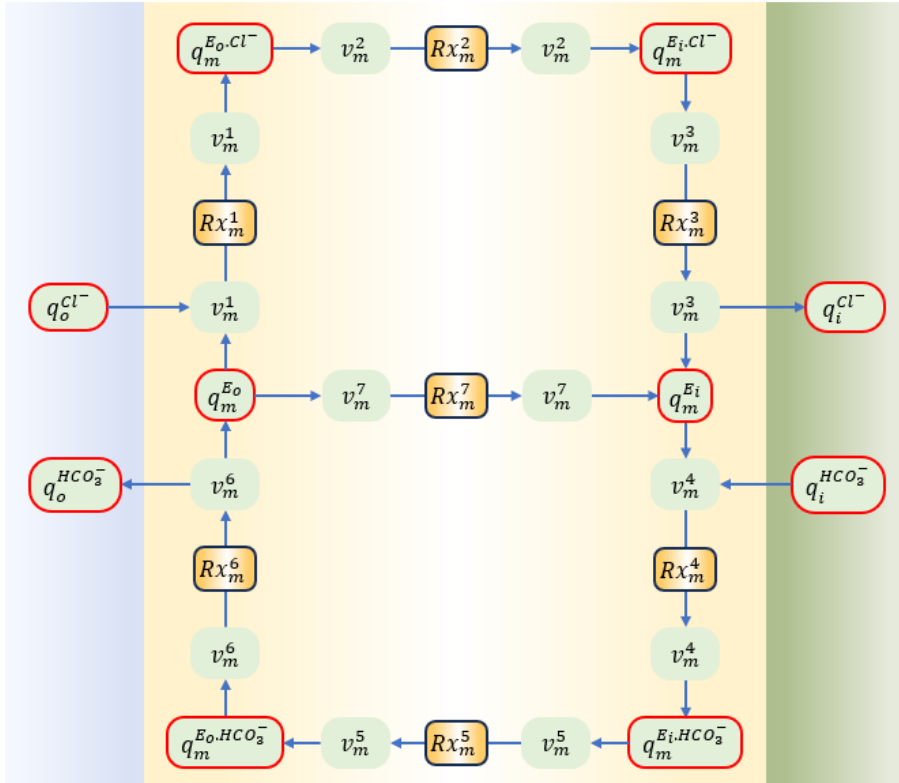
SLC4A3 (AE3) Chloride-bicarbonate exchange (SLCtables)

(see [model on PMR](#))

AE = 'anion exchanger'



SLC4A3 (AE3) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{\text{Cl}^-} - \bar{q}_m^{E_o, \text{Cl}^-} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o, \text{Cl}^-} - \bar{q}_m^{E_i, \text{Cl}^-} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_i, \text{Cl}^-} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{\text{Cl}^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i} \cdot \bar{q}_o^{\text{HCO}_3^-} - \bar{q}_i^{E_i, \text{HCO}_3^-} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_i^{E_i, \text{HCO}_3^-} - \bar{q}_i^{E_o, \text{HCO}_3^-} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_i^{E_o, \text{HCO}_3^-} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{\text{HCO}_3^-} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o, \text{Cl}^-} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i, \text{Cl}^-} = v_m^2 - v_m^3$$

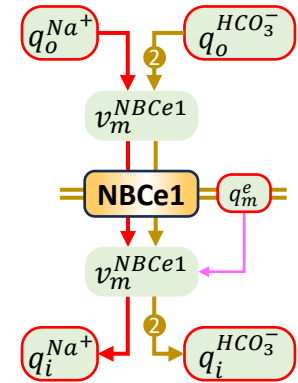
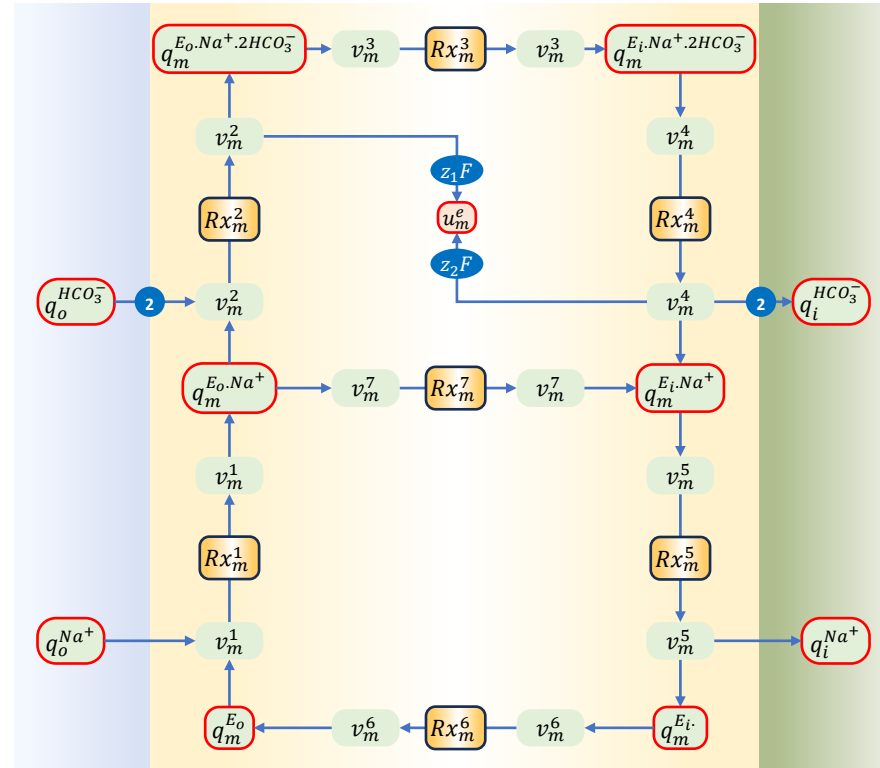
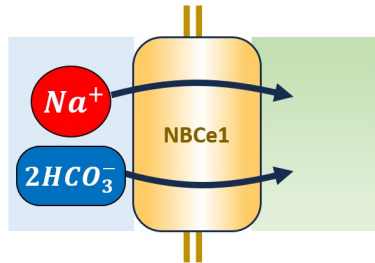
$$\frac{d}{dt} q_m^{E_i} = v_m^3 + v_m^7 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i, \text{HCO}_3^-} = v_m^4 - v_m^5$$

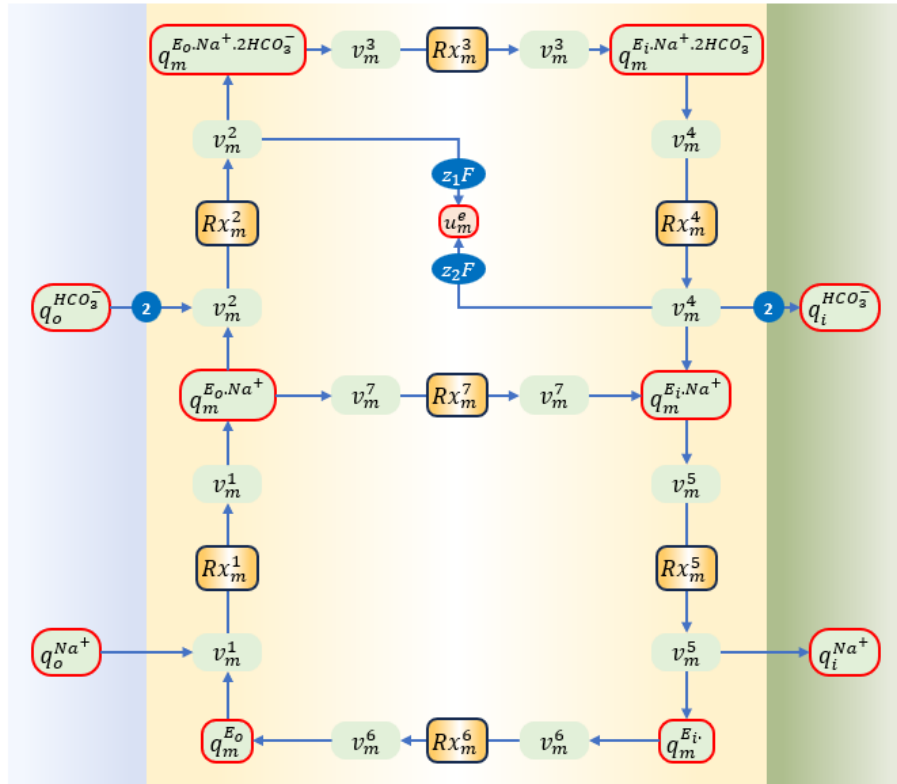
$$\frac{d}{dt} q_m^{E_o, \text{HCO}_3^-} = v_m^5 - v_m^6$$

SLC4A4 (NBCe1) Sodium-bicarbonate cotransport (SLCtables)

(see [model on PMR](#))



SLC4A4 (NBCe1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+}]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot \left(\bar{q}_o^{HCO_3^-} \right)^2 - \bar{q}_m^{E_o.Na^+.2HCO_3^-} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+.2HCO_3^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+} \cdot \left(\bar{q}_i^{HCO_3^-} \right)^2 \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} - \bar{q}_m^{E_o} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+} \right]$$

Mass balance:

$$\frac{d}{dt}q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt}q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt}q_m^{E_o.Na^+.2HCO_3^-} = v_m^2 - v_m^3$$

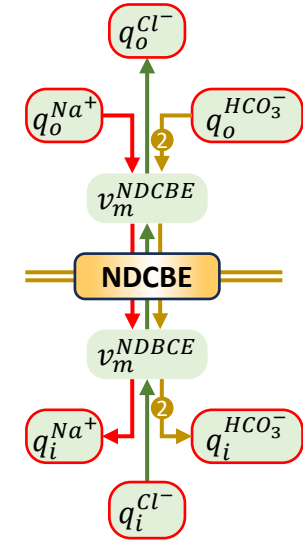
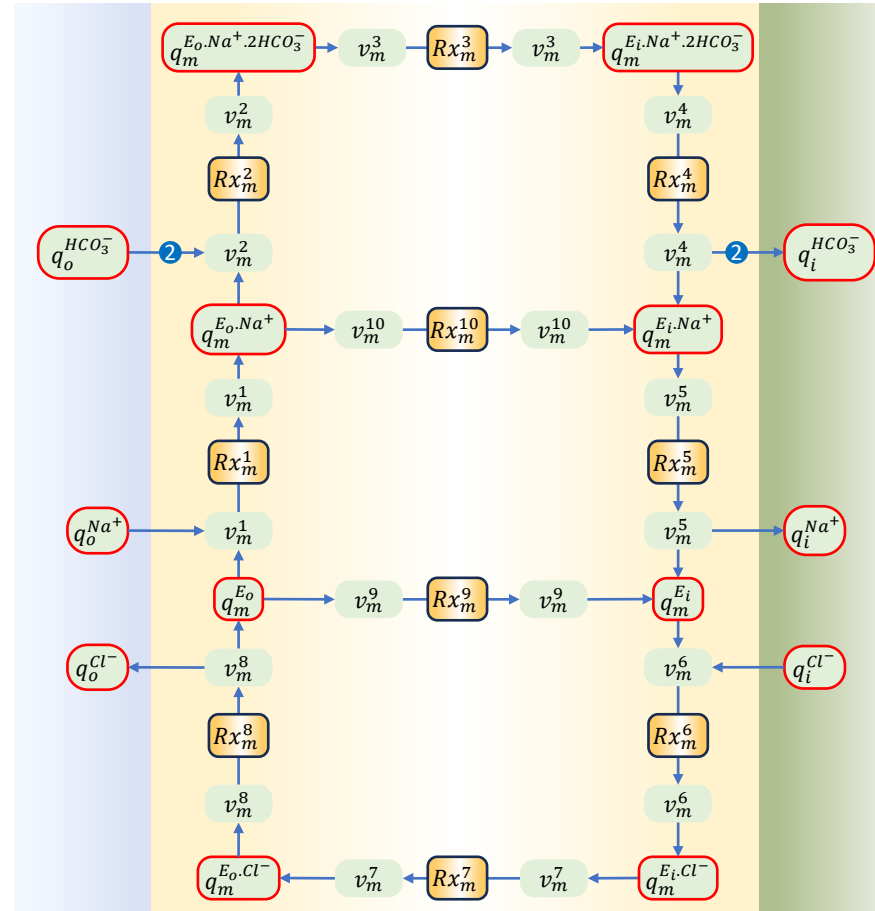
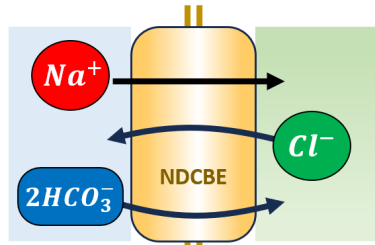
$$\frac{d}{dt}q_m^{Ei.Na^+.2HCO_3^-} = v_m^3 - v_m^4$$

$$\frac{d}{dt}q_m^{Ei.Na^+} = v_m^4 + v_m^7 - v_m^5$$

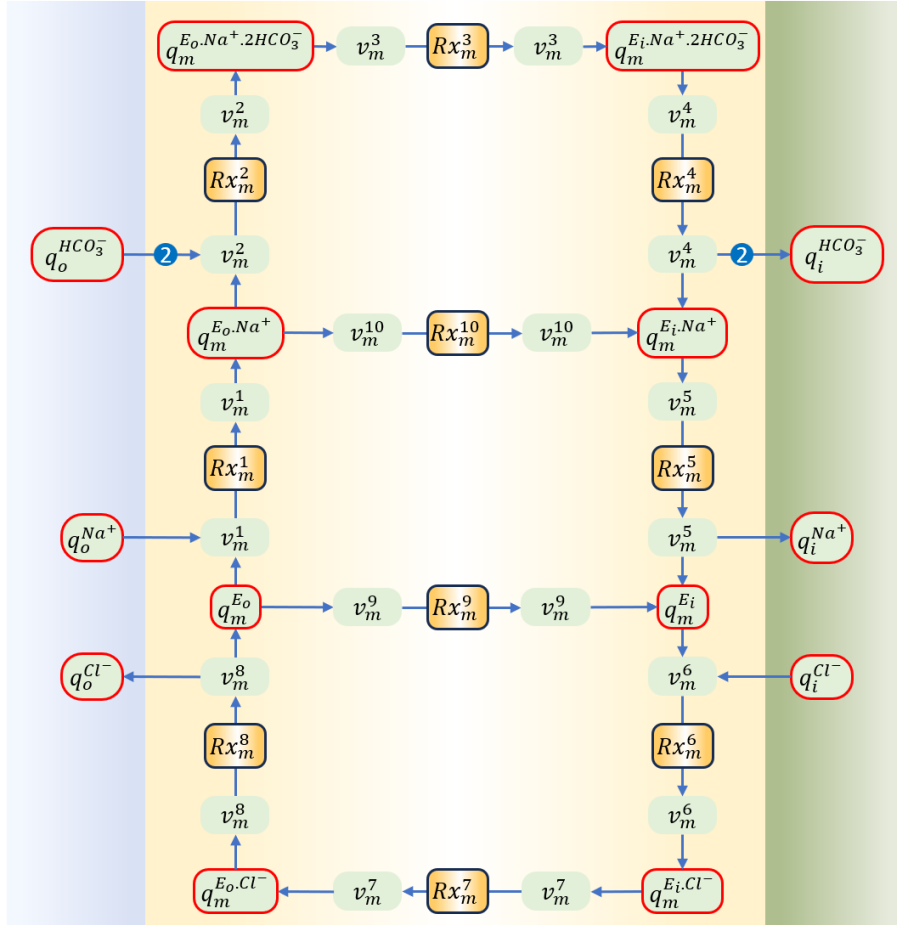
$$\frac{d}{dt}q_m^{E_i} = v_m^5 - v_m^6$$

SLC4A8 (NDCBE) Sodium-driven chloride-bicarbonate exchange ([SLCtables](#))

(see [model on PMR](#))



SLC4A8 (NDCBE) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot (q_o^{HCO_3^-})^2 - \bar{q}_m^{E_o.Na^+.2HCO_3^-} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+.2HCO_3^-} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.Na^+.2HCO_3^-} - \bar{q}_m^{E_i.Na^+} \cdot (q_i^{HCO_3^-})^2 \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+} \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot \bar{q}_i^{Cl^-} - \bar{q}_m^{E_i.Cl^-} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_i.Cl^-} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{Cl^-} \right]$$

$$v_m^8 = \kappa_8 \left[\bar{q}_m^{E_o} \cdot \bar{q}_o^{Cl^-} - \bar{q}_m^{E_o.Cl^-} \right]$$

$$v_m^9 = \kappa_9 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]$$

$$v_m^{10} = \kappa_{10} \left[\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^8 - v_m^1 - v_m^9$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^{10}$$

$$\frac{d}{dt} \bar{q}_m^{E_o.Na^+.2HCO_3^-} = v_m^2 - v_m^3$$

$$\frac{d}{dt} \bar{q}_m^{E_i.Na^+.2HCO_3^-} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^4 + v_m^{10} - v_m^5$$

$$\frac{d}{dt} q_m^{E_i} = v_m^5 + v_m^9 - v_m^6$$

$$\frac{d}{dt} q_m^{E_i.Cl^-} = v_m^6 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.Cl^-} = v_m^7 - v_m^8$$

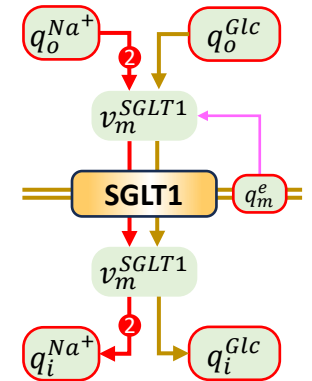
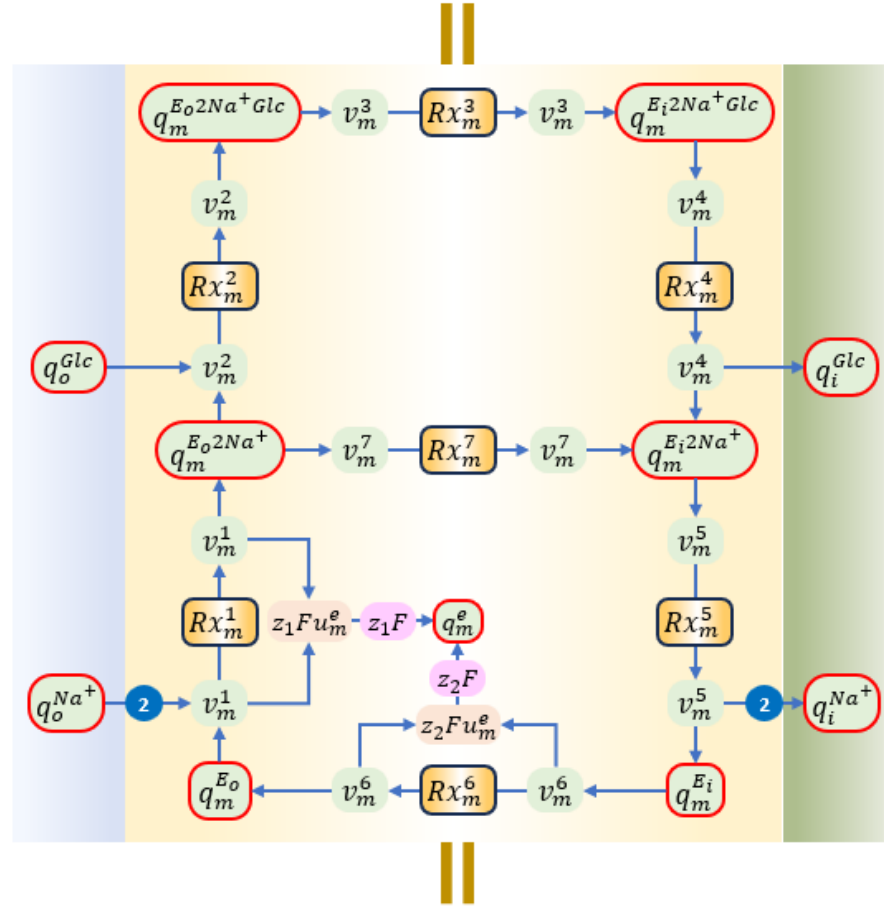
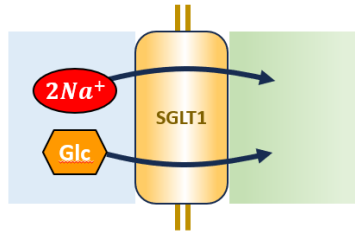
$$\frac{d}{dt} q_o^{Cl^-} = v_m^8, \quad \frac{d}{dt} q_i^{Cl^-} = -v_m^6$$

$$\frac{d}{dt} q_o^{Na^+} = -v_m^1, \quad \frac{d}{dt} q_i^{Na^+} = v_m^5$$

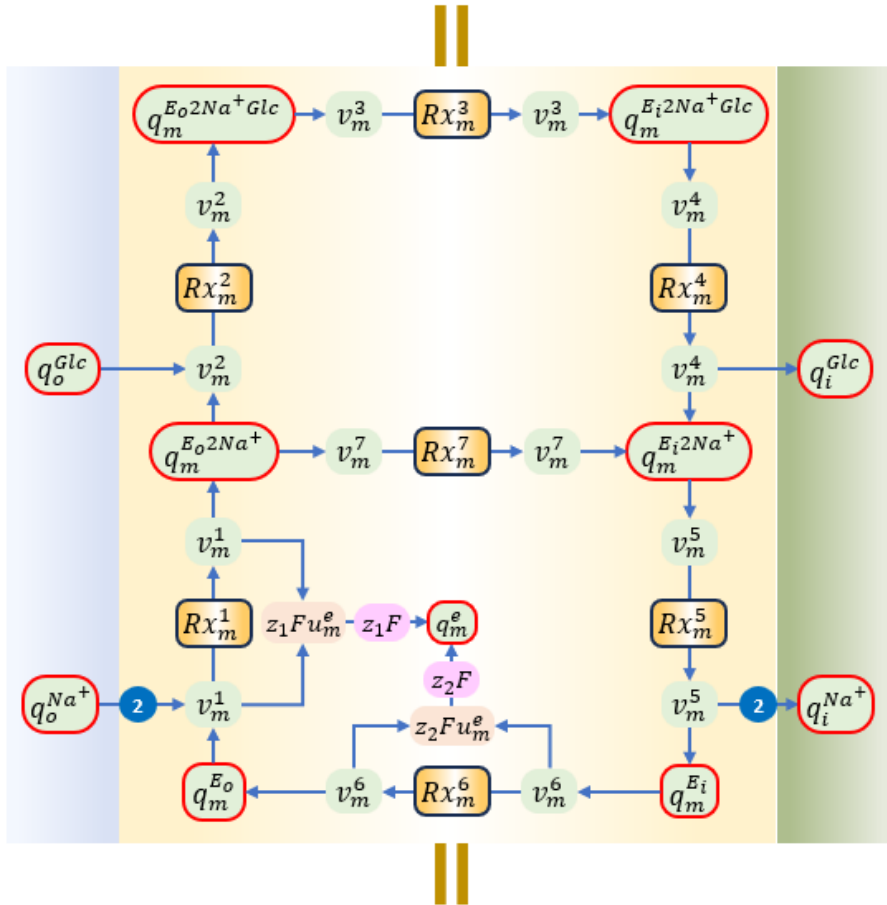
$$\frac{d}{dt} q_o^{HCO_3^-} = -2v_m^2, \quad \frac{d}{dt} q_i^{HCO_3^-} = 2v_m^4$$

SLC5A1 (SGLT1) Sodium-driven glucose transport (SLCtables)

(see [model on PMR](#))



SLC5A1 (SGLT1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^2 \cdot e^{\frac{-z_1 F u_m^e}{RT}} - \bar{q}_m^{E_o.2Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.Na^+} \cdot \bar{q}_o^{Glc} - \bar{q}_m^{E_o.2Na^+.Glc} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.2Na^+.Glc} - \bar{q}_m^{E_i.2Na^+.Glc} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_i.2Na^+.Glc} - \bar{q}_m^{E_i.2Na^+} \cdot \bar{q}_i^{Glc} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_i.2Na^+} - \bar{q}_m^{E_i} \cdot (\bar{q}_i^{Na^+})^2 \right]$$

$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i} \cdot e^{\frac{-z_2 F u_m^e}{RT}} - \bar{q}_m^{E_o} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_o.2Na^+} - \bar{q}_m^{E_i.2Na^+} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.2Na^+.Glc} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i.2Na^+.Glc} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^4 + v_m^7 - v_m^5$$

$$\frac{d}{dt} q_m^{E_i} = v_m^5 - v_m^6$$

$$\frac{d}{dt} q_o^{Na^+} = -2v_m^1$$

$$\frac{d}{dt} q_i^{Na^+} = 2v_m^5$$

$$\frac{d}{dt} q_o^{Glc} = -v_m^2$$

$$\frac{d}{dt} q_i^{Glc} = v_m^4$$

SLC5A1 (SGLT1) .. Reduced model

$$v_m^{SGLT1} = \frac{A_0}{A_1 A_2 + A_3 A_4} \cdot \left\{ \frac{\bar{q}_o^{Glc}}{\bar{q}_i^{Glc}} - \left(\frac{\bar{q}_i^{Na^+}}{\bar{q}_o^{Na^+}} \right)^2 \cdot e^{2Fu_m^e/RT} \right\}$$

$$A_0 = q_{tot} \kappa_3 K_1 \quad A_1 = \lambda_2 + \lambda_3 \cdot \bar{q}_o^{Glc} + \frac{e^{2z_1 Fu_m^e/RT}}{(\bar{q}_o^{Na^+})^2} \quad A_2 = \frac{1}{\bar{q}_i^{Glc}} + \lambda_1 \cdot (\bar{q}_i^{Na^+})^2 e^{z_2 Fu_m^e/RT}$$

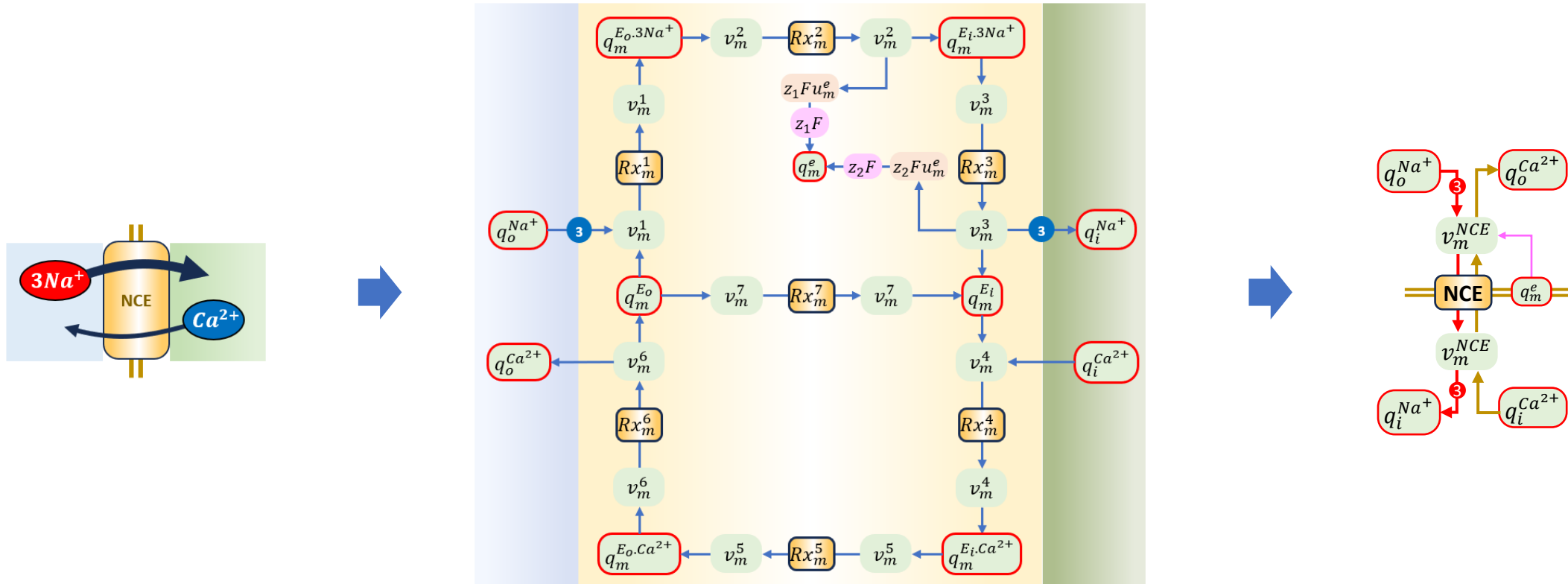
$$A_3 = \left(\lambda_4 + \frac{\lambda_5}{\bar{q}_i^{Glc}} \right) \cdot (\bar{q}_i^{Na^+})^2 + \frac{\lambda_6}{\bar{q}_i^{Glc}} \quad A_4 = \lambda_1 \bar{q}_o^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \frac{e^{2Fu_m^e/RT}}{(\bar{q}_o^{Na^+})^2}$$

$$\lambda_1 = \frac{\kappa_3}{\kappa_6}, \lambda_2 = \frac{K_1}{K_2}, \lambda_3 = \frac{K_1}{K_3}, \lambda_4 = \frac{K_1}{K_4}, \lambda_5 = \frac{K_1}{K_5}, \lambda_6 = \frac{K_1}{K_6},$$

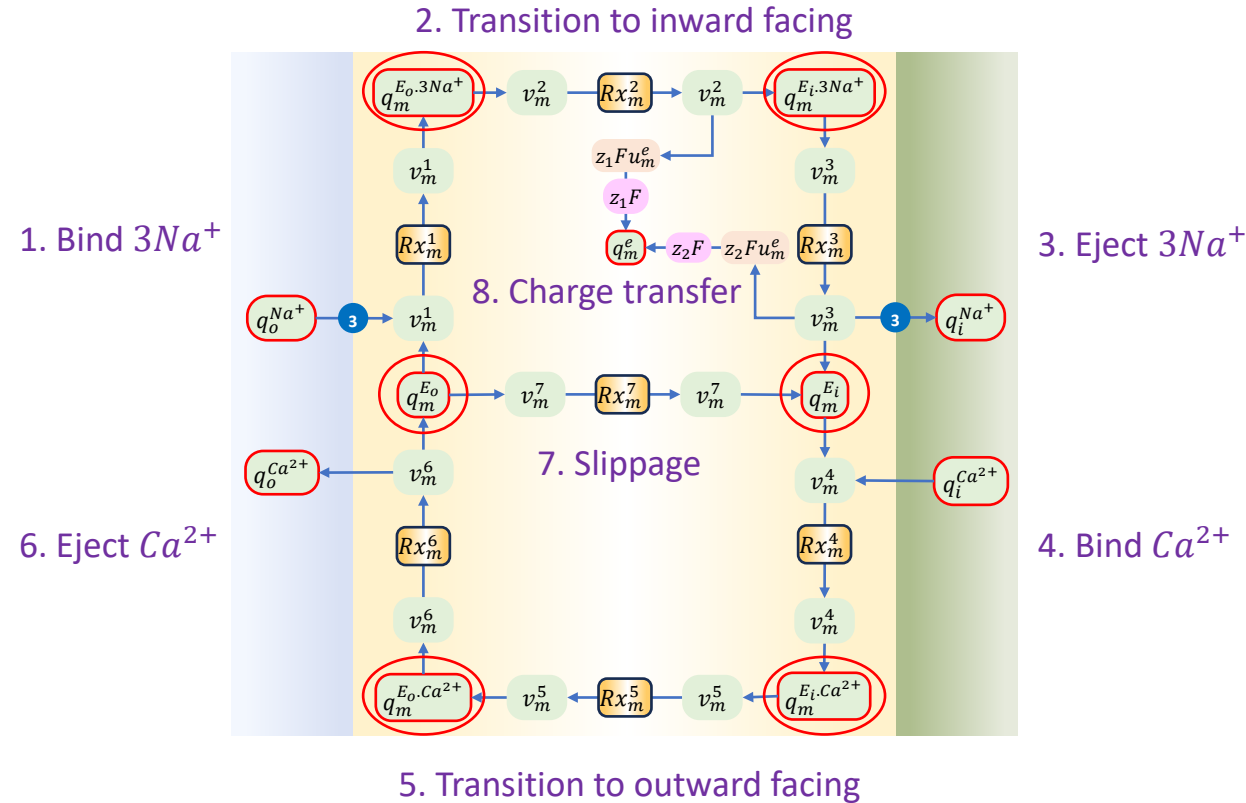
$$K_1 = K_m^{E_o}, K_2 = K_m^{E_o 2Na^+}, K_3 = K_m^{E_o 2Na^+ Glc}, K_4 = K_m^{E_i 2Na^+ Glc}, K_5 = K_m^{E_i 2Na^+}, K_6 = K_m^{E_i}$$

SLC8A1 (NCE/NCX) Sodium-calcium exchange (SLCtables)

(see [model on PMR](#))



SLC8A1 (NCE) .. continued



Reaction fluxes

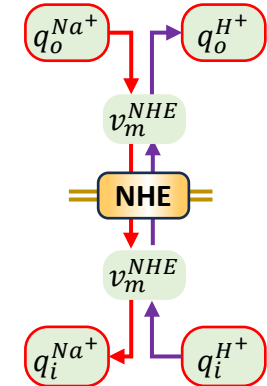
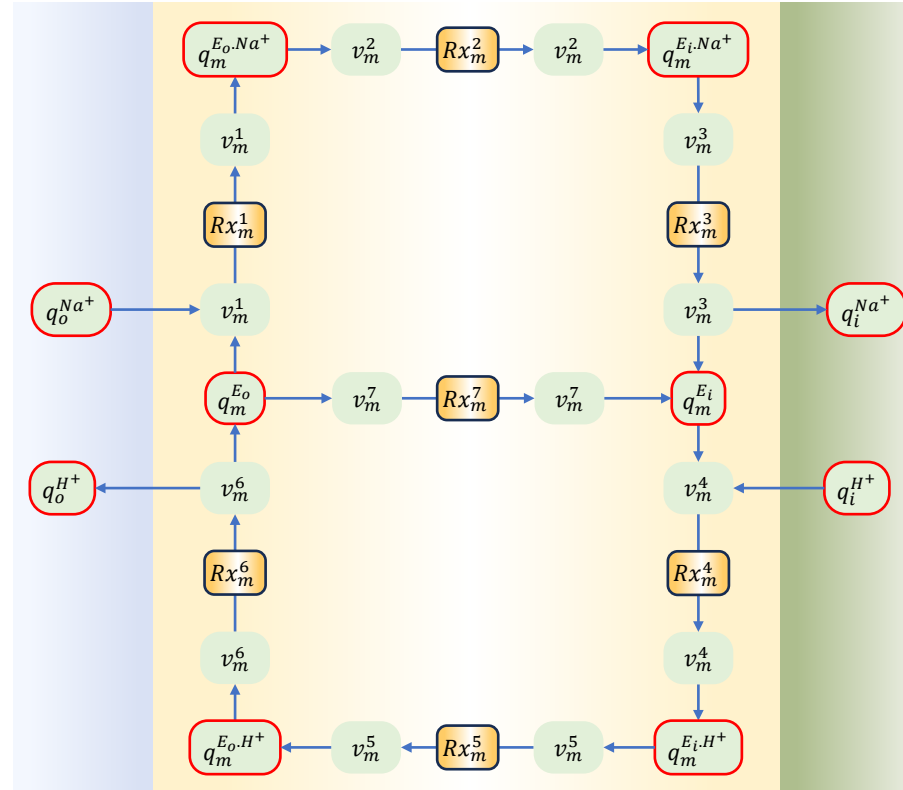
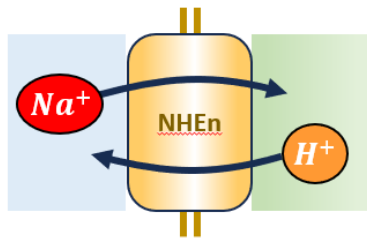
$$\begin{aligned}
 v_m^1 &= \kappa_1 \left[\bar{q}_m^{E_o} \cdot (\bar{q}_o^{Na^+})^3 - \bar{q}_m^{E_o \cdot 3Na^+} \right] \\
 v_m^2 &= \kappa_2 \left[\bar{q}_m^{E_o \cdot 3Na^+} - \bar{q}_m^{E_i \cdot 3Na^+} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right] \\
 v_m^3 &= \kappa_3 \left[\bar{q}_m^{E_i \cdot 3Na^+} - \bar{q}_m^{E_i} \cdot (\bar{q}_i^{Na^+})^3 \cdot e^{\frac{z_2 F u_m^e}{RT}} \right] \\
 v_m^4 &= \kappa_4 \left[\bar{q}_m^{E_i} \cdot \bar{q}_o^{Ca^{2+}} - \bar{q}_i^{E_i \cdot Ca^{2+}} \right] \\
 v_m^5 &= \kappa_5 \left[\bar{q}_i^{E_i \cdot Ca^{2+}} - \bar{q}_i^{E_o \cdot Ca^{2+}} \right] \\
 v_m^6 &= \kappa_6 \left[\bar{q}_i^{E_o \cdot Ca^{2+}} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{Ca^{2+}} \right] \\
 v_m^7 &= \kappa_7 \left[\bar{q}_m^{E_o} - \bar{q}_m^{E_i} \right]
 \end{aligned}$$

Mass balance:

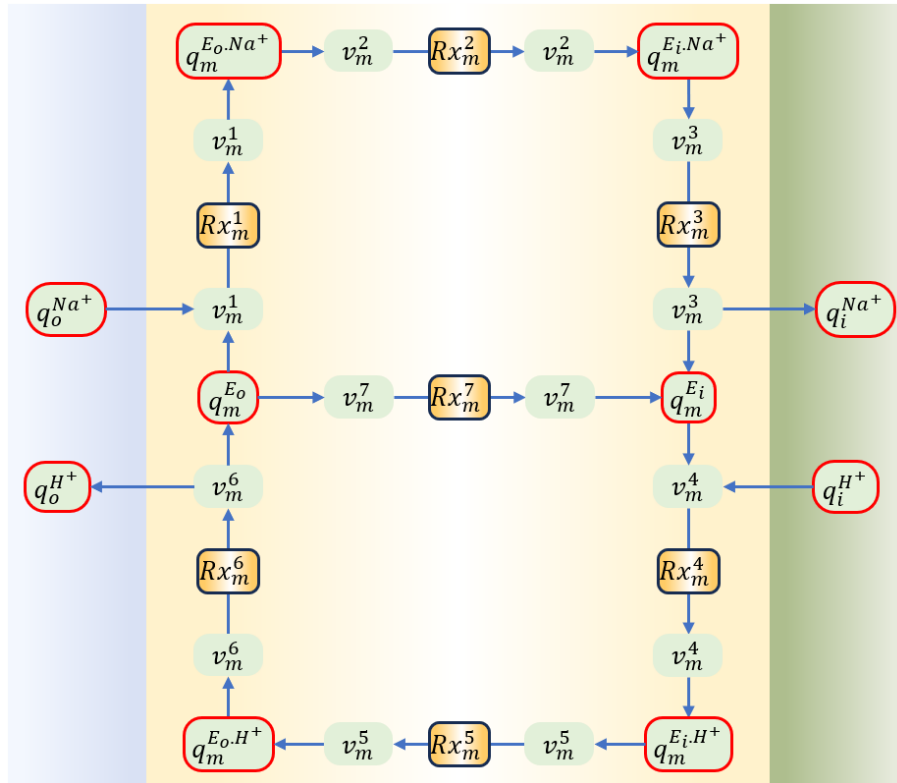
$$\begin{aligned}
 \frac{d}{dt} q_m^{E_o} &= v_m^6 - v_m^1 - v_m^7 \\
 \frac{d}{dt} q_m^{E_o \cdot 3Na^+} &= v_m^1 - v_m^2 \\
 \frac{d}{dt} q_m^{E_i \cdot 3Na^+} &= v_m^2 - v_m^3 \\
 \frac{d}{dt} q_m^{E_i} &= v_m^3 + v_m^7 - v_m^4 \\
 \frac{d}{dt} q_m^{E_i \cdot Ca^{2+}} &= v_m^4 - v_m^5 \\
 \frac{d}{dt} q_m^{E_o \cdot Ca^{2+}} &= v_m^5 - v_m^6
 \end{aligned}$$

SLC9A1 (NHE1) Sodium-hydrogen exchange (SLCtables)

(see [model on PMR](#))



SLC9A1 (NHE1) .. continued



Reaction fluxes

$$v_m^1 = \kappa_1 [\bar{q}_m^{E_o} \cdot \bar{q}_o^{Na^+} - \bar{q}_m^{E_o.Na^+}]$$

$$v_m^2 = \kappa_2 [\bar{q}_m^{E_o.Na^+} - \bar{q}_m^{E_i.Na^+}]$$

$$v_m^3 = \kappa_3 [\bar{q}_m^{E_i.Na^+} - \bar{q}_m^{E_i} \cdot \bar{q}_i^{Na^+}]$$

$$v_m^4 = \kappa_4 [\bar{q}_m^{E_i} \cdot \bar{q}_o^{H^+} - \bar{q}_i^{E_i.H^+}]$$

$$v_m^5 = \kappa_5 [\bar{q}_i^{E_i.H^+} - \bar{q}_i^{E_o.H^+}]$$

$$v_m^6 = \kappa_6 [\bar{q}_i^{E_o.H^+} - \bar{q}_m^{E_o} \cdot \bar{q}_o^{H^+}]$$

$$v_m^7 = \kappa_7 [\bar{q}_m^{E_o} - \bar{q}_m^{E_i}]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_o} = v_m^6 - v_m^1 - v_m^7$$

$$\frac{d}{dt} q_m^{E_o.Na^+} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i.Na^+} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_i} = v_m^3 + v_m^7 - v_m^4$$

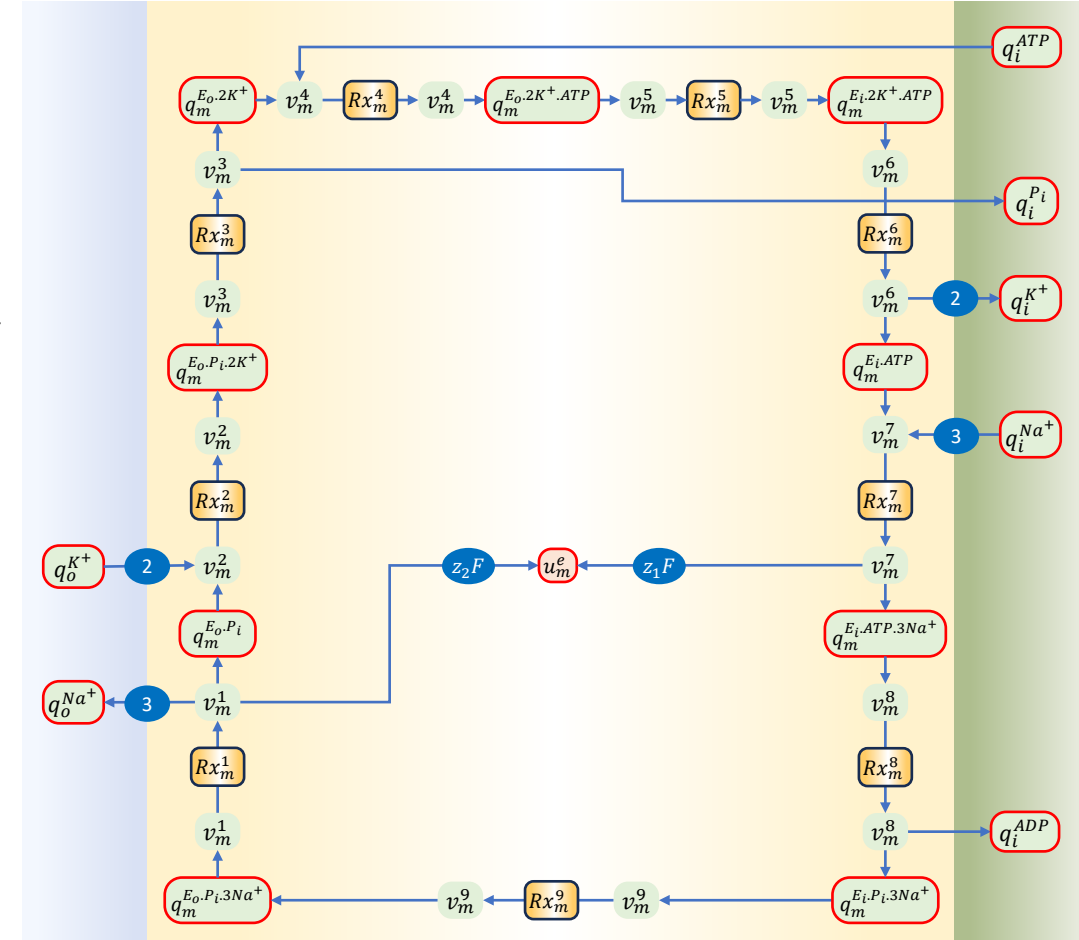
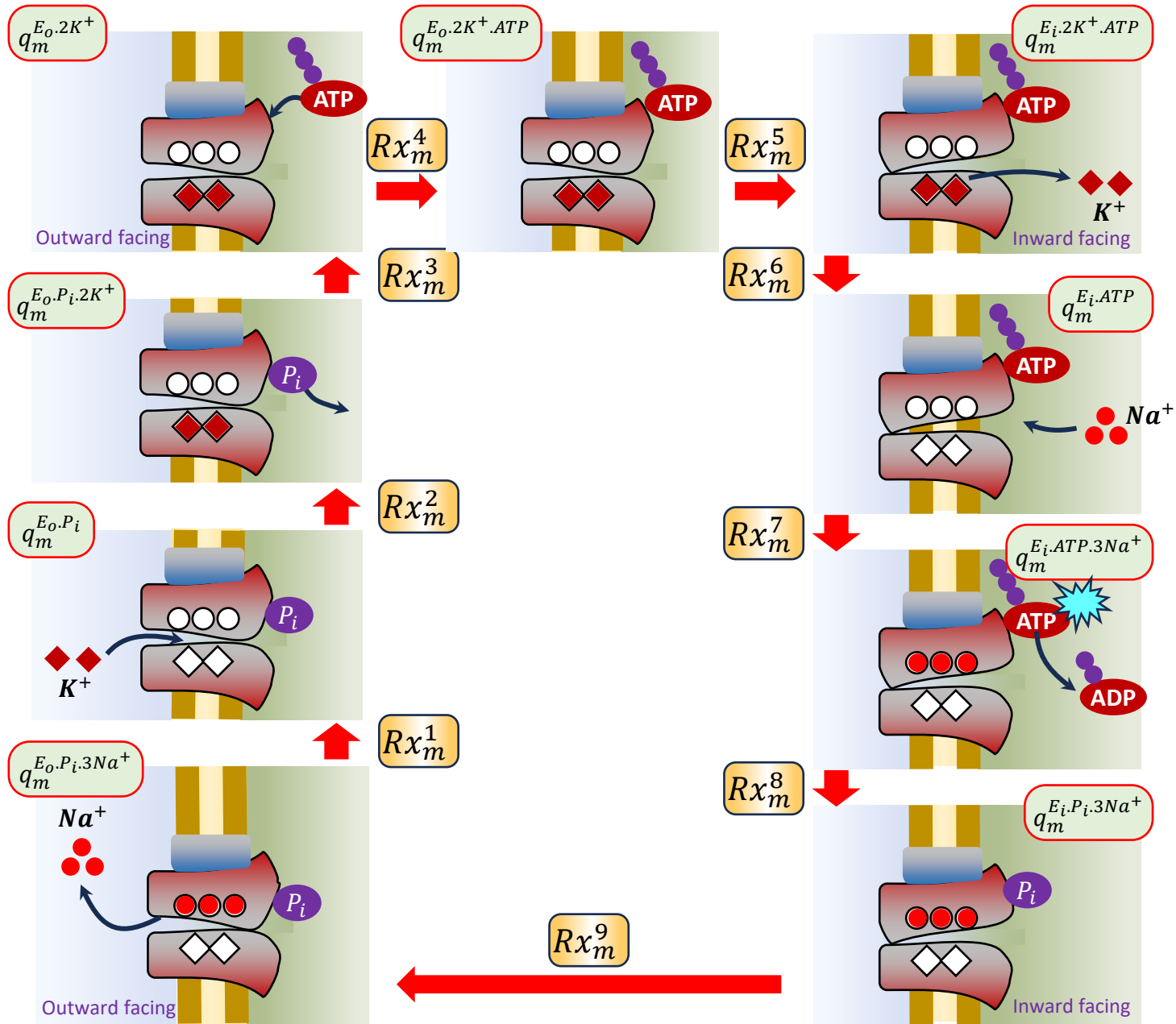
$$\frac{d}{dt} q_m^{E_i.H^+} = v_m^4 - v_m^5$$

$$\frac{d}{dt} q_m^{E_o.H^+} = v_m^5 - v_m^6$$

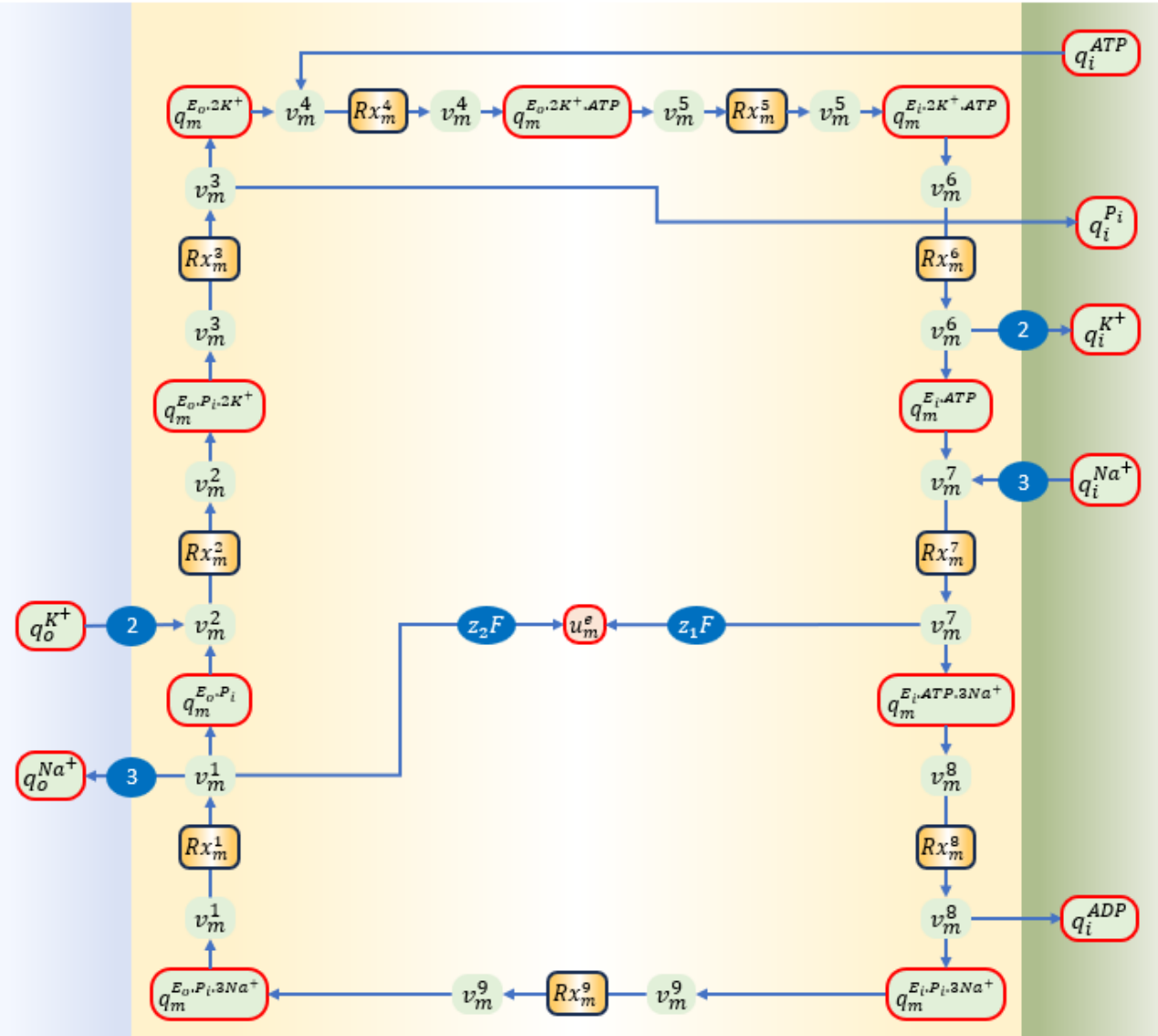
Lecture 5: *ATPase pumps (NKE, SERCA, Proton-pump)*

We start with the $\text{Na}^+\text{-K}^+$ exchanger (NKE), which is both dependent on ATP and is electrogenic - since 3 Na^+ ions are pumped out of the cell in exchange for 2 K^+ ions transferred into the cell – for every 1 ATP hydrolyzed. By maintaining the sodium gradient across cell membranes the NKE pump maintains the energy gradient used by many SLC transporters, and is responsible for nearly 10% of the body's resting metabolism. In this lecture we first explain the action of NKE in terms of a 9-state model and then derive an algebraic expression for the dependence of the steady state flux through the NKE pump on the quantities of substrate in the cytosol and the extracellular fluid. We also describe a 4-state model.

Sodium-potassium exchange pump (ATPase) – 9-state model



BG equations for 9-state NKE model



Mass balance:

$$\frac{d}{dt} q_m^{E_o.P_i.3Na^+} = v_m^9 - v_m^1$$

$$\frac{d}{dt} q_m^{E_o.P_i} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_o.P_i.2K^+} = v_m^2 - v_m^3$$

$$\frac{d}{dt} q_m^{E_o.2K^+} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_o.2K^+.ATP} = v_m^4 - v_m^5$$

$$\frac{d}{dt} q_m^{E_i.2K^+.ATP} = v_m^5 - v_m^6$$

$$\frac{d}{dt} q_m^{E_i.ATP} = v_m^6 - v_m^7$$

$$\frac{d}{dt} q_m^{E_i.ATP.3Na^+} = v_m^7 - v_m^8$$

$$\frac{d}{dt} q_m^{E_i.P_i.3Na^+} = v_m^8 - v_m^9$$

Reaction fluxes:

$$v_m^1 = \kappa_1 \left[\bar{q}_m^{E_o.P_i.3Na^+} - \bar{q}_m^{E_o.P_i} \cdot (\bar{q}_o^{Na^+})^3 \cdot e^{z_2 F u_m^e / RT} \right]$$

$$v_m^2 = \kappa_2 \left[\bar{q}_m^{E_o.P_i} \cdot (\bar{q}_o^{K^+})^2 - \bar{q}_m^{E_o.P_i.2K^+} \right]$$

$$v_m^3 = \kappa_3 \left[\bar{q}_m^{E_o.P_i.2K^+} - \bar{q}_m^{E_o.2K^+} \cdot \bar{q}_i^{P_i} \right]$$

$$v_m^4 = \kappa_4 \left[\bar{q}_m^{E_o.2K^+} \cdot \bar{q}_i^{ATP} - \bar{q}_m^{E_o.2K^+.ATP} \right]$$

$$v_m^5 = \kappa_5 \left[\bar{q}_m^{E_o.2K^+.ATP} - \bar{q}_m^{E_i.2K^+.ATP} \right]$$

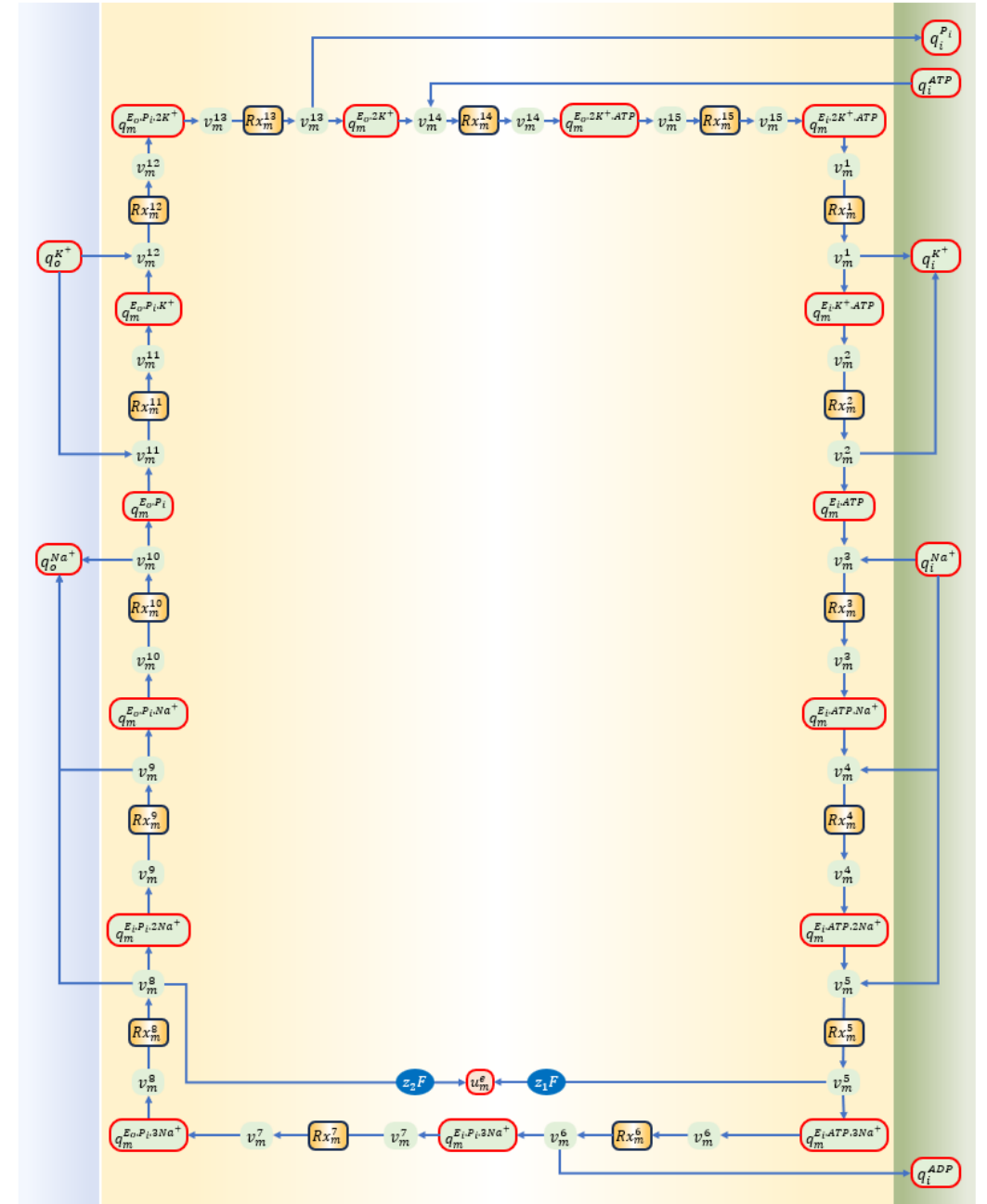
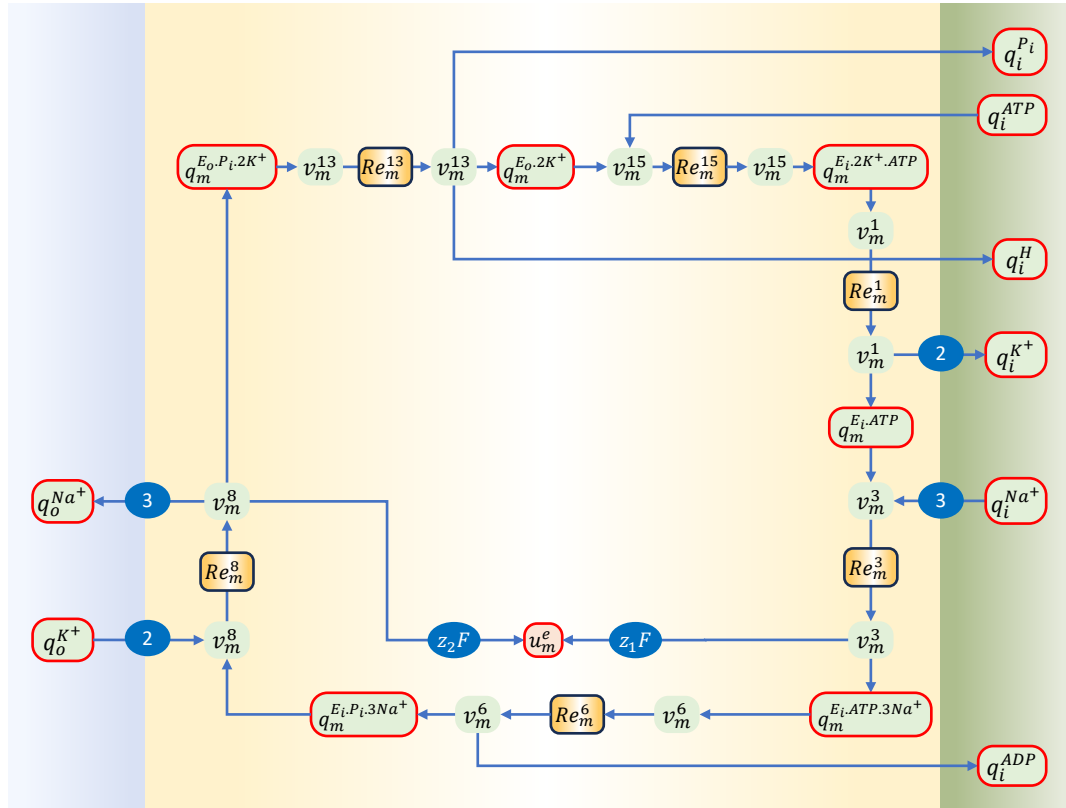
$$v_m^6 = \kappa_6 \left[\bar{q}_m^{E_i.2K^+.ATP} - \bar{q}_m^{E_i.ATP} \cdot (\bar{q}_i^{K^+})^2 \right]$$

$$v_m^7 = \kappa_7 \left[\bar{q}_m^{E_i.ATP} \cdot (\bar{q}_i^{Na^+})^3 - \bar{q}_m^{E_i.ATP.3Na^+} \cdot e^{z_1 F u_m^e / RT} \right]$$

$$v_m^8 = \kappa_8 \left[\bar{q}_m^{E_i.ATP.3Na^+} - \bar{q}_m^{E_i.P_i.3Na^+} \cdot \bar{q}_i^{ADP} \right]$$

$$v_m^9 = \kappa_9 \left[\bar{q}_m^{E_i.P_i.3Na^+} - \bar{q}_m^{E_o.P_i.3Na^+} \right]$$

Sodium-potassium exchange pump (ATPase) – 6 state model & 15-state model



The 15-state model fitted to kinetic data

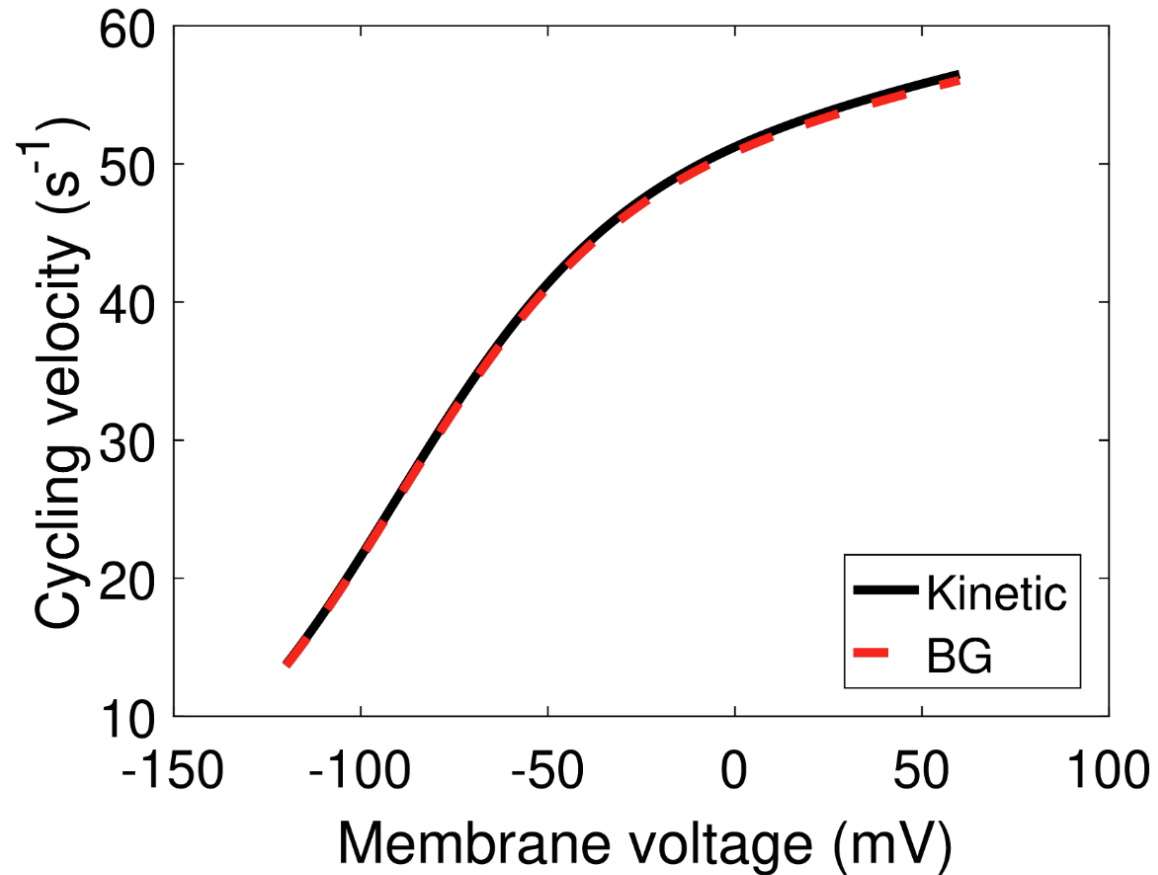
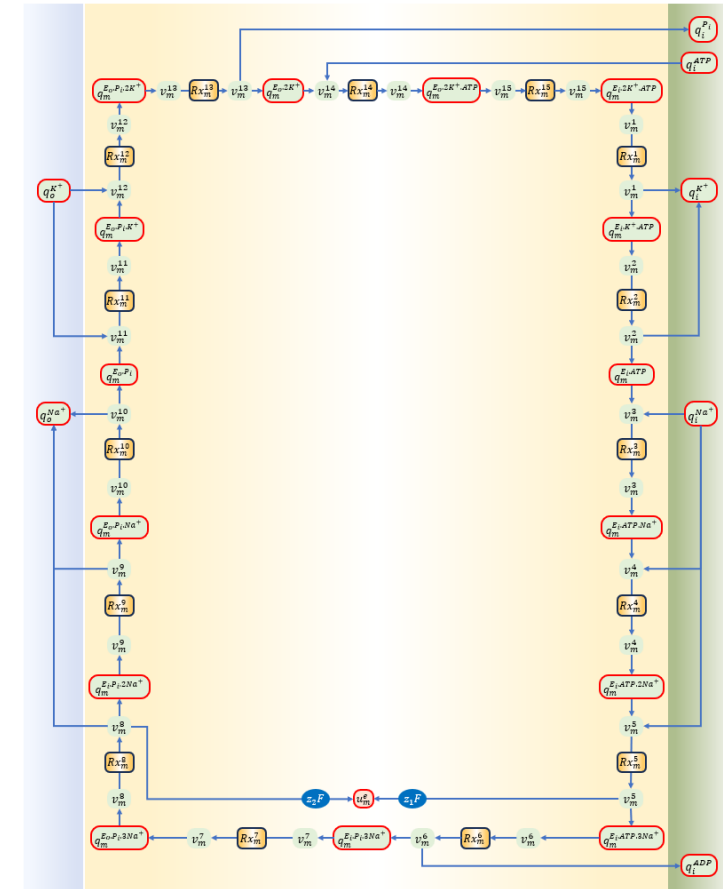
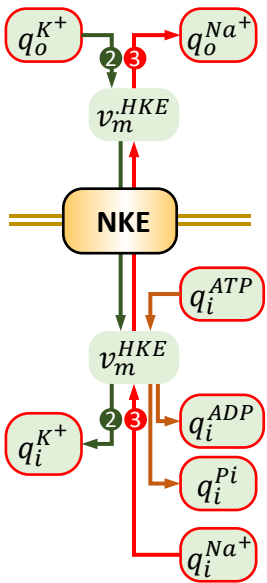
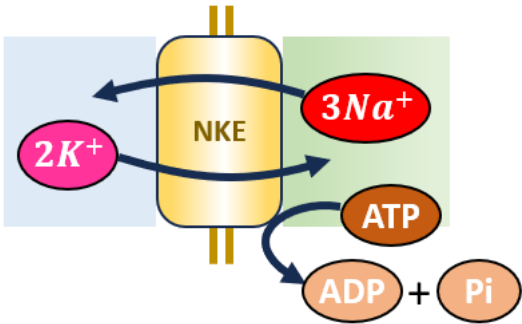
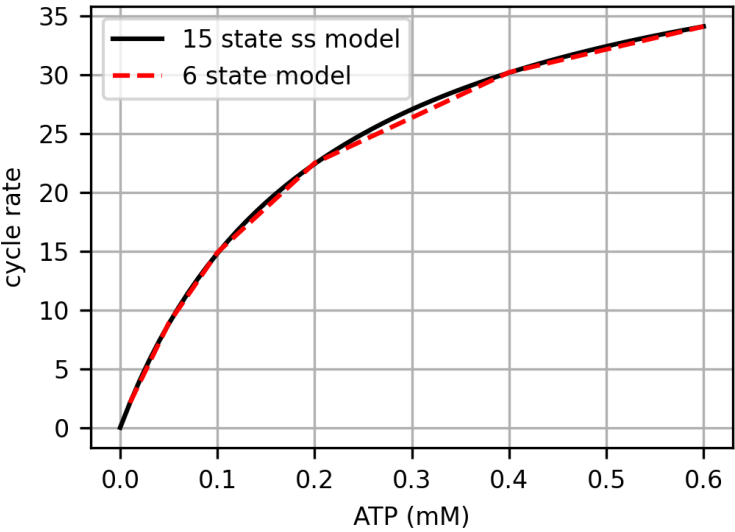
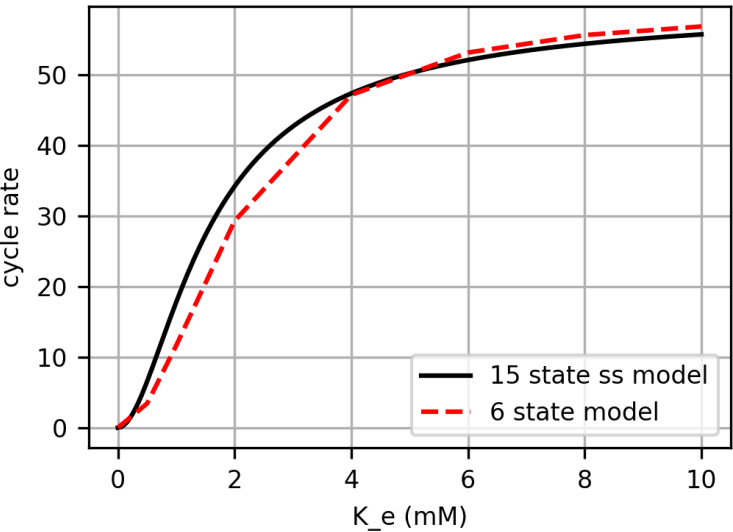
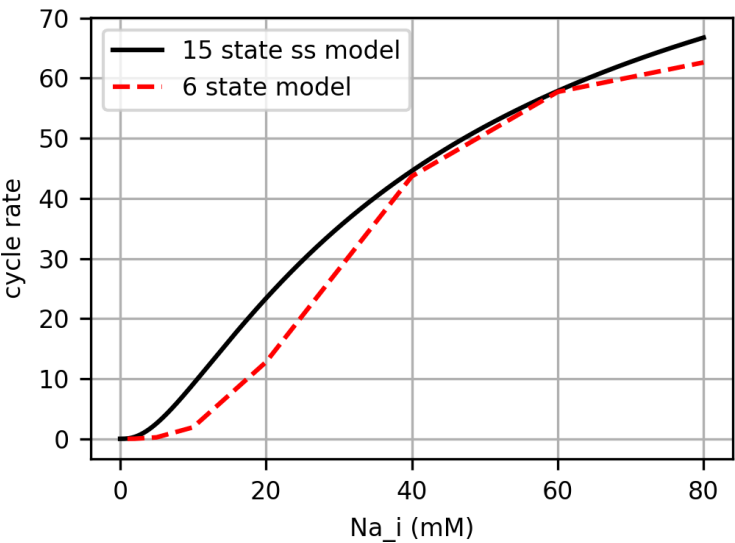
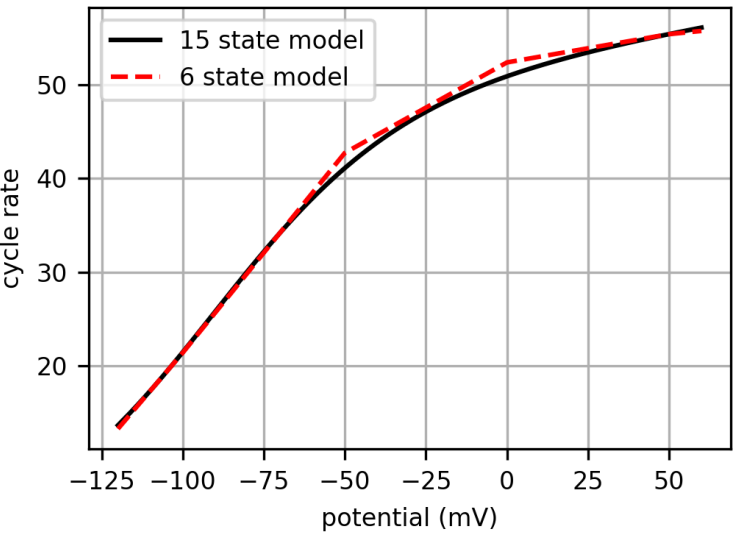


Figure 5: A comparison of the kinetic and bond graph cardiac Na^+/K^+ models. $[\text{Na}^+]_i = 50\text{mM}$, $[\text{Na}^+]_e = 150\text{mM}$, $[\text{K}^+]_i = 0\text{mM}$, $[\text{K}^+]_e = 5.4\text{mM}$, $\text{pH} = 7.4$, $[\text{Pi}]_{\text{tot}} = 0\text{mM}$, $[\text{MgATP}] = 10\text{mM}$, $[\text{MgADP}] = 0\text{mM}$, $T = 310\text{K}$. The bond graph model is formulated using concentration ratios thus zero concentrations were approximated by a concentration of 0.001mM to avoid numerical errors.

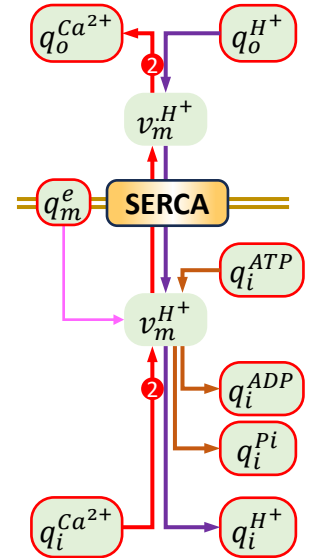
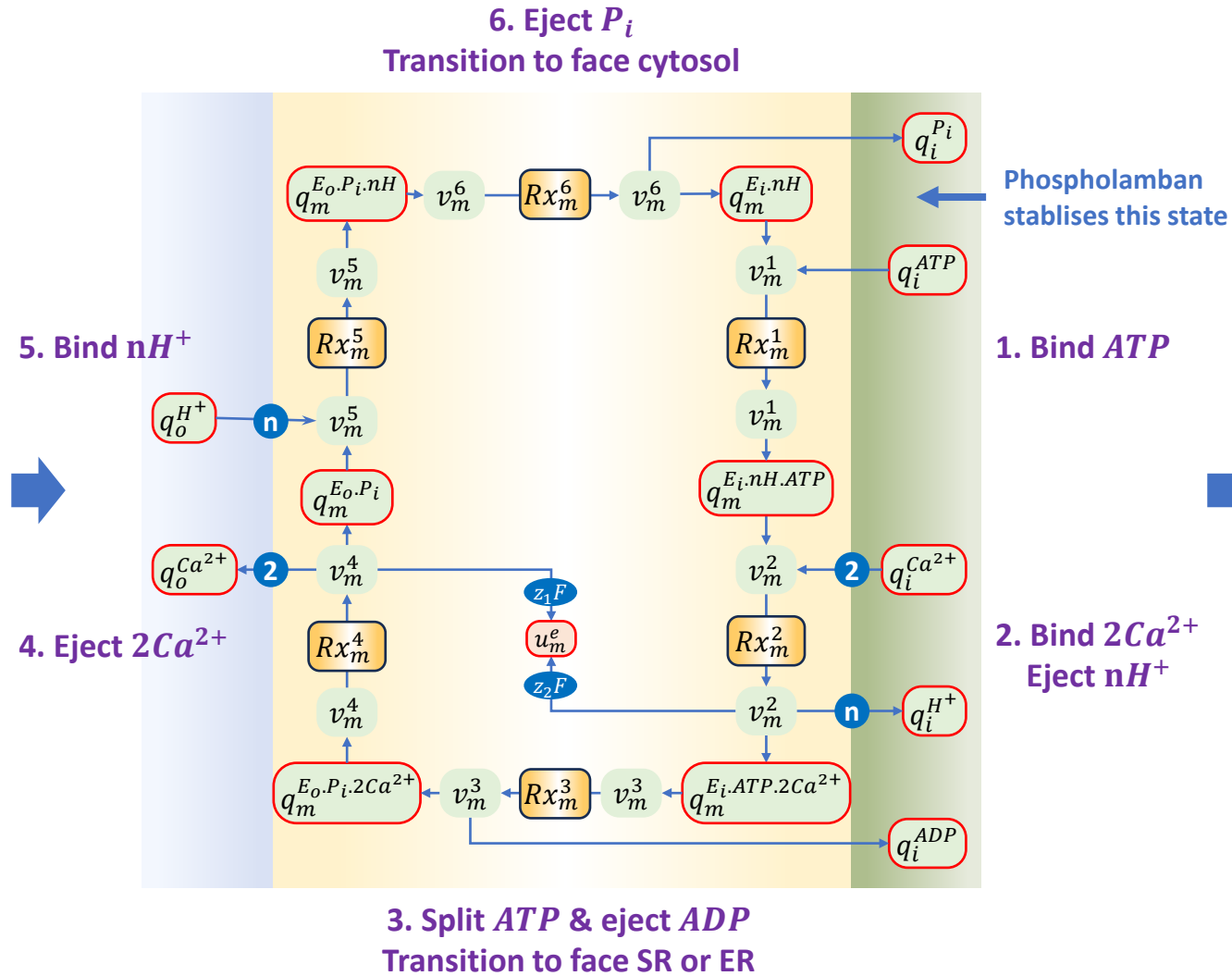
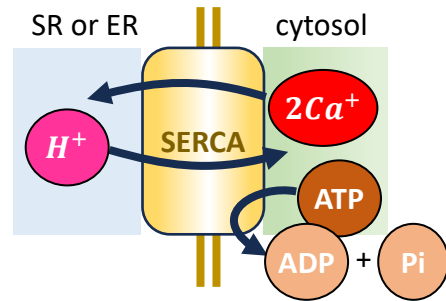


Comparing the 9-state model with the 15-state model

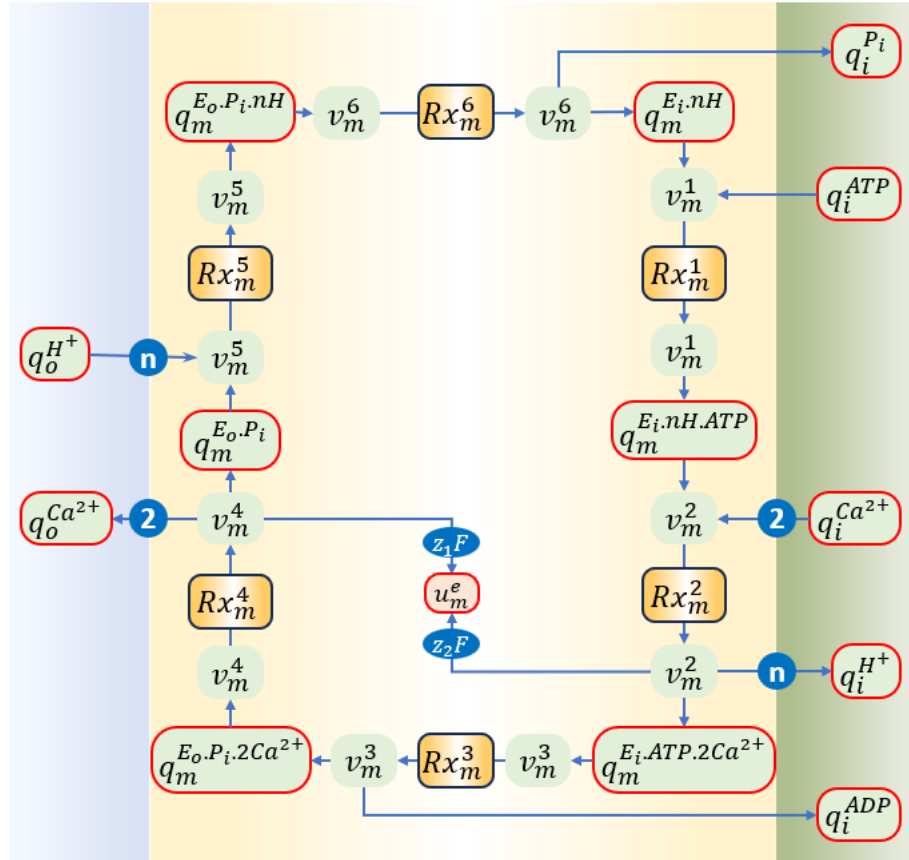


Calcium ATPase (SERCA)

(see [model on PMR](#))



SERCA .. continued



Reaction fluxes

$$v_1 = \kappa_1 \left[\bar{q}_m^{E_i.nH} \cdot \bar{q}_i^{ATP} - \bar{q}_m^{E_i.nH.ATP} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_m^{E_i.nH.ATP} \cdot \left(\bar{q}_i^{Ca^{2+}} \right)^2 - \bar{q}_m^{E_i.ATP.2Ca^{2+}} \cdot \left(\bar{q}_i^{H^+} \right)^n \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_i.ATP.2Ca^{2+}} - \bar{q}_m^{E_o.P_i.2Ca^{2+}} \cdot \bar{q}_i^{ADP} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_o.P_i.2Ca^{2+}} - \bar{q}_m^{E_o.P_i} \cdot \left(\bar{q}_o^{Ca^{2+}} \right)^2 \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_5 = \kappa_5 \left[\bar{q}_m^{E_o.P_i} \cdot \left(\bar{q}_o^{H^+} \right)^n - \bar{q}_m^{E_o.P_i.nH} \right]$$

$$v_6 = \kappa_6 \left[\bar{q}_m^{E_o.P_i.nH} - \bar{q}_m^{E_i.nH} \cdot \bar{q}_i^{P_i} \right]$$

Mass balance:

$$\frac{d}{dt} q_m^{E_i.nH} = v_m^6 - v_m^1$$

$$\frac{d}{dt} q_m^{E_i.nH.ATP} = v_m^1 - v_m^2$$

$$\frac{d}{dt} q_m^{E_i.ATP.2Ca^{2+}} = v_m^2 - v_m^3$$

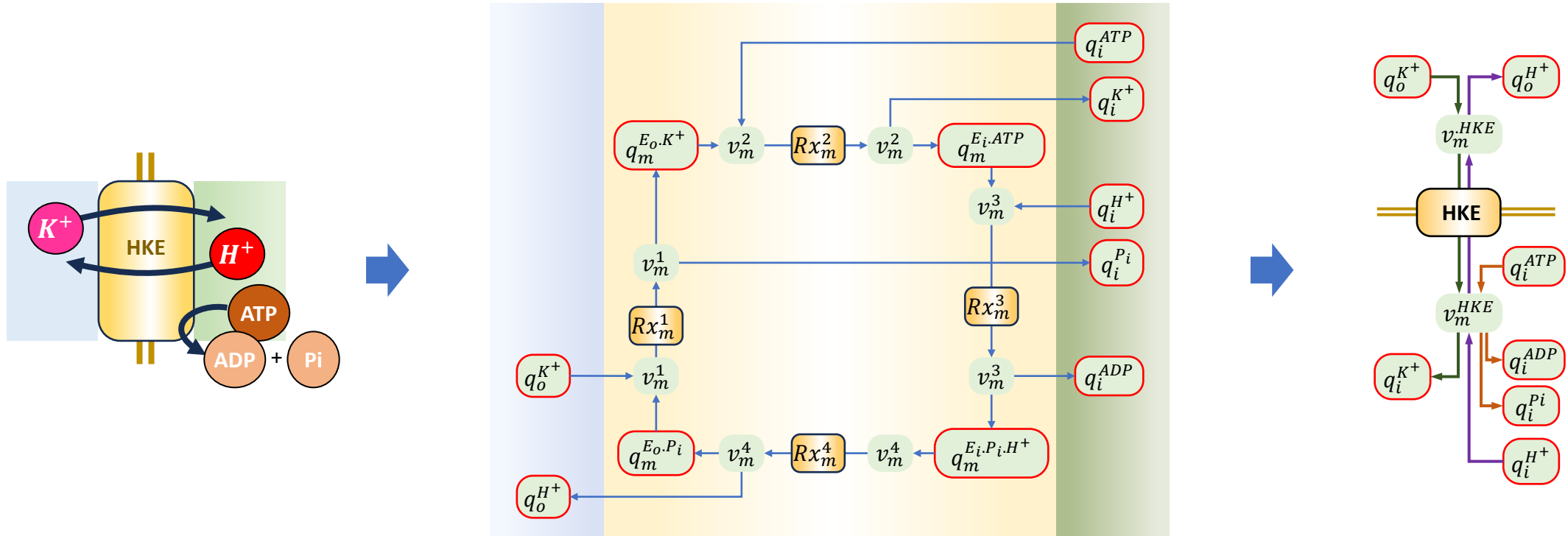
$$\frac{d}{dt} q_m^{E_o.P_i.2Ca^{2+}} = v_m^3 - v_m^4$$

$$\frac{d}{dt} q_m^{E_o.P_i} = v_m^4 - v_m^5$$

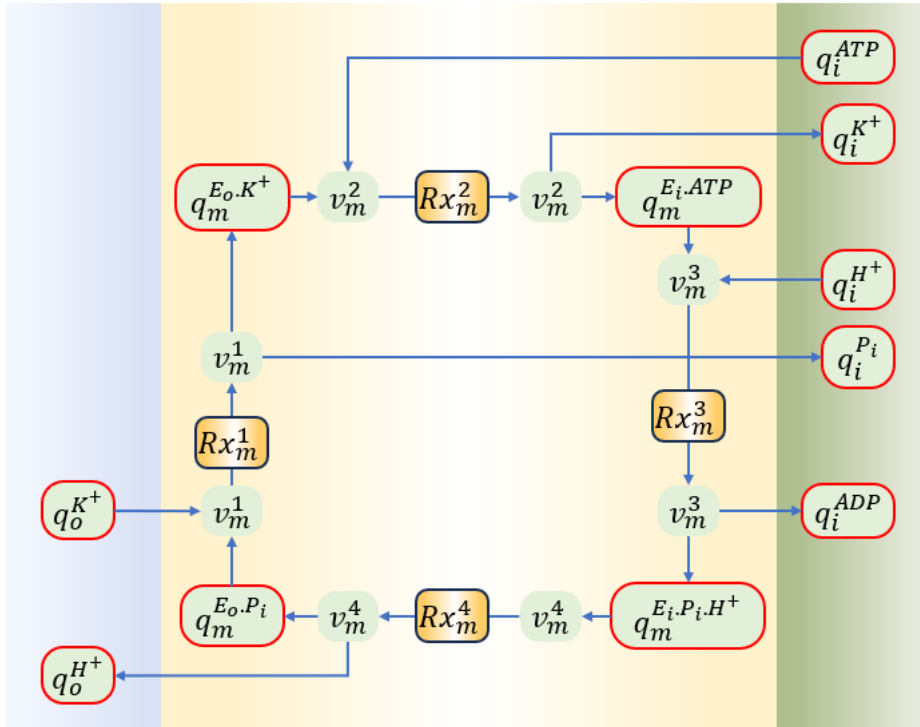
$$\frac{d}{dt} q_m^{E_o.P_i.nH} = v_m^5 - v_m^6$$

Hydrogen-potassium ATPase (HKE)

(see [model on PMR](#))



HKE .. continued



Reaction fluxes

$$v_1 = \kappa_1 \left[\bar{q}_m^{E_o.P_i} \cdot \bar{q}_o^{K^+} - \bar{q}_m^{E_o.K^+} \cdot \bar{q}_i^{P_i} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_m^{E_o.K^+} \cdot \bar{q}_i^{ATP} - \bar{q}_m^{E_i.ATP} \cdot \bar{q}_i^{K^+} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_i.ATP} \cdot \bar{q}_i^{H^+} - \bar{q}_m^{E_i.P_i.H^+} \cdot \bar{q}_i^{ADP} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i.P_i.H^+} - \bar{q}_m^{E_o.P_i} \cdot \bar{q}_o^{H^+} \right]$$

Mass balance:

$$\frac{d}{dt} \bar{q}_m^{E_o.P_i} = v_m^4 - v_m^1$$

$$\frac{d}{dt} \bar{q}_m^{E_o.K^+} = v_m^1 - v_m^2$$

$$\frac{d}{dt} \bar{q}_m^{E_i.ATP} = v_m^2 - v_m^3$$

$$\frac{d}{dt} \bar{q}_m^{E_i.P_i.H^+} = v_m^3 - v_m^4$$

$$\frac{d}{dt} \bar{q}_o^{K^+} = -v_m^1$$

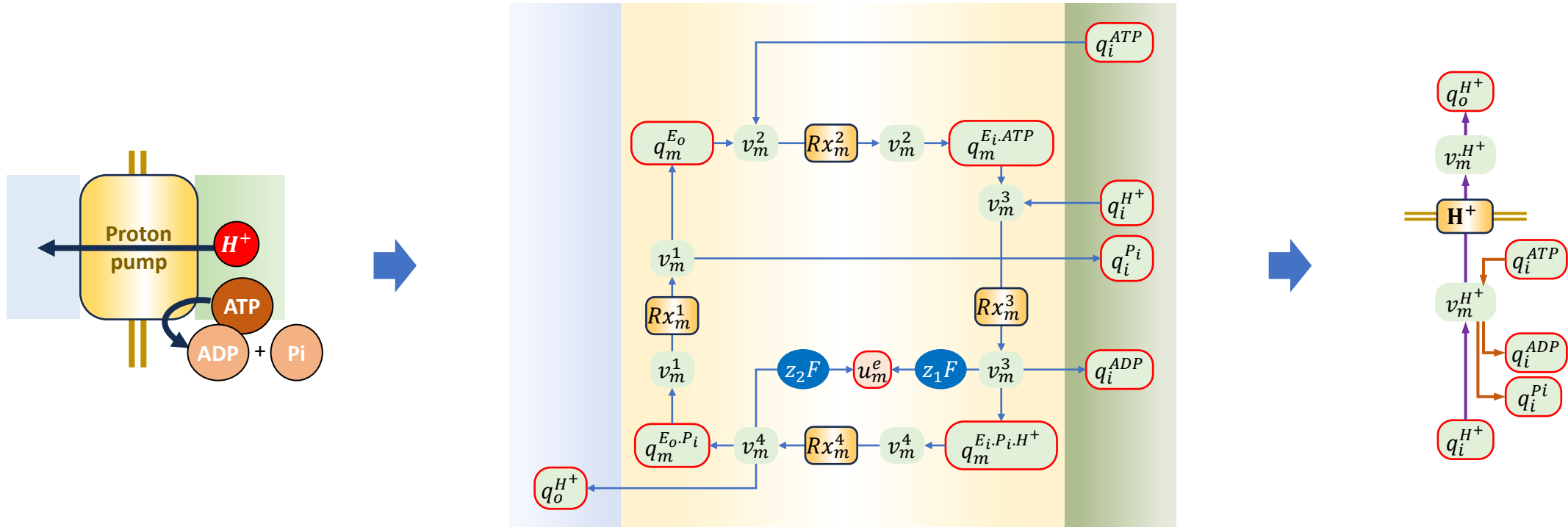
$$\frac{d}{dt} \bar{q}_i^{K^+} = v_m^2$$

$$\frac{d}{dt} \bar{q}_o^{H^+} = v_m^4$$

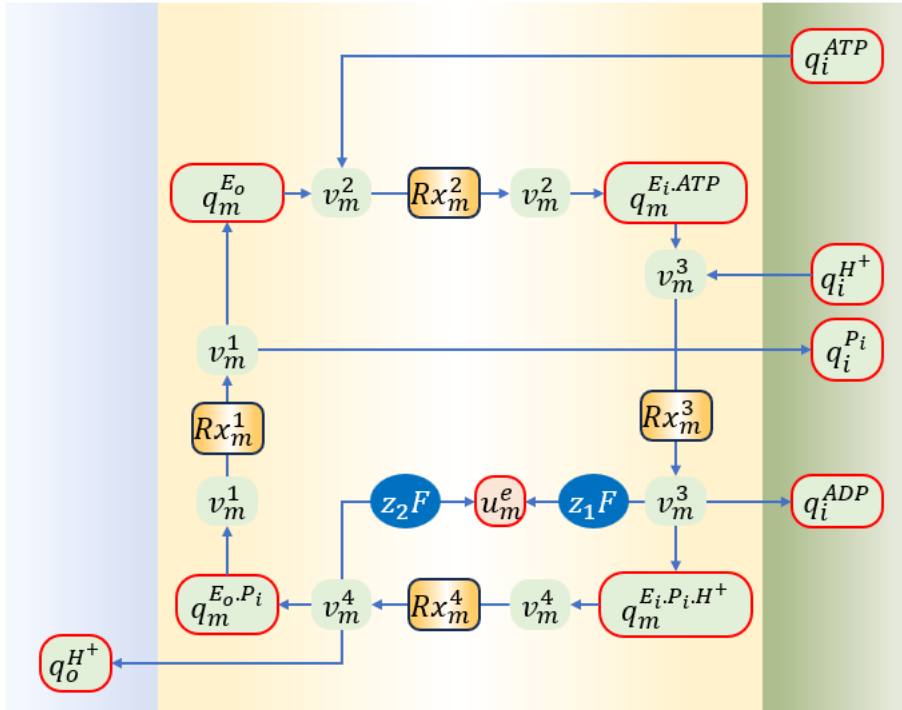
$$\frac{d}{dt} \bar{q}_i^{H^+} = -v_m^3$$

Hydrogen ATPase (Proton pump)

(see [model on PMR](#))



Proton pump .. continued



Reaction fluxes

$$v_1 = \kappa_1 \left[\bar{q}_m^{E_o.Pi} - \bar{q}_m^{E_o} \cdot \bar{q}_i^{Pi} \right]$$

$$v_2 = \kappa_2 \left[\bar{q}_m^{E_o} \cdot \bar{q}_i^{ATP} - \bar{q}_m^{E_i.ATP} \right]$$

$$v_3 = \kappa_3 \left[\bar{q}_m^{E_i.ATP} \cdot \bar{q}_i^{H^+} - \bar{q}_m^{E_i.Pi.H^+} \cdot \bar{q}_i^{ADP} \cdot e^{\frac{z_1 F u_m^e}{RT}} \right]$$

$$v_4 = \kappa_4 \left[\bar{q}_m^{E_i.Pi.H^+} - \bar{q}_m^{E_o.Pi} \cdot \bar{q}_o^{H^+} \cdot e^{\frac{z_2 F u_m^e}{RT}} \right]$$

Mass balance:

$$\frac{d}{dt} \bar{q}_m^{E_o.Pi} = v_m^4 - v_m^1$$

$$\frac{d}{dt} \bar{q}_m^{E_o} = v_m^1 - v_m^2$$

$$\frac{d}{dt} \bar{q}_m^{E_i.ATP} = v_m^2 - v_m^3$$

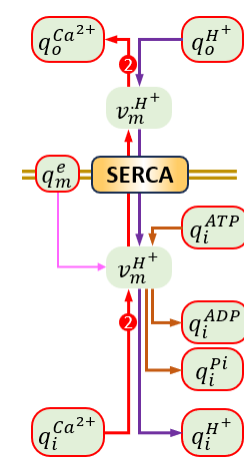
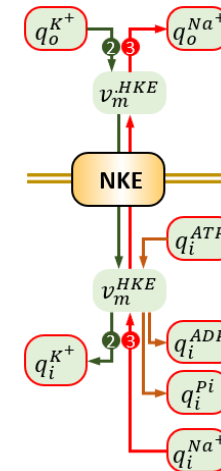
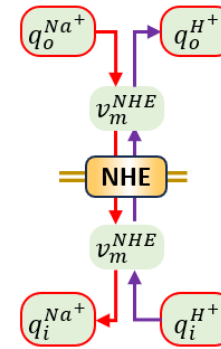
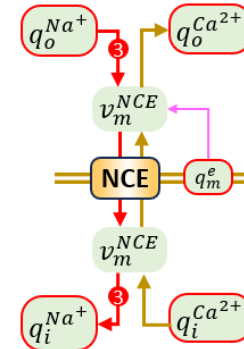
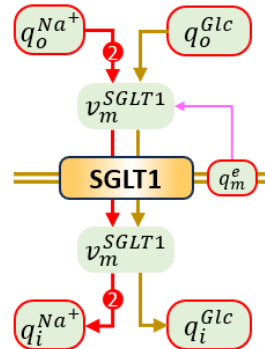
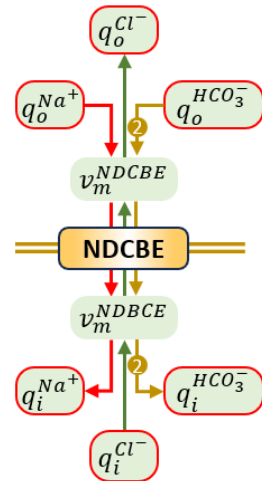
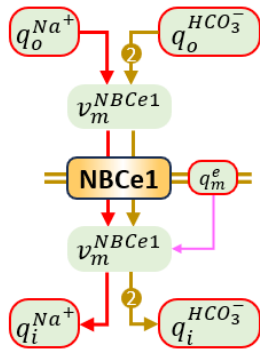
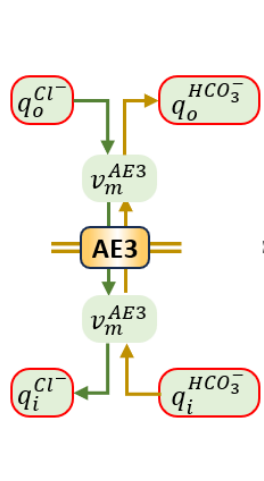
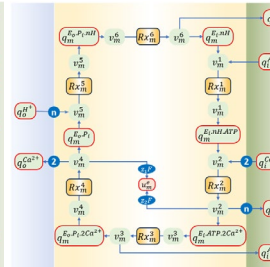
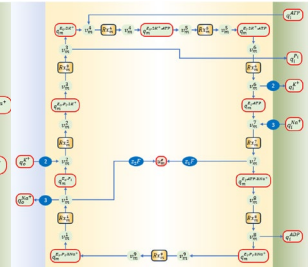
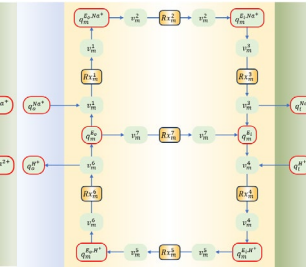
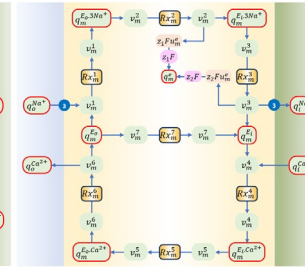
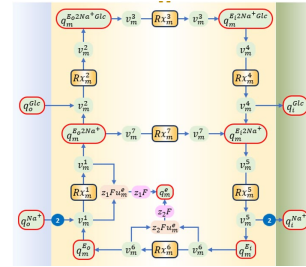
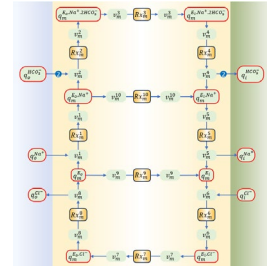
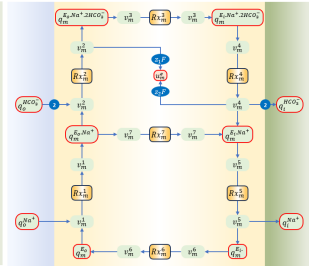
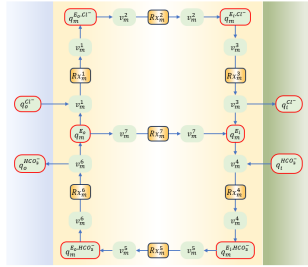
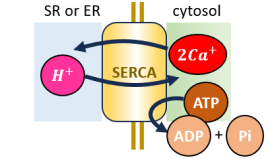
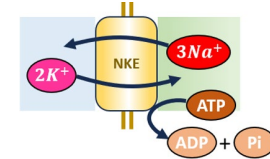
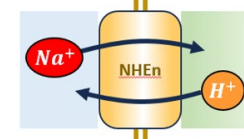
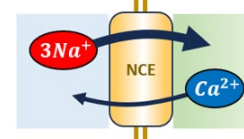
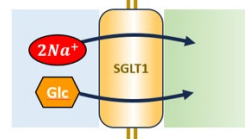
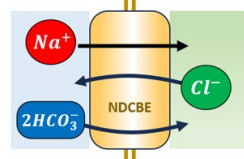
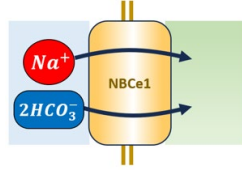
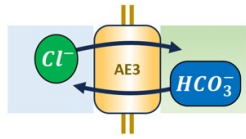
$$\frac{d}{dt} \bar{q}_m^{E_i.Pi.H^+} = v_m^3 - v_m^4$$

$$\frac{d}{dt} \bar{q}_o^{H^+} = v_m^4$$

$$\frac{d}{dt} \bar{q}_i^{H^+} = -v_m^3$$

Summary of membrane transporters

SLC4A3 (AE3) SLC4A4 (NBCe1) SLC4A8 (NDCBE) SLC5A1 (SGLT1) SLC8A1 (NCE) SLC9A1 (NHE1) NKE ATPase SERCA ATPase



Lecture 6: *Ion channels and action potentials*

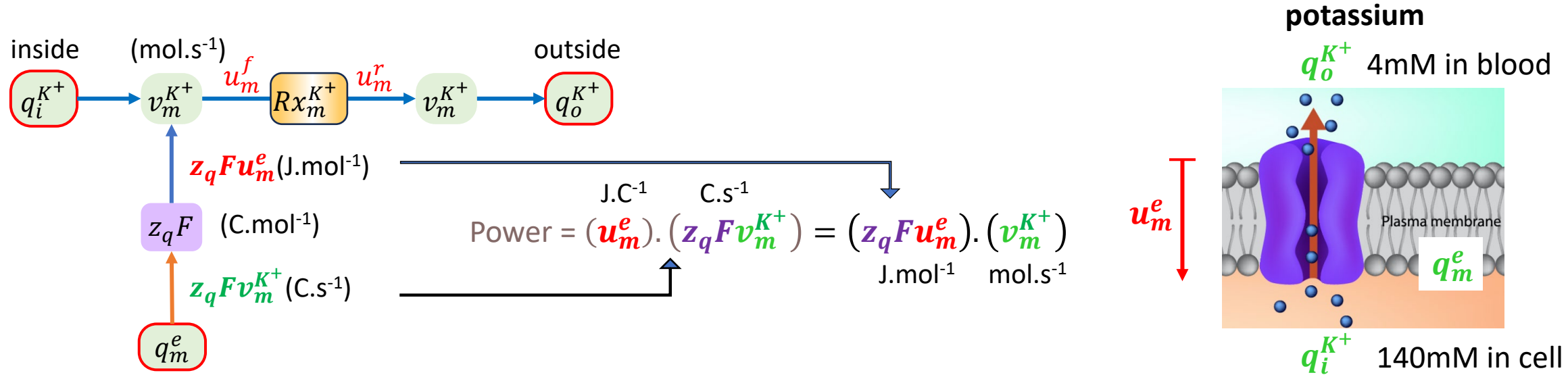
Ion channels support the flow of ions (especially Na^+ , K^+ , Ca^{2+} and Cl^-) across cell membranes at much greater rates than membrane transporters (SLC proteins or ATPase pumps). They regulate the passage of these ions with 'gating processes' that can depend on membrane voltage or ligand binding. We first consider the balance of the chemical and electrical potential differences that drive ion movement (the 'Nernst-Planck equation'), and the membrane potential at which this current is zero (the 'equilibrium' or 'Nernst' potential). Then we consider the way that bond graph models can represent the gating processes and how these give rise to the action potentials seen in neurons.

Ion channel families

- Voltage-gated Na^+ channels
- Voltage-gated K^+ channels
- Inwardly-rectifying K^+ channels
- Ca^{2+} -activated K^+ Channels
- Voltage-gated Ca^{2+} channels
- Ligand-gated Ca^{2+} Channels

See [Channelpedia](#)

The Nernst-Planck equation for moving an ion through a membrane channel



$$(u_i^{K^+}) \quad \frac{dq_i^{K^+}}{dt} = -v_m^{K^+}, \quad (u_o^{K^+}) \quad \frac{dq_o^{K^+}}{dt} = -v_m^{K^+}, \quad (u_m^e) \quad \frac{dq_m^e}{dt} = -z_q F v_m^{K^+},$$

$$(v_m^{K^+}) \quad u_m^f = u_i^{K^+} + z_q F u_m^e, \quad u_m^r = u_o^{K^+}$$

$$(Re_m^{K^+}) \quad v_m^{K^+} = \kappa_m^{K^+} \left(e^{u_m^f/RT} - e^{u_m^r/RT} \right) = \kappa_m^{K^+} \left(\bar{q}_i^{K^+} \cdot e^{z_q F u_m^e/RT} - \bar{q}_o^{K^+} \right) \quad (\text{mol.s}^{-1})$$

The K^+ ion moving outwards down its concentration gradient is opposed by an inward electrical current flow driven by a voltage gradient (inside potential < outside potential). Since the outside potential is defined as a zero reference, the inside must be negative. For sodium moving in the inward direction, the inside potential becomes increasingly positive.

The Nernst equation

Chemical flux v_m^i (mol.s⁻¹) or electrical current flow v_m^e (C.s⁻¹) is zero when

$$v_m^i = \kappa_m^i \left(\bar{q}_1^i \cdot e^{\frac{z_q F u_m^e}{RT}} - \bar{q}_2^i \right) = 0$$

i.e. when $\bar{q}_1^i e^{\frac{z_q F u_m^e}{RT}} = \bar{q}_2^i$

i.e. $e^{\frac{z_q F u_m^e}{RT}} = \frac{\bar{q}_2^i}{\bar{q}_1^i}$ or $\frac{z_q F u_m^e}{RT} = \ln \left(\frac{\bar{q}_2^i}{\bar{q}_1^i} \right)$

or $u_m^e = \frac{RT}{z_q F} \ln \left(\frac{\bar{q}_2^i}{\bar{q}_1^i} \right) = \frac{RT}{z_q F} \ln \left(\frac{c_2^i}{c_1^i} \right)$ called the 'equilibrium' or 'reversal' or 'Nernst' potential

Note that $\frac{RT}{F} \approx 25\text{mV}$ so a 20-fold ($\approx e^3$) concentration difference gives $\frac{RT}{F} \cdot \ln(20) = \frac{RT}{F} \cdot 3 \approx 75\text{mV}$

Constant field model of ion permeation (Goldman, Hodgkin, Katz or GHK model)

Current through channel from Nernst-Planck is $I = z\varphi_m^{ion}RT \cdot \frac{dc_m^{ion}}{dx} + zF \cdot z\varphi_m^{ion}c_m^{ion} \cdot \frac{du_m^e}{dx}$,

where c_m^{ion} & u_m^e are the ion concentration and electrical potential along the channel, respectively, z is the ion valence, and φ_m^{ion} is the ionic mobility within the membrane.

The constant field model assumes $\frac{du_m^e}{dx} = \frac{u_m^e}{d_m}$, where u_m^e is the membrane potential and d_m is the membrane thickness,

and the above eqn reduces to $\frac{dc_m^{ion}}{dx} + ac_m^{ion} = b$, where $a = \frac{zF \cdot u_m^e}{RT \cdot d_m}$ and $b = \frac{I}{z\varphi_m^{ion}RT}$

The solution is $c_m^{ion}(x) = \frac{b}{a} + \left(\beta c_o^{ion} - \frac{b}{a}\right) \cdot e^{-ax}$, and $\frac{dc_m^{ion}}{dx} = (b - a\beta c_o^{ion}) \cdot e^{-ax}$,

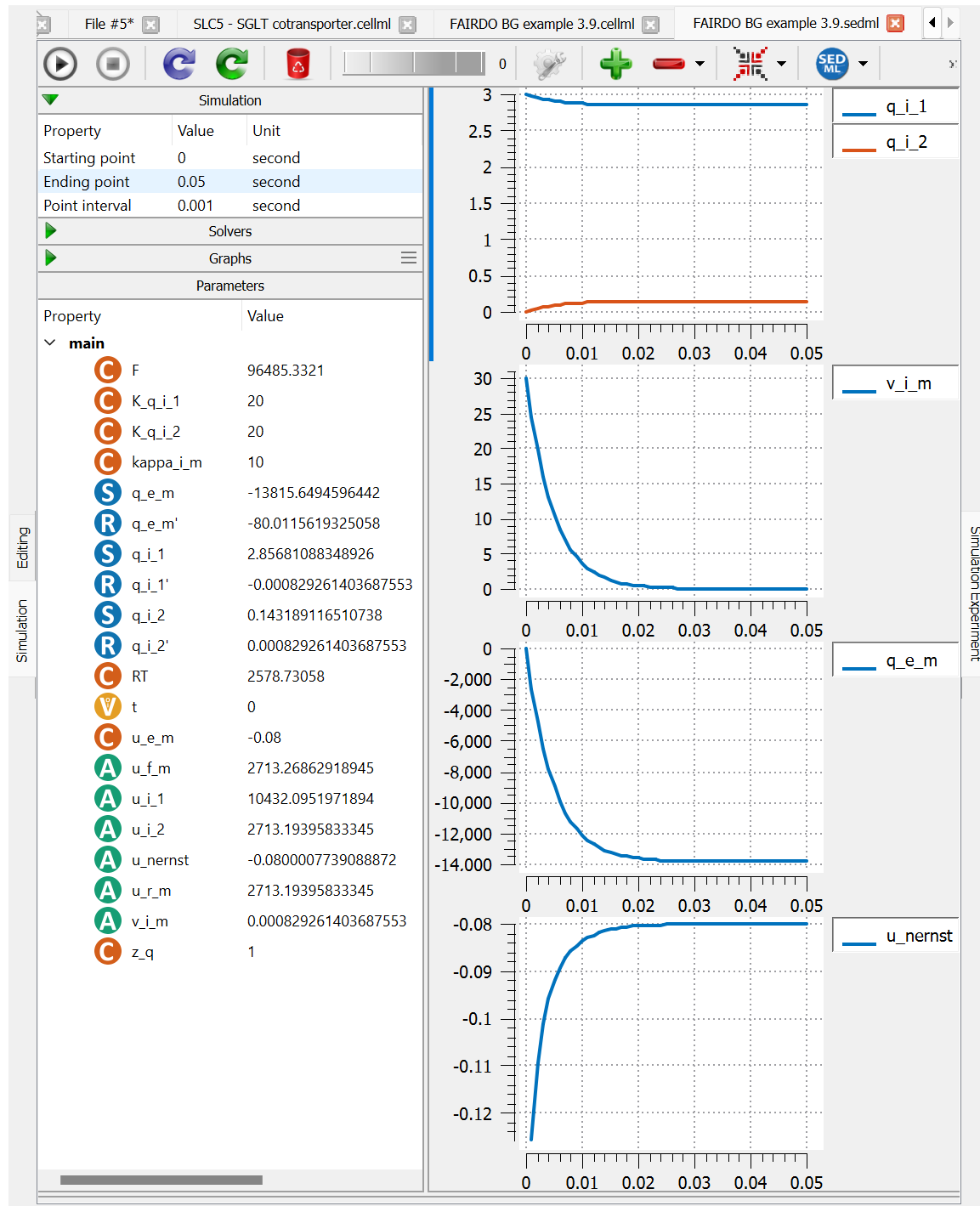
where c_o^{ion} is the ion concentration outside the cell.

The ion concentration inside the cell is $c_m^{ion}(d_m) = \beta c_i^{ion} = \frac{b}{a} + \left(\beta c_o^{ion} - \frac{b}{a}\right) \cdot e^{-a \cdot d_m}$.

Defining $\xi = d_m \cdot a = \frac{zFu_m^e}{RT}$ and $P = \frac{\beta\varphi_m^{ion}RT}{d_mF}$ gives $c_i^{ion} = \frac{I \cdot RT}{z^2F^2 \cdot u_m^e \cdot P} (1 - e^{-\xi}) + c_o^{ion} e^{-\xi}$

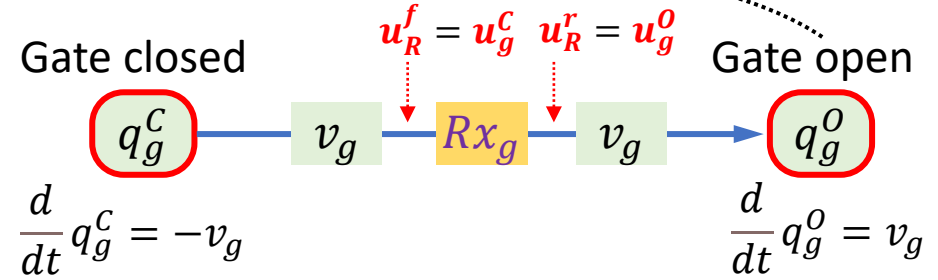
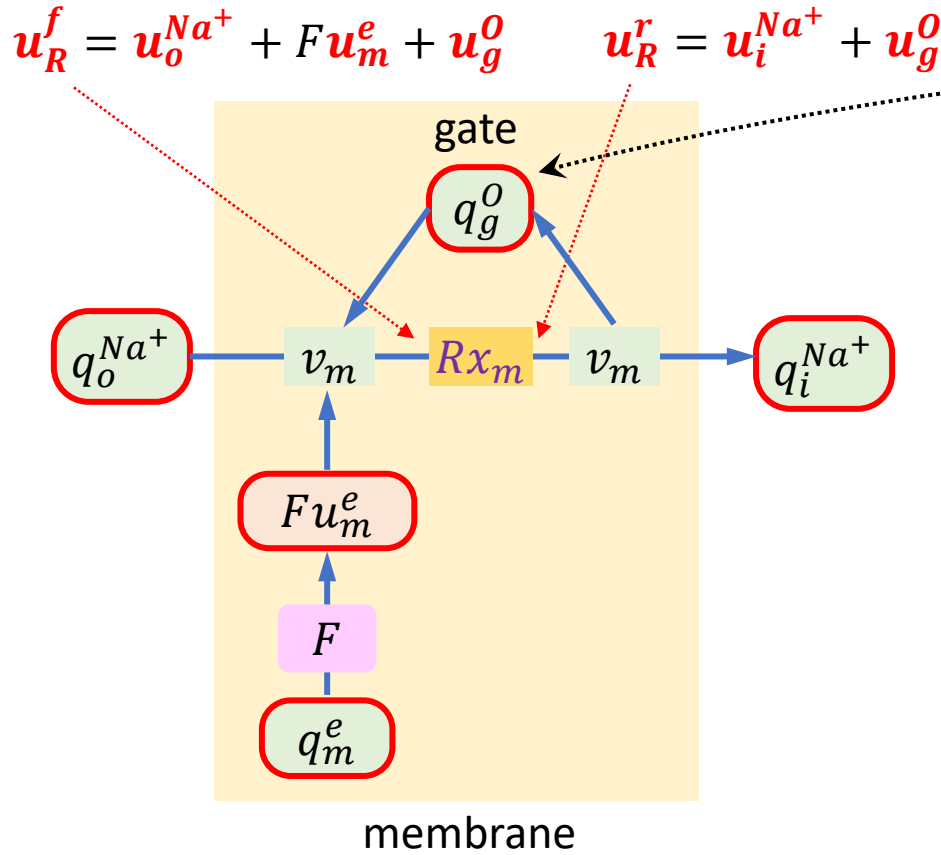
or $I = \frac{z^2F^2P \cdot u_m^e}{RT} \cdot \frac{c_i^{ion} - c_o^{ion} e^{-\xi}}{1 - e^{-\xi}}$ or $I = zPF\xi \cdot \frac{c_i^{ion} e^{\xi} - c_o^{ion}}{e^{\xi} - 1}$

([Link to PMR model](#))



← $u_m^e = -80\text{mV}$

Ion channel gating



Fraction of gates open $\frac{q_g^O}{q_g^{tot}} = p$; Fraction of gates closed $\frac{q_g^C}{q_g^{tot}} = 1 - p$

$$\mathbf{u}_g^C = \frac{RT}{F} \ln(K_g^C q_g^C) = \frac{RT}{F} \ln(K_g^C q_g^{tot} \cdot (1 - p)); \quad \mathbf{u}_g^O = \frac{RT}{F} \ln(K_g^O q_g^{tot} \cdot p)$$

$$\frac{dp}{dt} = \frac{v_g}{q_g^{tot}} = \frac{\kappa_g}{q_g^{tot}} [K_g^C q_g^{tot} \cdot (1 - p) - K_g^O q_g^{tot} \cdot p]$$

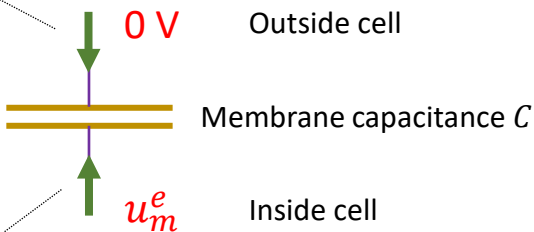
$\frac{dp}{dt} = \alpha_g(1 - p) - \beta_g p$	$\alpha_g = \kappa_g K_g^C$ is rate at which closed gates open $\beta_g = \kappa_g K_g^O$ is rate at which open gates close
---	--

α_g and β_g are often dependent on $\mathbf{u}_m^e$ and on ligand binding.

Chemical flux is $v_m = \kappa_m^* \cdot p \left(K_o^{Na^+} q_o^{Na^+} \cdot e^{\frac{nF u_m^e}{RT}} - K_i^{Na^+} q_i^{Na^+} \right) \text{ mol.s}^{-1}$ and charge flux is $F v_m \text{ C.s}^{-1}$

where $\kappa_m^* = \kappa_m K_g^0 q_g^{tot}$ is the open channel permeability.

Inward Na^+ flux $v_m^{Na^+}$ (current $Fv_m^{Na^+}$) moves +ve charge into the cell and makes u_m^e more positive.



Outward K^+ flux $v_m^{K^+}$ (current $Fv_m^{K^+}$) moves +ve charge out of the cell and makes u_m^e more negative.

Outward Cl^- flux $v_m^{Cl^-}$ (current $-Fv_m^{Cl^-}$) moves -ve charge out of the cell and makes u_m^e more positive.

$$v_m^{Na^+} = \kappa_m^{Na^+} \left(\bar{q}_o^{Na^+} - \bar{q}_i^{Na^+} \cdot e^{\frac{z^{Na^+} F u_m^e}{RT}} \right)$$

Reversal potential: $u_m^e = \frac{RT}{z^{Na^+} \cdot F} \ln \left(\frac{\bar{q}_o^{Na^+}}{\bar{q}_i^{Na^+}} \right) = \frac{RT}{F} \ln \left(\frac{c_o^{Na^+}}{c_i^{Na^+}} \right) = 25 \cdot \ln \left(\frac{140}{12} \right) = 61mV$

$$v_m^{K^+} = \kappa_m^{K^+} \left(\bar{q}_i^{K^+} \cdot e^{\frac{z^{K^+} F u_m^e}{RT}} - \bar{q}_o^{K^+} \right)$$

Reversal potential: $u_m^e = \frac{RT}{z^{K^+} \cdot F} \ln \left(\frac{\bar{q}_o^{K^+}}{\bar{q}_i^{K^+}} \right) = \frac{RT}{F} \ln \left(\frac{c_o^{K^+}}{c_i^{K^+}} \right) = 25 \cdot \ln \left(\frac{5}{150} \right) = -85mV$

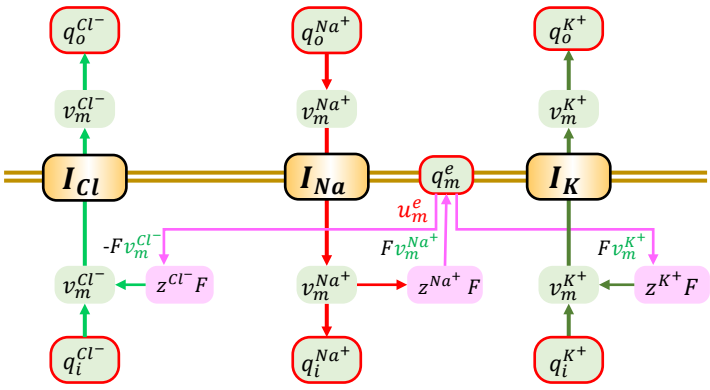
$$v_m^{Cl^-} = \kappa_m^{Cl^-} \left(\bar{q}_i^{Cl^-} \cdot e^{\frac{z^{Cl^-} F u_m^e}{RT}} - \bar{q}_o^{Cl^-} \right)$$

Reversal potential: $u_m^e = \frac{RT}{z^{Cl^-} \cdot F} \ln \left(\frac{\bar{q}_o^{Cl^-}}{\bar{q}_i^{Cl^-}} \right) = \frac{RT}{-F} \ln \left(\frac{c_o^{Cl^-}}{c_i^{Cl^-}} \right) = -25 \cdot \ln \left(\frac{100}{20} \right) = -40mV$

Current flow into the membrane:

$$\frac{dq_m^e}{dt} = C \frac{du_m^e}{dt} = Fv_m^{Na^+} - Fv_m^{K^+} + Fv_m^{Cl^-}$$

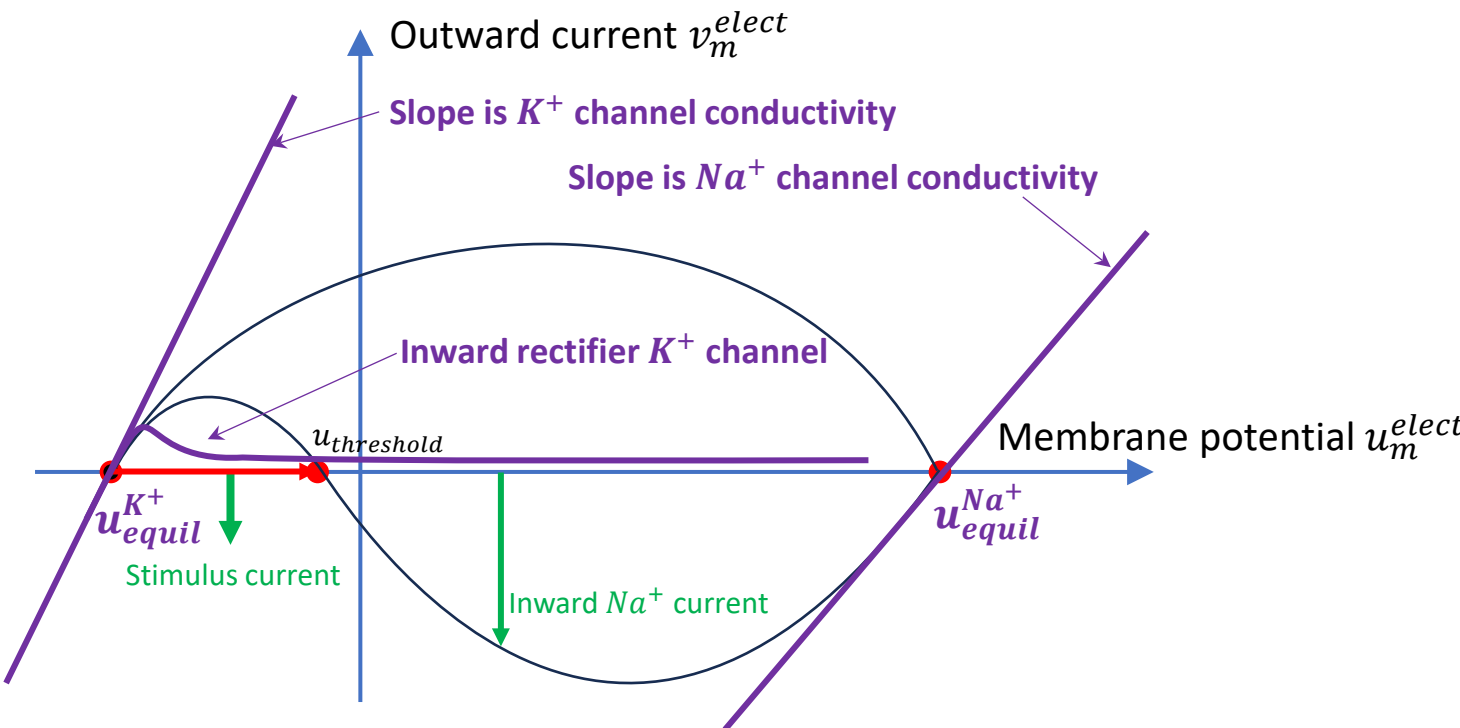
or $\frac{du_m^e}{dt} = \frac{F}{C} (v_m^{Na^+} - v_m^{K^+} + v_m^{Cl^-})$



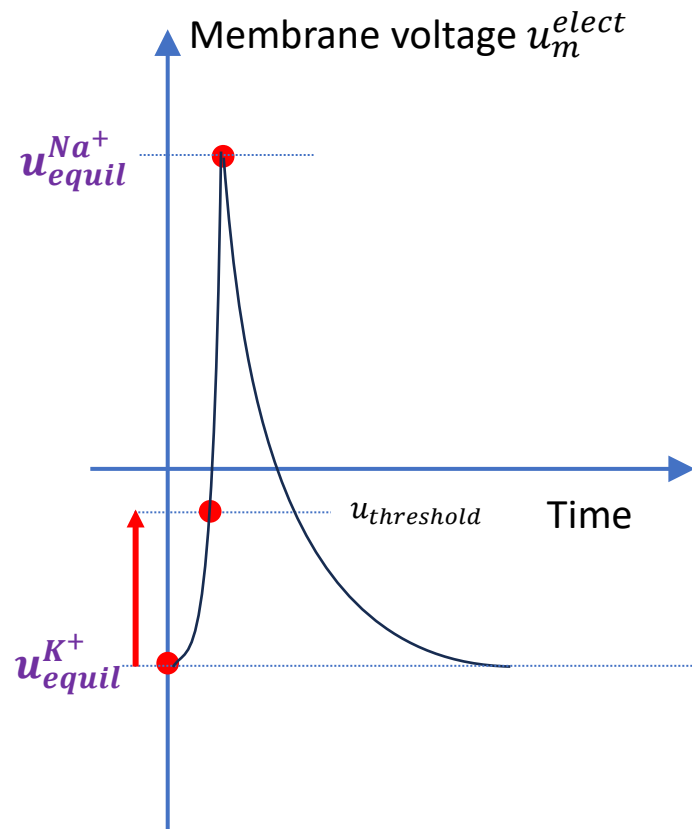
Generating an action potential

The ‘inward rectifier’ K^+ channel used in cardiac myocytes to prevent major K^+ leakage out of cardiac myocytes during systole when the membrane voltage is $\sim 0\text{mV}$. Called a ‘pact with the devil’ by Denis Noble because the high impedance means that the heart is susceptible to small perturbations that can cause arrhythmias.

Current-voltage relations

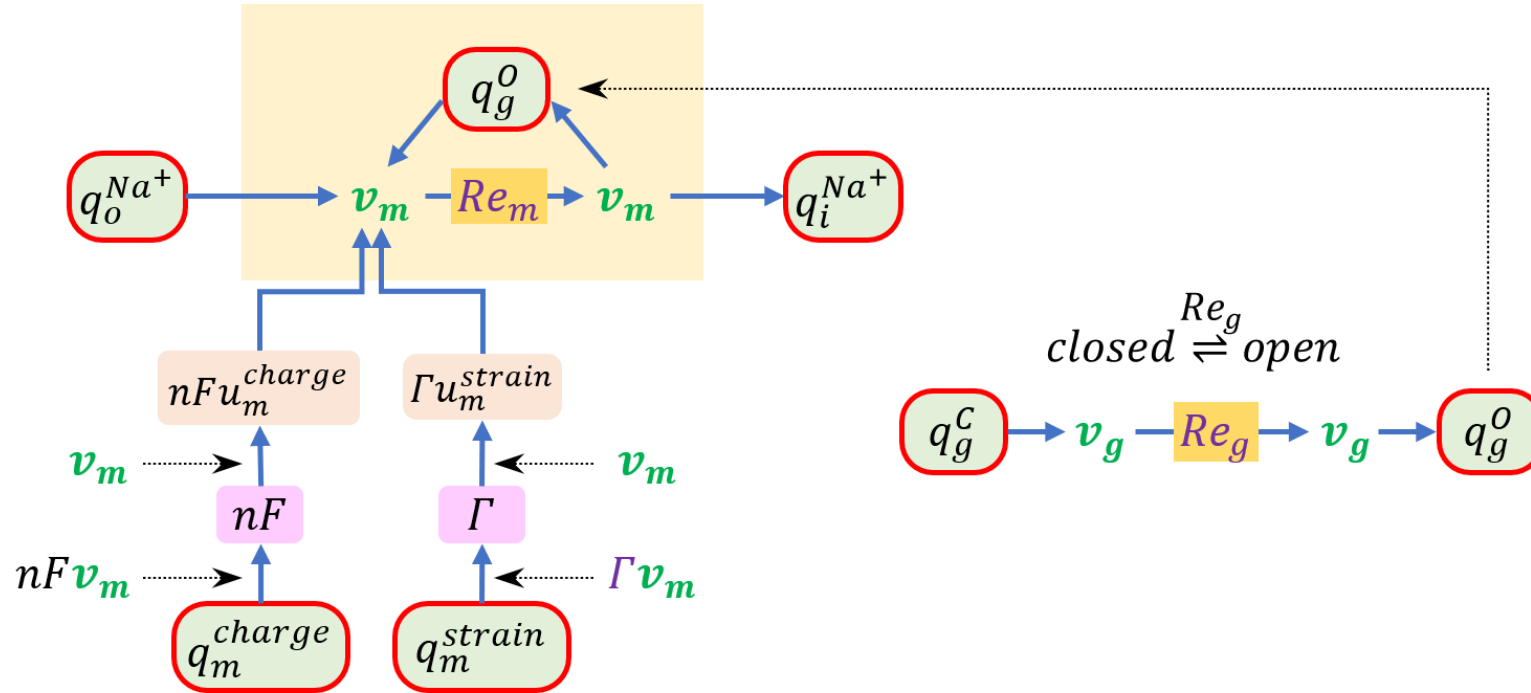


Action potential



Equilibrium (Nernst) potentials for K^+ ($\sim -85\text{mV}$) and Na^+ ($\sim 135\text{mV}$)
 Current-voltage trajectory during the action potential. Threshold for opening Na^+ channel $\sim -35\text{mV}$.
 K^+ open. Inject sufficient **current** to push intracellular **potential** up to the threshold. Then massive inward flux of Na^+ ions.
 Na^+ channel gates close, allowing potential to return to resting value

A strain-dependent gated ion channel



$$v_m = \kappa_m^* \cdot p \left(K_o^{Na^+} q_o^{Na^+} \cdot e^{\frac{nF u_m^{charge}}{RT}} \cdot e^{\frac{\Gamma u_m^{strain}}{RT}} - K_i^{Na^+} q_i^{Na^+} \right)$$

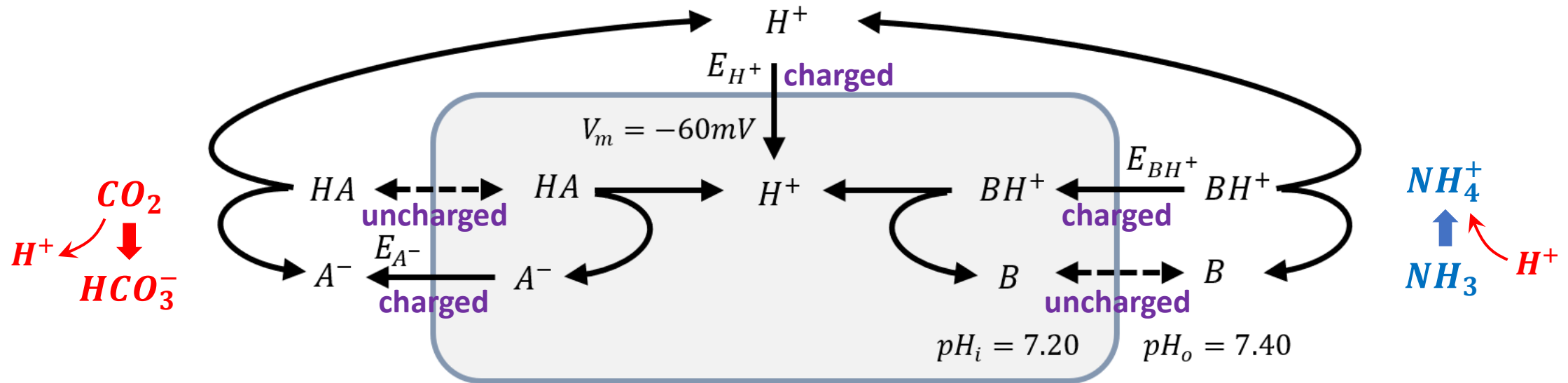
Lecture 7: *Cell models*

A major goal for these lectures is getting to the point where you can understand how to build **composite bond graph models** of intracellular processes based on the underlying physics of all the relevant components. In this lecture we provide examples of subcellular processes involving multiple proteins. The first is an example of building a composite model for intracellular pH regulation. The second is a model of glucose transport. The third is an example of bicarbonate control in neurons. The fourth is a vascular smooth muscle cell. These composite cell models require bond graph models of enzyme-catalysed reactions (Lecture 3), facilitated diffusion of a chemical species crossing a cell membrane (Lecture 4), ATPase pumps (Lecture 5) and electro-diffusion in ion channels (Lecture 6).

pH buffering with weak acids and weak bases

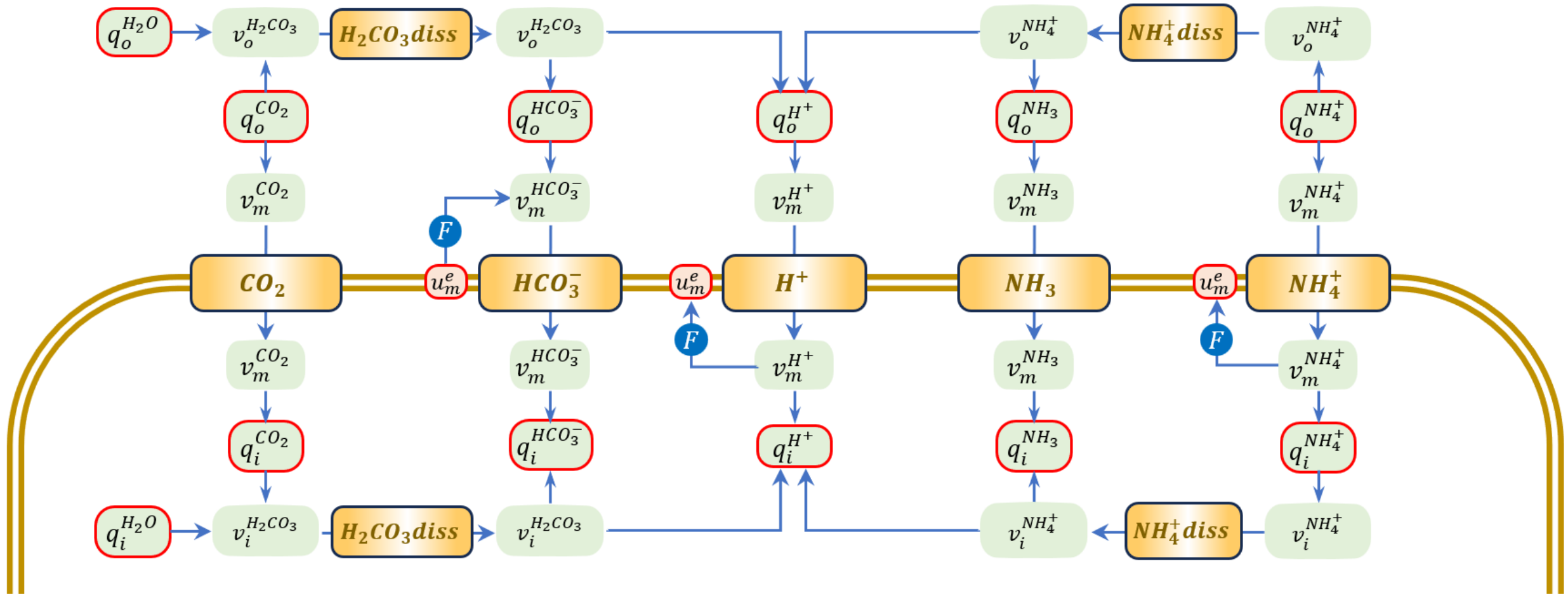
CO_2 as a weak acid (HA) that combines with H_2O to give up a proton (H^+) to become HCO_3^- (A^-)

NH_3 is a weak base (B) that accepts a proton (H^+) to become NH_4^+ (BH^+)

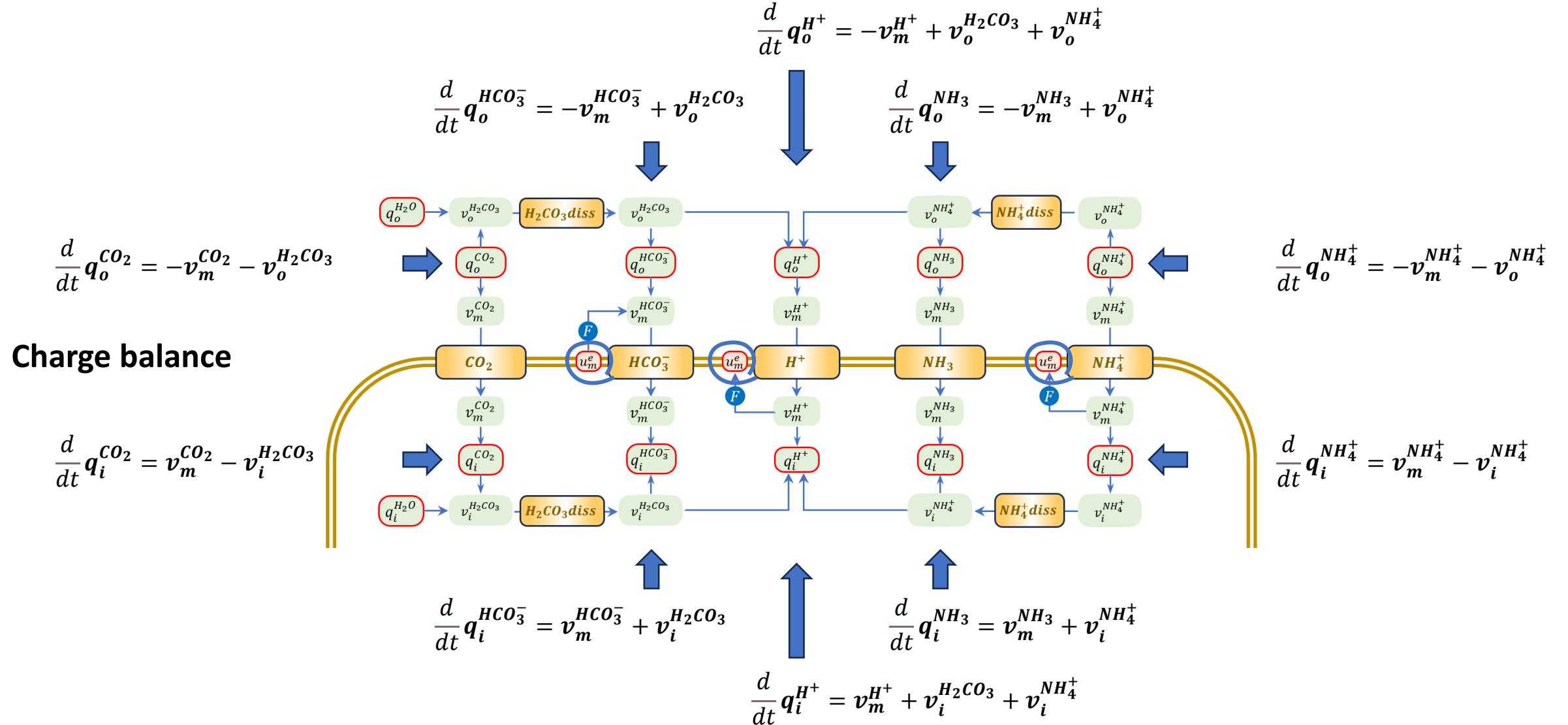


i.e. We need 2 uncharged membrane transporters, 3 ion channels and 4 catalysed reactions.

Bond graph model for pH regulation



Mass balance at 0-nodes



Fluxes

Diffusion of uncharged species:

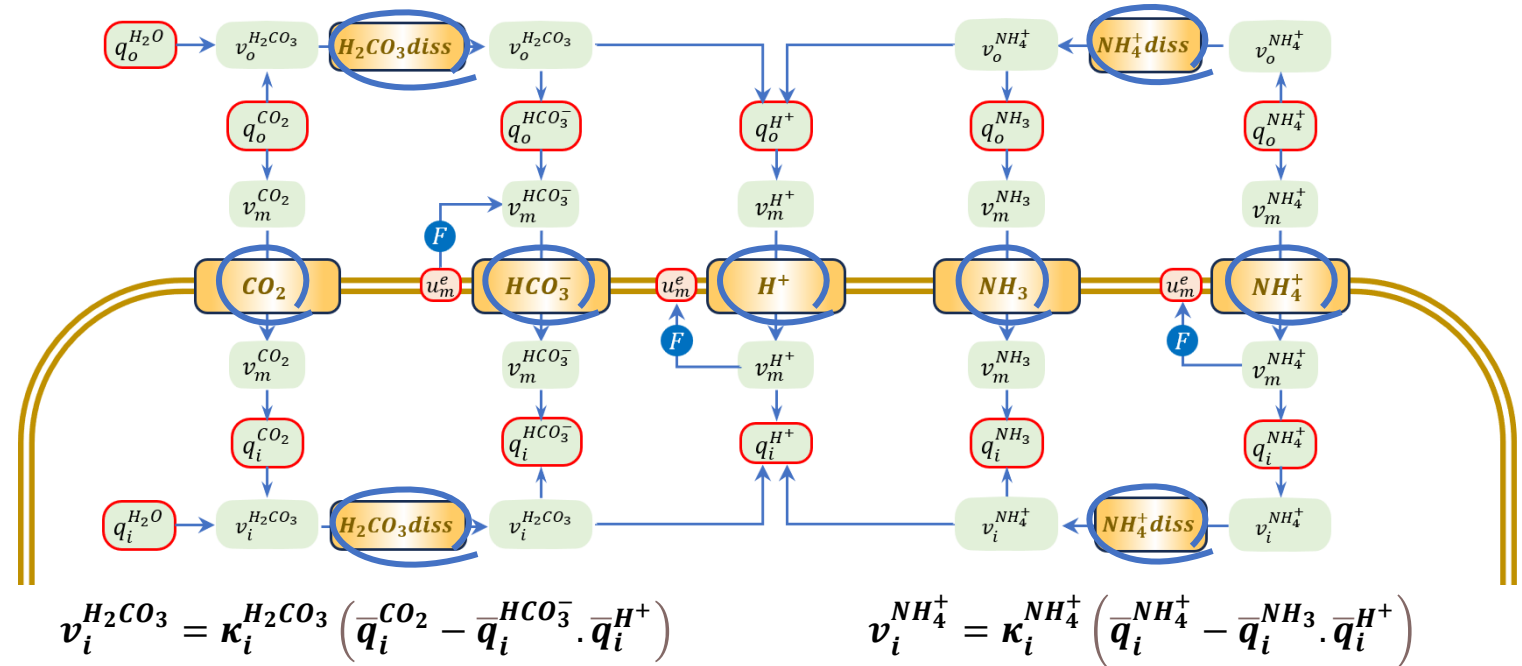
$$v_m^{CO_2} = \kappa_m^{CO_2} (\bar{q}_o^{CO_2} - \bar{q}_i^{CO_2}).$$

$$v_m^{NH_3} = \kappa_m^{NH_3} (\bar{q}_o^{NH_3} - \bar{q}_i^{NH_3}).$$

Dissociation reactions:

$$v_o^{H_2CO_3} = \kappa_o^{H_2CO_3} (\bar{q}_o^{CO_2} - \bar{q}_o^{HCO_3^-} \cdot \bar{q}_o^{H^+})$$

$$v_o^{NH_4^+} = \kappa_o^{NH_4^+} (\bar{q}_o^{NH_4^+} - \bar{q}_o^{NH_3} \cdot \bar{q}_o^{H^+})$$



Transport of charged species:

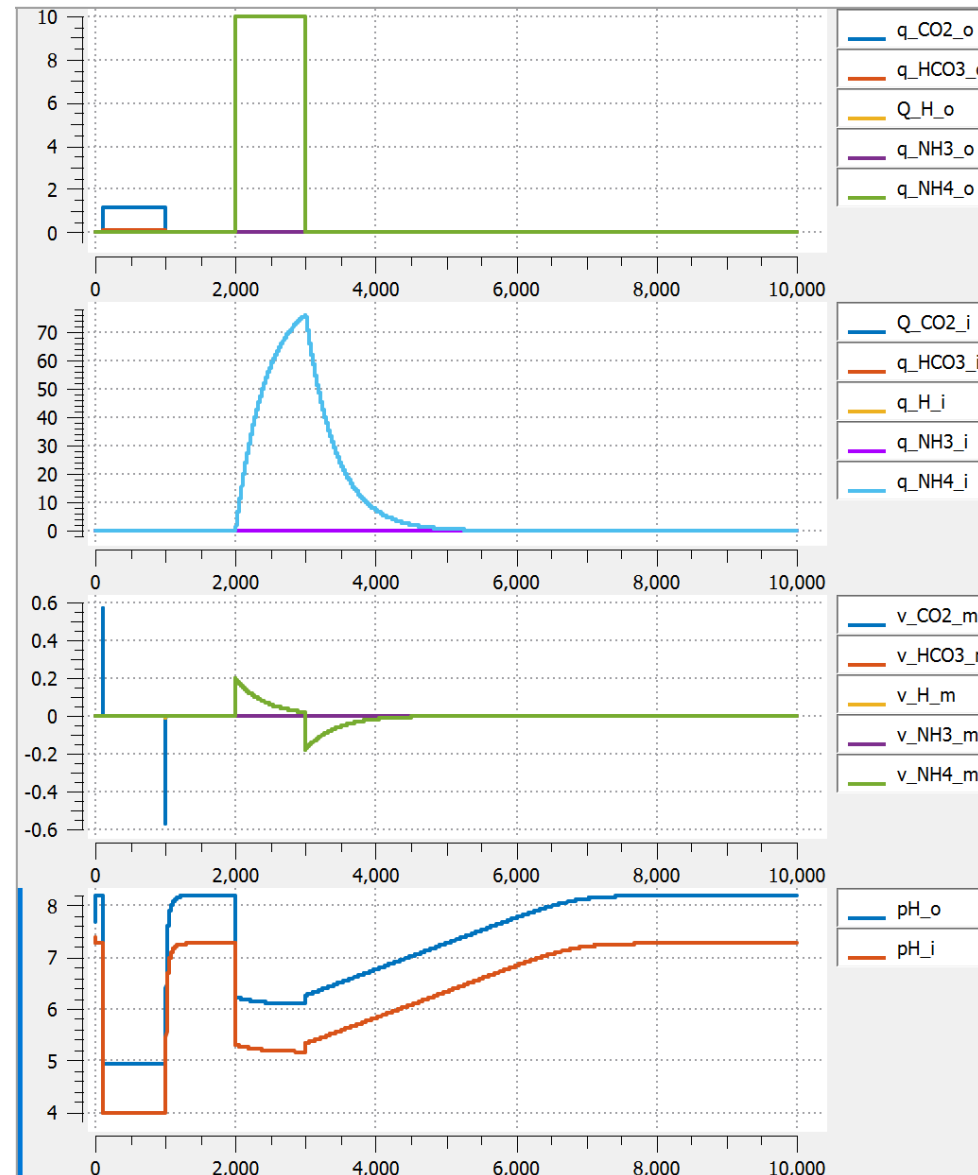
$$v_m^{HCO_3^-} = \kappa_m^{HCO_3^-} (\bar{q}_o^{HCO_3^-} \cdot \varepsilon - \bar{q}_i^{HCO_3^-})$$

$$v_m^{H^+} = \kappa_m^{H^+} (\bar{q}_o^{H^+} - \bar{q}_i^{H^+} \cdot \varepsilon)$$

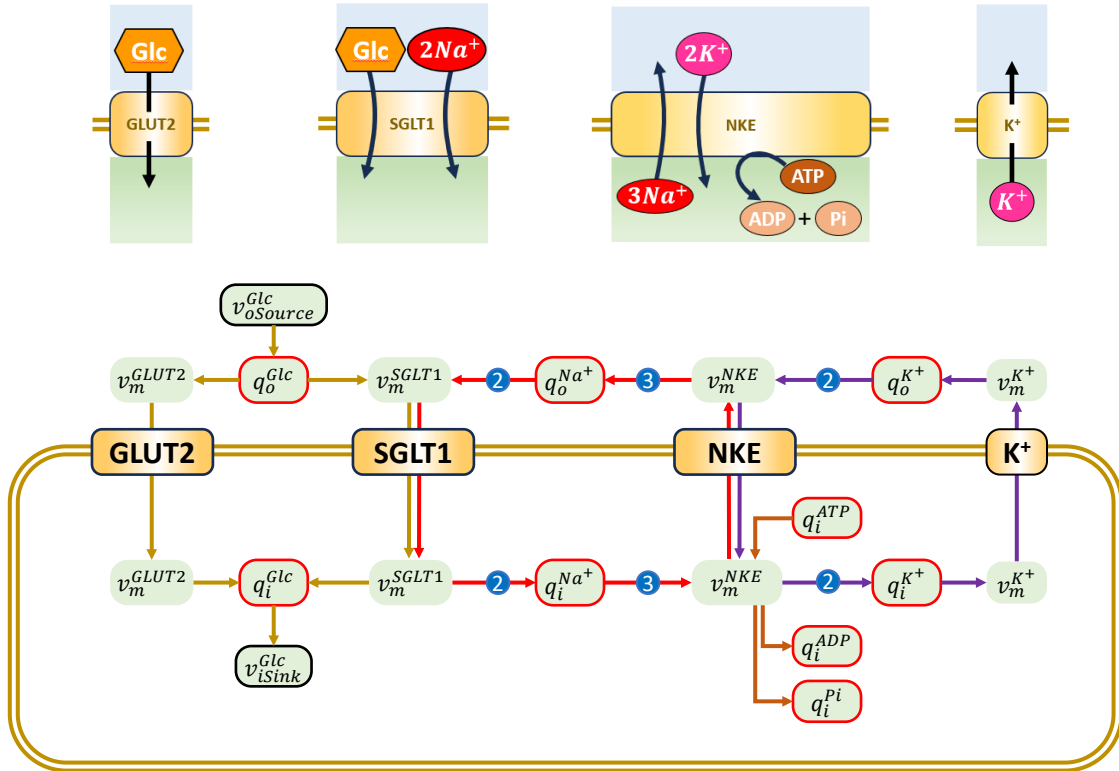
$$v_m^{NH_4^+} = \kappa_m^{NH_4^+} (\bar{q}_o^{NH_4^+} - \bar{q}_i^{NH_4^+} \cdot \varepsilon)$$

$$\varepsilon = e^{z_q F u_m^e / RT}$$

Solution in OpenCOR



Intracellular regulation of Glucose, Na⁺ & K⁺



Mass balance:

$$\frac{d}{dt} q_i^{Glc} = v_m^{GLUT2} + v_m^{SGLT1} - v_{iSink}^{Glc}$$

$$\frac{d}{dt} q_o^{Glc} = -v_m^{GLUT2} - v_m^{SGLT1} + v_{oSink}^{Glc}$$

$$\frac{d}{dt} q_i^{Na^+} = 2v_m^{SGLT1} - 3v_m^{NKE}$$

$$\frac{d}{dt} q_o^{Na^+} = -2v_m^{SGLT1} + 3v_m^{NKE}$$

$$\frac{d}{dt} q_i^{K^+} = 2v_m^{NKE} - 2v_m^{K^+}$$

$$\frac{d}{dt} q_o^{K^+} = -2v_m^{NKE} + 2v_m^{K^+}$$

Transmembrane fluxes

$$v_m^{GLUT2} = v_{max}^{GLUT2} \cdot (\bar{q}_o^{Glc} - \bar{q}_i^{Glc}) / \left(1 + \frac{\bar{q}_o^{Glc}}{k_m^1} + \frac{\bar{q}_i^{Glc}}{k_m^2} + \frac{\bar{q}_o^{Glc} \bar{q}_i^{Glc}}{k_m^3} \right)$$

$$v_m^{SGLT1} = a_1 \left\{ (\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} - (\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{2Fu_m^e/RT} \right\} / A$$

where

$$A = \left[e^{2z_1 Fu_m^e/RT} + (\bar{q}_o^{Na^+})^2 (a_2 + a_3 \cdot \bar{q}_o^{Glc}) \right] \left[(\bar{q}_i^{Na^+})^2 \cdot \bar{q}_i^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \Lambda \right] + \left[a_6 + (\bar{q}_i^{Na^+})^2 (a_5 + a_4 \cdot \bar{q}_i^{Glc}) \right] \left[(\bar{q}_o^{Na^+})^2 \cdot \bar{q}_o^{Glc} \cdot e^{z_2 Fu_m^e/RT} + \Lambda \cdot e^{2Fu_m^e/RT} \right]$$

$$v_m^{NKE} = \kappa_2 \cdot q_{tot} \left\{ \frac{1}{\bar{q}_i^{ADP}} \left(\frac{\bar{q}_i^{Na^+}}{\bar{q}_o^{Na^+}} \right)^3 \cdot e^{-Fu_m^e/RT} - \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} \cdot \left(\frac{\bar{q}_i^{K^+}}{\bar{q}_o^{K^+}} \right)^2 \right\} / \left(A_1 \left\{ \frac{1}{\bar{q}_i^{ADP}} \left(\frac{\bar{q}_i^{Na^+}}{\bar{q}_o^{Na^+}} \right)^3 \cdot e^{-Fu_m^e/RT} - \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} \cdot \left(\frac{\bar{q}_i^{K^+}}{\bar{q}_o^{K^+}} \right)^2 \right\} + A_2 A_3 \right)$$

with

$$A_1 = \left\{ a_1 + a_2 \frac{(\bar{q}_o^{K^+})^2}{\bar{q}_i^{Pi}} \right\} \cdot \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} \cdot (\bar{q}_o^{K^+})^{-2} \quad (\text{mol})$$

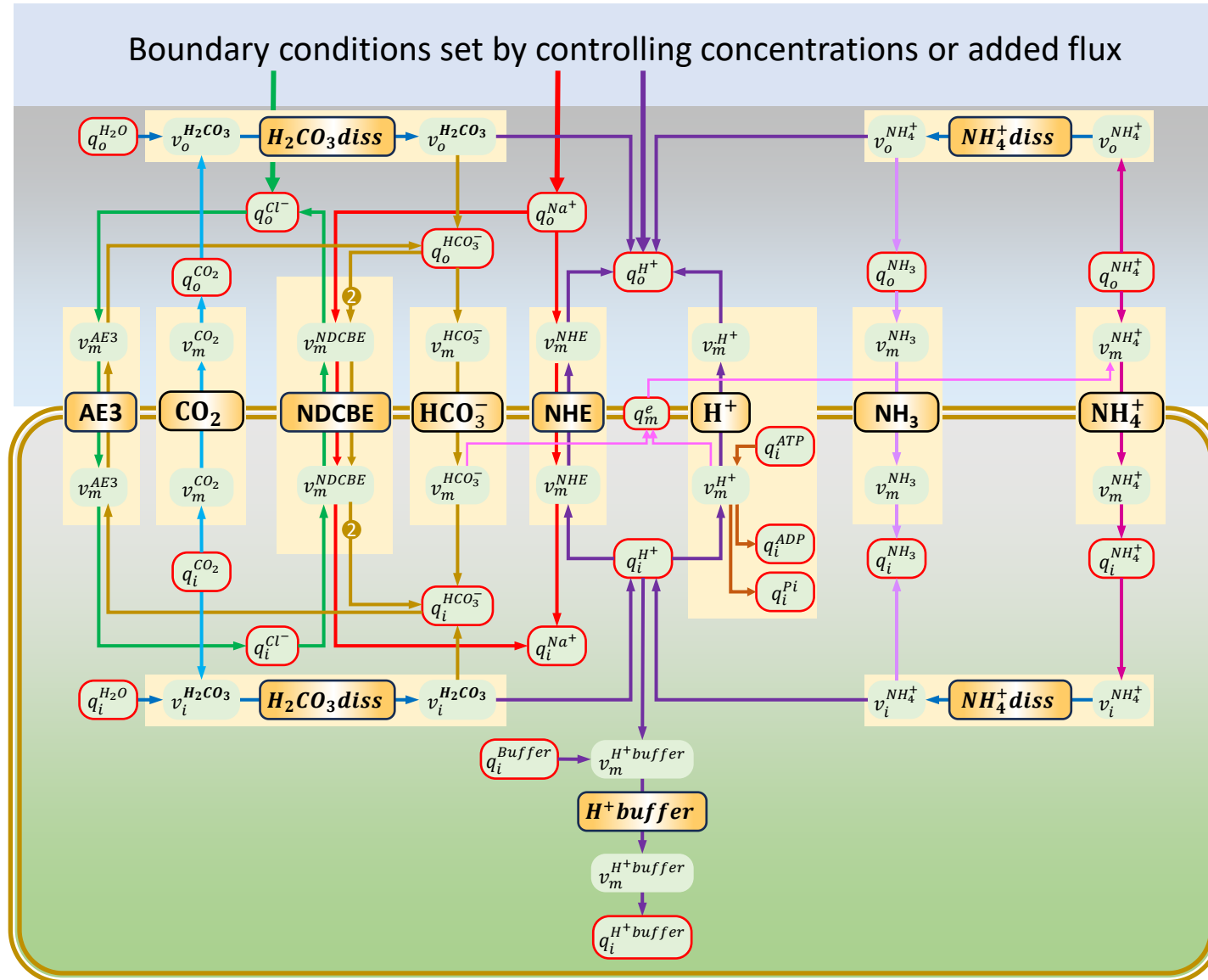
$$A_2 = a_1 \cdot \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} \cdot \left(\frac{\bar{q}_i^{K^+}}{\bar{q}_o^{K^+}} \right)^2 + a_2 \frac{(\bar{q}_i^{K^+})^2}{\bar{q}_i^{Pi}} \cdot \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} + a_3 + a_4 \frac{(\bar{q}_i^{Na^+})^3}{\bar{q}_i^{ADP}} \cdot e^{-z_1 Fu_m^e/RT} \quad (\text{units mol})$$

$$A_3 = \frac{\kappa_2}{\kappa_4} \cdot e^{-z_2 Fu_m^e/RT} \cdot (\bar{q}_o^{Na^+})^{-3} + \frac{\bar{q}_i^{Pi}}{\bar{q}_i^{ATP}} \cdot (\bar{q}_o^{K^+})^{-2} \quad (\text{units dimensionless})$$

$$a_1 = \frac{1}{K_m^{E_o, Pi}}, \quad a_2 = \frac{1}{K_m^{E_o, 2K^+}}, \quad a_3 = \frac{1}{K_m^{E_i, ATP}} \quad \text{and} \quad a_4 = \frac{1}{K_m^{E_i, Pi, 3Na^+}} \quad (\text{all with units of mol}).$$

$$v_m^K = \kappa_m^K \left(\bar{q}_i^{K^+} \cdot e^{Fu_m^e/RT} - \bar{q}_o^{K^+} \right)$$

Control of bicarbonate in neurons



1. Anion exchange (chloride-bicarbonate)
2. CO_2 leakage
3. Sodium-driven chloride-bicarbonate exchange
4. Bicarbonate leakage
5. Sodium-hydrogen exchange
6. Proton pump
7. NH_3 leakage
8. Electro-diffusion of ammonium
9. Carbonic anhydrase reaction
10. Ammonium reaction
11. Sodium gradient
12. Chloride gradient
13. Bicarbonate flux
14. Proton flux
15. NH_3 flux
16. Ammonium flux
17. Water flux
18. CO_2 flux
19. Charge balance
20. Boundary conditions

Mass balance:

$$\frac{d}{dt}q_o^{Glc} = -v_m^{GLUT2} - v_m^{SGLT1}$$

$$\frac{d}{dt}q_i^{Glc} = v_m^{GLUT2} + v_m^{SGLT1} - v_c^{Resp}$$

$$\frac{d}{dt}q_o^{Na^+} = 3v_m^{NKE} - 2v_m^{SGLT1} - 3v_m^{NCX} - v_m^{NSC}$$

$$\frac{d}{dt}q_i^{Na^+} = -3v_m^{NKE} + 2v_m^{SGLT1} + 3v_m^{NCX} + v_m^{NSC}$$

$$\frac{d}{dt}q_o^{K^+} = -2v_m^{NKE} + v_m^{K,b} + v_m^{Kv2.1} + v_m^{Kv1.5} + v_m^{NSC} + v_m^{BKCa}$$

$$\frac{d}{dt}q_i^{K^+} = 2v_m^{NKE} - v_m^{K,b} - v_m^{Kv2.1} - v_m^{Kv1.5} - v_m^{NSC} - v_m^{BKCa}$$

$$\frac{d}{dt}q_o^{Ca^{2+}} = v_m^{NCX} - v_m^{Cav1.2} - v_m^{Ca,b} + v_m^{BKCa}$$

$$\frac{d}{dt}q_i^{Ca^{2+}} = -v_m^{NCX} + v_m^{Cav1.2} + v_m^{Ca,b} - v_m^{SERCA} + v_{DS}^{Ca^{2+}}$$

$$\frac{d}{dt}q_{DS}^{Ca^{2+}} = v_m^{RyR} - v_{DS}^{Ca^{2+}} - v_m^{BKCa}$$

$$\frac{d}{dt}q_{SR}^{Ca^{2+}} = -v_m^{RyR} + v_m^{SERCA}$$

$$\frac{d}{dt}q_i^{ATP} = 36v_c^{Resp} - v_m^{NKE} - v_m^{SERCA}$$

$$\frac{d}{dt}q_i^{ADP} = -36v_c^{Resp} + v_m^{NKE} + v_m^{SERCA}$$

$$\frac{d}{dt}q_i^{Pi} = -36v_c^{Resp} + v_m^{NKE} + v_m^{SERCA}$$

$$\frac{d}{dt}q_o^{O_2} = -6v_m^{Resp}$$

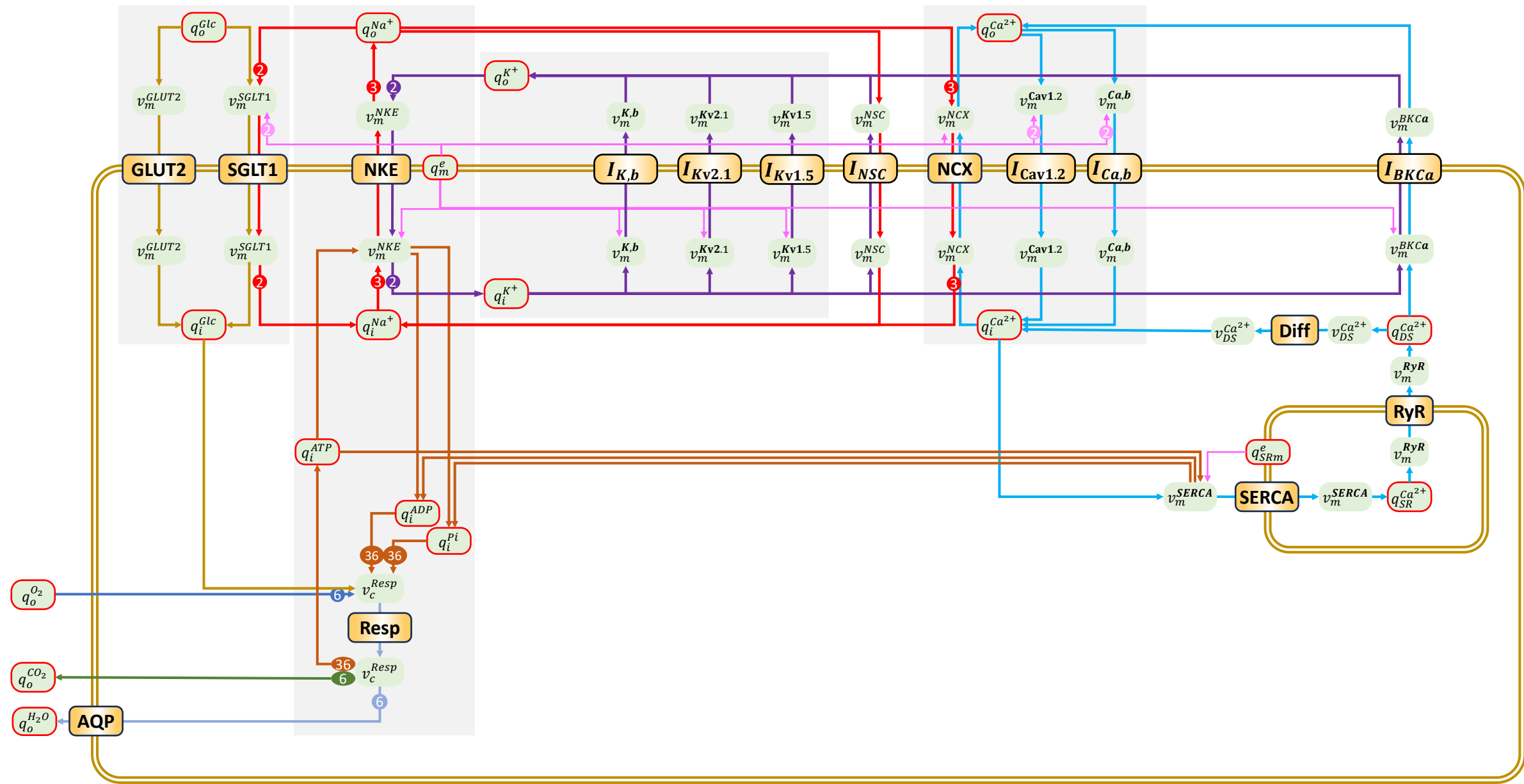
$$\frac{d}{dt}q_o^{CO_2} = 6v_m^{Resp}$$

$$\frac{d}{dt}q_o^{H_2O} = 6v_m^{Resp}$$

Charge balance:

$$\frac{d}{dt}q_m^e = C \frac{du_m^e}{dt} = F(v_m^{BKCa} + 2v_m^{SGLT1} + v_m^{NCX} + 2v_m^{Cav1.2} + 2v_m^{Ca,b} - v_m^{NKE} - v_m^{K,b} - v_m^{Kv2.1} - v_m^{Kv1.5})$$

Vascular smooth muscle cell



Lecture 8: *Myofilament contractile proteins*

Here we consider the mechanics and energetics of cardiac muscle fibres and the control of myofilament force generation by intracellular calcium.

Calcium (Ca^{2+}) binding to TroponinC (TnC) on the actin filament (reaction Re^1) initiates an interaction with Tropomyosin (Tm) (reaction Re^2) that uncovers crossbridge (XB) binding sites on actin (A) to yield a complex A^* that the myosin heads $M.ADP.P_i$ can bind to (reaction Re^4). The bound XB complex $A^*M.ADP.P_i$ (shown here without the TnC and bound Ca^{2+}) releases first P_i (reaction Re^5) and then ADP (reaction Re^6) before binding another ATP (reaction Re^7) and detaching from actin to generate $M.ATP$. The ATP splits (reaction Re^8) to form the primed XB head $M.ADP.P_i$ (i.e., the biochemical splitting reaction generates stored elastic energy in the XB head). The chemical power associated with the XB strain energy is transferred to mechanical power manifest as myofilament force $T = u_m^T$ and myofilament shortening velocity $V = v_m^T$. We also define a nonlinear function $T = f(T_0, V)$, where $T_0(\lambda)$ is the tension-length relation under tetanised (i.e., fully activated) conditions (λ is myofilament stretch). This active force is balanced against the inertial (L), viscous damping (R), and elastic storage (C) components of the muscle. Note that calcium release from A^* (reaction Re^3) is force-dependent and the binding of myosin to actin (reaction Re^4) is length-dependent.

XB mechanics

